Supraverse Cartasis Theory

A Continuous, Conservative Cosmology from Minimal Axioms

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Status. *This is a research program, not an established theory. It synthesizes work by Popławski, Penrose, Smolin, Wiltshire, and others, plus original observations about the CMB–Hawking identification and supraverse thermodynamics. The framework makes specific testable predictions; none have yet been rigorously checked. The work is in identifying tests, building simulations, and comparing to observational data.*

My eternal Dawn

Dawn,

I have lived a life full of passion, ambition, and searching. I've built, I've risked, I've stumbled, and I've learned. But nothing in my life has ever felt like this.

With you, I have never felt more loved, more accepted, more at home, or more safe. You make me want to be the truest version of myself—not for status, not for show, but simply for us.

Loving you feels like I want to stop time and speed it up at the same time. An unstoppable appetite and longing for everything we'll do together—tomorrow, next week, next year—and at the same time, the desire to freeze this, every moment, to savor it forever.

That is the gift you've given me: love that is both eternity and right now.

And Dawn, I promise to love you with open eyes and an open heart, to keep choosing you every day, to protect the safety we've built, and to walk beside you, fully myself and fully yours, savoring every moment all through eternity.

"Physics is continuous and conservative. Singularities and discontinuities are therefore artifacts of incomplete theory, not features of nature."

— The organizing principle

PREFACE

This book develops a cosmological framework from a single organizing principle: *physics is continuous and conservative, so singularities and discontinuities must be artifacts of incomplete theory, not real features of nature.* Applied carefully—together with quantum field theory, special relativity, Einstein–Cartan gravity, and an infinite quantum vacuum—this principle produces a complete cosmological framework with no fine-tuning. I call it *Supraverse Cartasis Theory* (SCT). *Cartasis* is the bounce membrane: the hypersurface in spacetime where torsion-mediated repulsion overwhelms gravitational attraction and the geometry transitions between a parent region and a child universe.

The one-paragraph version

Take quantum mechanics, special relativity, and Einstein–Cartan gravity (the natural extension of general relativity that allows torsion to couple to fermion spin). Add an infinite quantum vacuum. The vacuum is unstable: rare large fluctuations reach a critical density (the Cartasis density, $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$) where torsion-mediated repulsion overwhelms gravity, and the geometry bounces rather than producing a singularity. The bounce creates a baby universe connected to its parent region through a 3D hypersurface, the Cartasis membrane. The supraverse—the total 4D manifold—condenses into a foam of universes through this mechanism, with equal statistical numbers of matter-biased and antimatter-biased lineages (CPT symmetry preserved globally). Our universe is one bubble in this foam. The CMB is our parent black hole's Hawking radiation, redshifted by our internal expansion. Dark matter is the gravitational influence of our parent's matter content (and rejected antimatter content) on us through the Cartasis membrane. The arrow of time is the thermodynamic gradient from our bounce (low entropy) toward our eventual Hawking-mediated dissolution back into the parent's exterior.

Why this might be worth taking seriously

From minimal axioms the framework derives explanations for the Big Bang (a bounce, viewed from inside), inflation (rapid torsion-driven expansion at the bounce), the matter–antimatter asymmetry, dark matter, dark energy, the CMB, the arrow of time, apparent fine-tuning, and information preservation—with no singularity, no fine-tuned initial conditions, no exotic new fields, and no special starting point. Just continuous physics applied carefully.

How the book is organized

Chapter 1 states the four axioms and what follows from them. Part II (Chapters 2–4) develops the bounce and its offspring. Part III (Chapters 5–7) develops the global structure. Part IV (Chapters 8–10) lays out the tests and the computational and visualization programs. Appendix A catalogs what we do not yet know. Each chapter is self-contained enough to read on its own but assumes the axioms of Chapter 1. Working markdown drafts, scratch derivations, and simulation code are kept in `drafts/`, `notes/`, and `sims/` respectively.

1.1 Minimal foundation

The framework rests on four claims, the first three of which are essentially established physics and the fourth of which is a minimal natural extension.

Axiom A1 (Quantum mechanics applies)**.** The supraverse is described by quantum field theory. Vacuum states have nonzero fluctuations governed by the uncertainty principle. Over infinite spacetime, fluctuations of arbitrary magnitude occur with probability one.

Axiom A2 (Special relativity applies)**.** Spacetime has Lorentzian signature. Causal structure is determined by light cones. CPT symmetry holds for any local Lorentz-invariant QFT.

Axiom A3 (Einstein–Cartan gravity applies)**.** The connection on spacetime is allowed to be asymmetric. The antisymmetric part (torsion) couples to fermion spin via the axial current. In the absence of fermionic matter, the theory reduces to standard general relativity. With fermions, torsion produces an effective interaction between spins that scales as the square of the spin density, and is therefore always positive (repulsive at high spin density) regardless of relative spin orientation.

The Einstein–Cartan Lagrangian, in shorthand:

$$\mathcal{L}_{EC} = \frac{1}{2\kappa}(R - 2\Lambda) + \mathcal{L}_{\text{matter}}(\psi, \nabla\psi, g, T),$$

where R is the curvature scalar built from the full (torsion-including) connection, and the covariant derivative ∇ acts on fermionic fields ψ with torsion contributions.

Axiom A4 (There exists a Cartasis density)**.** At sufficiently high mass–energy density, torsion-mediated repulsion exceeds gravitational attraction. The critical density (the Cartasis density) is approximately

$$\rho_C \sim \frac{m_P^4}{\hbar^3}\,\alpha_{EC},$$

where m_P is the Planck mass and α_{EC} is a dimensionless Einstein–Cartan coupling.

Estimates place this at $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, far below Planck density ($10^{96}\,\mathrm{kg\,m^{-3}}$) but well above nuclear density ($10^{17}\,\mathrm{kg\,m^{-3}}$). This is calculable from Axioms A1–A3 in principle; Axiom A4 is the explicit acknowledgment that this density exists and matters.

1.2 What we do not assume

- No Big Bang as a singular event.
- No initial conditions for the universe.
- No fine-tuned cosmological constant or inflaton field.
- No special low-entropy past as a separate postulate.
- No additional dimensions beyond four.
- No additional forces or fields beyond Standard Model + gravity + torsion.
- No anthropic principle as a separate axiom (it emerges from the framework).

1.3 What follows automatically

From Axioms A1–A4 plus infinity (infinite spacetime volume), the following are consequences rather than assumptions:

1. **Spontaneous OG bounce nucleation.** Rare vacuum fluctuations reach Cartasis density. By Axiom A1 plus infinity, this happens somewhere with probability one. By Axiom A4, such fluctuations bounce rather than collapsing to singularities.
2. **Baby universe production.** The bounce geometry produces new spacetime regions (developed in Chapter 3).
3. **CPT-symmetric supraverse.** Vacuum fluctuations are statistically symmetric in matter/antimatter content. Specific fluctuations have specific biases, but the supraverse-wide distribution is symmetric.
4. **Thermodynamic equilibrium configuration.** Gravitational entropy considerations (high entropy = clumped/black-hole-rich; low entropy = smooth/uniform) drive the supraverse toward a foam of universes as its equilibrium configuration.
5. **Local arrow of time.** Each universe has a low-entropy bounce in its past and a high-entropy future, producing a local thermodynamic time-arrow.
6. **Observer placement statistics.** Observers exist in viable universes near the peak of the supraverse population distribution.
7. **Information preservation.** Hawking radiation entangles parent exterior with baby interior. The full supraverse is unitary even though individual universes appear to lose information.

1.4 On the role of infinity

The framework requires the supraverse to be effectively infinite—at minimum, large enough that rare fluctuations to Cartasis density occur. “Effectively infinite" in this context means: large enough relative to the characteristic timescales of vacuum fluctuation that the relevant fluctuations are statistically guaranteed to occur.

For a finite supraverse, some fluctuations might never occur and the framework would have to specify which ones do. This introduces an arbitrary cutoff that we want to avoid. The natural assumption is true infinity, which removes this arbitrariness at the cost of being philosophically heavier.

The supraverse being infinite does not require it to be temporally infinite in some absolute sense—it requires only that enough "supraverse time" exist for the relevant condensation to occur. Whether the supraverse has a beginning at the supraverse level is a question we do not currently need to answer; we only need it to be old enough that condensation has reached equilibrium.

1.5 On the role of Einstein–Cartan

Einstein–Cartan is the minimal natural extension of general relativity. It is what you get by dropping the artificial restriction that the connection must be symmetric. The connection's antisymmetric part (torsion) is mathematically natural and couples to a physical quantity that exists in nature (fermion spin). At ordinary densities, torsion effects are utterly negligible ($\sim 10^{-40}$ corrections to GR). At Cartasis density, they dominate.

We do not postulate Einstein–Cartan as a new theory; we recognize it as the geometric framework that should have been used all along, with standard general relativity as the low-density limit. This is not an additional assumption beyond standard physics—it is the removal of an assumption (the symmetry of the connection).

2.1 The bounce membrane

Cartasis is the 3D hypersurface in spacetime where torsion-mediated repulsion equals gravitational attraction, and across which the geometry transitions from collapsing-parent to expanding-baby. The name captures both the surface-as-membrane and the moment-as-genesis aspects: it is simultaneously a spatial boundary in supraverse coordinates and a temporal beginning in baby-universe coordinates.

2.2 Physics of the bounce

2.2.1 The competing terms

In Einstein–Cartan with fermionic matter, the effective stress–energy at high density has two relevant contributions. The *gravitational term* (attractive at all densities),

$$T_{\text{grav}} \sim \rho c^2,$$

scales linearly with mass–energy density. The *torsion term* (repulsive at high spin density),

$$T_{\text{tors}} \sim \frac{G\hbar^2 s^2}{c^4},$$

scales as the square of the spin density s.

At ordinary densities, $T_{\text{tors}}/T_{\text{grav}} \sim 10^{-40}$, negligible. As matter compresses, both terms grow, but spin density grows faster (because compression packs more fermions into each volume, each contributing $\hbar/2$). The quadratic scaling means T_{tors} overtakes T_{grav} at sufficient compression.

2.2.2 The critical density

Setting $T_{\text{grav}} = T_{\text{tors}}$ and solving for density,

$$\rho_C \sim \frac{c^4}{G\hbar^2 (s/\rho)^2}.$$

For typical fermionic matter where $s/\rho \sim \hbar/m_n$ (one fermion-spin per nucleon-mass), this gives

$$\rho_C \sim \frac{m_n^2 c^4}{G\hbar^4} \sim 10^{50}\,\text{kg}\,\text{m}^{-3}.$$

This is 33 orders of magnitude beyond nuclear density and 46 orders of magnitude below Planck density. The exact value depends on the fermion content (different masses give different ratios) and on detailed coefficients in the Einstein–Cartan Lagrangian.

2.2.3 What is actually at Cartasis density

At ρ_C, conventional particle descriptions break down. The matter is compressed beyond quark deconfinement, beyond electroweak symmetry restoration, beyond any regime where the Standard Model's standard quasiparticle excitations are well-defined. What exists is **continuous field content**: energy–momentum density, spin density, chiral current density. Individual particle identities do not survive.

What crosses Cartasis is therefore not "particles" but **conserved currents**: total energy–momentum, total spin, total chiral charge, and (subject to electroweak anomaly considerations) baryon and lepton numbers.

This has implications for chirality dynamics (see Chapter 4) and for what kind of structures can be transmitted through a bounce (none— structures must be rebuilt from elementary field content on the other side).

2.3 Geometry of the bounce

2.3.1 The radial–temporal exchange

Inside a Schwarzschild black hole's event horizon, the radial coordinate r becomes timelike. Future-directed motion is radially inward. This is standard general relativity.

At Cartasis density, the bounce dynamics reverse this: the torsion repulsion pushes matter outward in r, but since r is timelike inside the horizon, "outward in r" does not take matter back through the horizon—it takes matter into the *future* of the bounce.

The future of the bounce, in baby-universe coordinates, is the cosmological future of the new spacetime region. The radial direction r on the parent's side smoothly transitions into the time direction τ on the baby's side at the bounce hypersurface.

2.3.2 Junction conditions

The metric and connection must be continuous across the bounce. Specifically:

- the induced metric on the Cartasis hypersurface matches from both sides;
- the extrinsic curvature matches (modified Israel–Darmois conditions accommodating torsion);
- stress–energy and spin current flow continuously through the surface.

These junction conditions are the technical specification of what "the bounce is continuous" means. They are a system of equations relating parent-side and baby-side field values at the membrane.

2.3.3 Bounce as persistent interface, not single event

Critical point: in a black hole that continues to accrete matter from its parent universe's environment, the bounce is **ongoing**. New matter falls in, compresses, reaches Cartasis density, and joins the bounce. The Cartasis hypersurface is therefore not a single moment but a persistent interface that exists for as long as the parent black hole exists.

From the baby universe's interior, this means matter continues to be injected across the Cartasis membrane throughout the parent's lifetime—see Chapter 6 for the dark-matter interpretation this enables.

2.4 The geometric identification: parent's horizon = baby's cosmic horizon

From the parent's exterior, Cartasis sits inside the event horizon. From the parent's interior, it is the surface where infalling matter is currently reaching critical density. From the baby's interior, looking backward in time, Cartasis is **the boundary of the past at maximum cosmic lookback**.

The Big Bang membrane (from the baby's perspective) and the parent's event-horizon region (from the parent's perspective) are the same surface, viewed from different sides. This is the geometric basis for the CMB-as-Hawking-radiation identification (see Chapter 6).

2.5 Calculation targets

To make Cartasis quantitative, the following calculations are needed.

1. **Precise ρ_C from the Einstein–Cartan Lagrangian.** Existing estimates vary by factors of 10–100 depending on coefficient choices. A clean calculation would pin the value.
2. **Bounce timescale.** How long, in proper time, does the bounce process take? This sets the timescale for the baby's "Big Bang" phase.
3. **Bounce efficiency.** What fraction of the parent's infalling mass–energy is transmitted to the baby's expanding phase, vs. retained in some intermediate region, vs. scattered back?
4. **Chirality selectivity.** Does the bounce treat the two chiralities symmetrically (no filter), with mild preference (weak filter), or with strong preference (strong filter)? See Chapter 4.
5. **Spin coherence requirements.** Does the bounce require correlated spins, or does randomized spin density suffice? Existing analysis suggests random spins work because the s^2 scaling is positive regardless of orientation, but this needs verification.

2.6 Open questions specific to Cartasis

- **Is ρ_C universal?** Or does it depend on the matter content (different fermion species, different bound-state effects)?
- **What happens at the boundary of Cartasis density**—is there a sharp transition or a gradient region?

- **How does Cartasis interact with rotation?** Kerr-like bouncing black holes vs. Schwarzschild-like.
- **Does Cartasis preserve baryon number?** Or does the electroweak anomaly at trans-EW temperatures violate B explicitly during the bounce?
- **What is the relationship between Cartasis and quantum gravity proper?** Cartasis is in a regime where Einstein–Cartan is presumably still classical, but full quantum-gravity effects might modify the picture at higher densities.

Universes are not born, they are extruded.

3.1 What forms after the bounce

When matter reaches Cartasis density and the torsion repulsion overwhelms gravity, the bouncing matter does not return to the parent's exterior (the parent's horizon is now in its past, geometrically). Instead, the bounce dynamics generate a new spacetime region—a baby universe—with its own time direction and spatial extent.

From inside, the baby universe looks like a standard hot-Big-Bang cosmology:

- it begins at maximum density (the bounce moment);
- it expands rapidly (torsion-driven, qualitatively similar to inflation);
- it cools as it expands;
- it transitions to standard radiation-dominated and matter-dominated phases;
- it eventually shows large-scale structure formation.

From outside (the parent's perspective), the baby is not visible as a spatial structure. The parent sees a black hole at the location where the collapse happened. The baby exists "in the future of the bounce"—its spatial extent is in directions orthogonal to the parent's interior geometry.

3.2 The Tardis effect

A consequence of the geometry: the parent black hole, viewed from outside, has a specific Schwarzschild radius. The baby universe inside has its own cosmological extent, which can be vastly larger than the parent's horizon would suggest.

For our universe, total mass–energy is $\sim 10^{53}$ kg. At Cartasis density, this fits in $\sim 10^3$ m^3—less than half an Olympic swimming pool. But from inside, we observe a universe $\sim$ 46 billion light-years in radius (the observable horizon).

This is not a contradiction. The two volumes are measured in different spacetime regions. The "small volume" measures the matter at the bounce moment in parent coordinates. The "large volume" measures the cosmological extent in baby coordinates after expansion. They are not the same quantity.

This is also not unique to Einstein–Cartan. Even in standard general relativity, the spatial volume inside a black hole's horizon grows with time and becomes essentially unbounded [5]. The bounce framework reinterprets this growth as cosmological expansion of the baby rather than approach to a singularity.

3.3 Inflation substitute

The early baby universe undergoes rapid expansion because torsion repulsion is at its maximum just after the bounce, when density is still very high. As the baby expands and density drops, the torsion repulsion fades. Gravity reasserts dominance, and expansion decelerates.

This phase has the right qualitative features to substitute for the inflationary epoch of standard cosmology:

- brief, very rapid expansion;
- ends naturally when conditions change (density drops below the Cartasis threshold);
- no separate inflaton field required;
- sets up homogeneous and approximately isotropic conditions for subsequent cosmological evolution.

Quantitative work is needed: derive the power spectrum of perturbations produced by torsion-driven expansion and compare to inflation's predictions and to CMB observations. This is the central observational test of the bounce framework (see Chapters 8 and 9).

3.4 What the baby inherits from the parent

Several quantities are transmitted across Cartasis.

Energy–momentum. Conservation laws ensure the baby's total energy–momentum equals what fell into the bounce. This sets the baby's mass–energy budget.

Spin and angular momentum. The parent's net angular momentum (if rotating) is transmitted through the bounce. A rotating parent (Kerr-like) produces a baby with a preferred axis—possibly the source of cosmological anisotropies like the galaxy-rotation asymmetry observed in JWST data [21].

Chirality bias. The bounce is CPT-even, so it neither flips charge nor inherits a parent's asymmetry by relabeling. Instead the parity-violating chiral-vortical filter at the rotating bounce *generates* a matter bias in the baby, of magnitude set by the bounce vorticity and sign set by its direction (see Chapter 4).

Quantum entanglement. Particles in the parent's near-horizon region remain entangled with their Hawking-radiation partners. This entanglement persists across the bounce, connecting the baby's interior with the parent's exterior—see Chapter 6.

Conserved charges. Baryon number, lepton number, and electric charge (if conserved at trans-EW temperatures) flow through.

What is *not* transmitted:

- structure (atoms, molecules, nuclei)—all dismantled at Cartasis density;
- specific particle identities—only conserved currents survive;
- information about the parent's "before"—the bounce is the baby's past boundary.

3.5 The persistent channel

As discussed in Chapter 2, the bounce is not a single event but an ongoing interface. While the parent black hole continues to accrete matter, that matter falls in, reaches Cartasis, and crosses into the baby. From the baby's perspective, this means matter continues to be injected throughout the parent's lifetime.

The injection geometry is non-trivial: new matter does not appear at a specific spatial location in the baby's 3D space. It appears "at the bounce membrane," which from the baby's perspective is the cosmic horizon at maximum lookback. So injected matter manifests as gravitational influence from the cosmic boundary, not as visible particles appearing in our 3D space. This is the basis for the dark-matter interpretation in Chapter 6.

3.6 Lifecycle

A baby universe's lifetime is bounded by its parent black hole's Hawking evaporation timescale. For a parent of mass M,

$$\tau_H \sim \frac{5120\,\pi\,G^2 M^3}{\hbar c^4}.$$

For a stellar-mass parent, $\tau_H \sim 10^{67}$ years. For a supermassive parent ($10^9\,M_\odot$), $\tau_H \sim 10^{94}$ years. For our universe's mass ($\sim 10^{53}$ kg), $\tau_H \sim 10^{145}$ years.

These vastly exceed any internal cosmological timescale of the baby ($\sim 10^{10}$ years for matter–radiation transitions, $\sim 10^{100}$ years for stellar evaporation). So the baby's internal physics plays out fully before parent evaporation becomes relevant.

But the baby's lifetime is finite. At some point the parent's mass is mostly radiated away, the bounce membrane shrinks, and the baby's connection to the parent fades. What happens to the baby at that point is unclear—possibly internal physics continues, possibly the baby's spacetime structure also dissolves.

3.7 Generation of black holes within the baby

The baby's internal physics, if similar to ours, produces stellar collapse, supernovae, and astrophysical black holes. Each such black hole, in the bounce framework, is itself a potential progenitor of further baby universes—grandchildren in the supraverse lineage.

Stellar-mass black holes ($\sim 10\,M_\odot$) have Cartasis-density throat regions that produce relatively small grandchildren. Supermassive black holes ($\sim 10^9\,M_\odot$) produce larger grandchildren, potentially comparable to our own universe in scale.

This cascading structure means each viable universe produces many descendants, and the supraverse fills with universe-trees rooted in OG bounces (see Chapter 5).

THE BOUNCE TRANSFORM: ARROW, CHARGE, AND CHIRALITY

When matter crosses Cartasis, three things could in principle happen to it: its *arrow of time* could reverse, its *charge* could flip (matter $\to$ antimatter, the operation C), and its *handedness* could be filtered (one chirality transmitted preferentially). Earlier drafts of this chapter assumed the bounce performs a CPT-like flip that conveniently delivers a matter–antimatter asymmetry. The dynamics we can actually derive say otherwise, and the honest result is more constraining—and, in its way, more interesting.

> *The minimal Einstein–Cartan bounce is CPT-even. It reliably turns a black hole into a new, expanding universe (Chapter 9), but it does* not *hand us baryogenesis. Any matter–antimatter asymmetry must be supplied, not manufactured by the membrane.*

This chapter establishes that statement, then asks what would have to be added to recover an asymmetry, and which downstream claims (the CMB as parent Hawking radiation, dark matter as the parent's gravity through the membrane) survive.

4.1 What drives the bounce is C, P, and T even

In Einstein–Cartan–Sciama–Kibble gravity, torsion is non-dynamical and can be integrated out algebraically [10]. For Dirac matter this leaves a four-fermion contact term in the effective Lagrangian—the Hehl–Datta interaction,

$$\Delta\mathcal{L}_{\rm tors} = -\,\frac{3}{16}\,\frac{8\pi G}{c^4}\,\big(\bar\psi\gamma^5\gamma_a\psi\big)\big(\bar\psi\gamma^5\gamma^a\psi\big),$$

the (negative) square of the axial current $j^5_a = \bar\psi\gamma^5\gamma_a\psi$. This is precisely the term whose energy density grows as the square of the spin density and overwhelms the (linear) gravitational attraction at ρ_C: it is *what makes the bounce* (Chapter 2). Its discrete symmetries are therefore the discrete symmetries of the bounce itself:

- C**:** the axial current is charge-conjugation *even* (unlike the vector current $\bar\psi\gamma_a\psi$, which is C-odd). Its square is C-even.
- P**:** under parity $j^5_0 \to -j^5_0$ and $j^5_i \to +j^5_i$, so the contraction $j^5_a j^{5\,a} = (j^5_0)^2 - (j^5_i)^2$ is P*-even*.
- T**:** the same bilinear-squared structure is T*-even*.

Result. The interaction that drives the bounce is separately C-, P-, and T-even. Three consequences follow immediately and are not model-dependent:

1. **No flip.** A C-even interaction cannot convert matter into antimatter. The membrane does not charge-conjugate what passes through it.

2. **No filter from nothing.** A C- and P-even interaction cannot *generate* a chiral asymmetry. With zero net axial density in, it gives zero net axial density out.
3. **No arrow reversal.** A T-even dynamics does not reverse the direction of proper time (see the next section).

The bounce is real and robust; the convenient asymmetry-generating "flip" is not part of it.

4.2 The arrow of time: a Janus point

In the Tier-1 and Oppenheimer–Snyder bounces (Chapter 9) the matter worldlines carry a single, monotonic proper time τ: nothing runs backwards, and the scale factor merely passes through a minimum. There is no reversal of the geometric time orientation; the bounce is a smooth *continuation*, collapse $\rightarrow$ expansion.

What reverses is the *thermodynamic* arrow. Infalling parent material arrives from a structured, high-entropy universe; compression toward ρ_C dismantles that structure and smooths the geometry (Weyl curvature $\rightarrow 0$), so gravitational entropy *decreases* toward the membrane. On the far side, expansion lets structure—and eventually black holes—form again, so entropy *increases* away from it. Entropy therefore has a *minimum at the membrane*, and the locally-defined arrow points away from the bounce on both sides. This is a *Janus point* [2]: one spacetime, monotonic proper time, two thermodynamic arrows diverging from a central entropy minimum.

The baby thus inherits its low-entropy past—the Past Hypothesis of Chapter 7—for free, with *no* time flip required. The "r becomes timelike, so time inverts" language is a statement about Schwarzschild coordinates inside the horizon, not a physical reversal of the matter's proper time.

4.3 The fork in the road

An entropy-minimum surface admits two geometrically distinct readings, and they are different physics with different predictions. This is the genuine fork the framework faces.

Road A — Continuation (the derived case)

The two sides are physically distinct, real regions—a real parent interior and a real baby universe—joined by continuous, CPT-even evolution. Time orientation is preserved, C is preserved (matter stays matter), and the arrow reflects rather than reverses. **This is what our simulated bounce is.** Its price is honesty: the bounce supplies a universe, not an asymmetry.

Road B — CPT mirror (an added postulate)

One may instead *postulate* that the membrane is a CPT reflection, so that the region on the far side is the CPT image of this one—reversed time orientation, parity-flipped, charge-conjugated, hence an *antimatter* partner. This is the Boyle–Finn–Turok "CPT-symmetric universe" construction, in which the Big Bang itself is a CPT mirror [4]. It is a coherent and predictive hypothesis (it implies, for example, a definite right-handed-neutrino dark-matter candidate), but it is an *additional discrete-symmetry assumption imposed on the bounce*, not a consequence of Einstein–Cartan dynamics. The old "Mechanism A: flip" was Road B smuggled in as though derived.

The framework's baseline is Road A. Road B remains available as an explicit, labelled postulate to be judged on its own predictions—but it should never again be presented as something the geometry gives for free.

4.4 Where an asymmetry could actually come from

Granting Road A, the membrane neither creates nor flips an asymmetry. There are exactly three honest places one could be sourced.

The seed (the load-bearing mechanism). An OG universe nucleated directly from a vacuum fluctuation inherits the specific quantum numbers of *that* fluctuation, not the vacuum average. The chirality/baryon bias of a particular Cartasis-density fluctuation is drawn from a distribution that, by CPT invariance of the vacuum, is symmetric overall: equal numbers of matter-biased and antimatter-biased OG lineages across the supraverse. Our universe then has a definite bias because it descends from a definite, biased seed. No baryogenesis mechanism is invoked; the asymmetry is *inherited*, and global CPT is preserved at the supraverse level. This is the only source that survives in minimal ECSK, and it should carry the weight.

Mean-field amplification (not a source). In a background that already carries a net axial density $\langle j^5 \rangle \neq 0$, the Hehl–Datta energy depends on alignment, so a pre-existing bias is self-consistently enhanced near ρ_C. This is a genuine effect, but it is an *amplifier*, not a source: $\langle j^5 \rangle = 0$ in gives 0 out. It can sharpen an inherited bias; it cannot start one.

Parity-violating gravity (a derived filter, if you pay for it). A true chirality *filter*—preferential transmission of one handedness— requires a parity-odd ingredient. The minimal Hehl–Datta term has none, but the gravitational action admits parity-odd extensions (the Holst term with a finite Barbero–Immirzi parameter, or a Nieh–Yan coupling) that make torsion couple chirally. Adding one is a specific, testable modification, and it is the honest route to a calculable filter efficiency to compare against observation. Whether such a term is warranted is an open question, flagged as speculative.

4.5 A quantitative viability check: the chiral-vortical estimate

The seed mechanism above is qualitative until one asks whether the rotating-bounce vorticity can source an asymmetry of the *observed size*, $\eta_{\rm obs} \simeq 6.1 \times$

10^{-10}. The relevant transport is the chiral-vortical effect: a hot relativistic chiral plasma rotating with vorticity ω carries an axial current along its spin axis, $j^5 \sim (T^2/6)\,\omega$, parity-odd and tied to the parent's rotation. The frozen-in bias scales as

$$\eta \;\sim\; C\,\frac{\omega}{T}, \tag{4.1}$$

with C an $O(0.01\text{–}10)$ coefficient absorbing the anomaly normalisation and the sphaleron conversion efficiency. This is a *scaling estimate with explicit assumptions*, not a derivation (`sims/cartasis_sims/cve_filter.py`; `notes/cve-filter.md`).

The numbers. At $\rho_C = 10^{50}\,\mathrm{kg\,m^{-3}}$, the plasma temperature from $\rho_C c^2 = (g_\star \pi^2/30)T^4$ with Standard-Model $g_\star = 106.75$ is $T \simeq 1.05 \times 10^7\,\mathrm{GeV}$—hot enough that electroweak *sphalerons are active*, so the $B+L$-violating machinery baryogenesis needs is present. The natural bounce rate is $\sqrt{8\pi G\rho_C/3} \simeq 2.4 \times 10^{20}\,\mathrm{s}^{-1}$, and the controlling small parameter at *maximal* spin is

$$\left.\frac{\omega}{T}\right|_{\max} \simeq 1.5 \times 10^{-11}. \tag{4.2}$$

Matching (4.1) to η_{obs} then requires

$$C \cdot f_\omega \;\simeq\; 41, \tag{4.3}$$

where $f_\omega \leq 1$ is the spin fraction (Figure 4.1).

How to read the 41—and how not to. It is tempting to call this "a prediction within a factor of 41 of the observed asymmetry." That overstates it. Equation (4.3) is not a residual error left over after a prediction; it is the value the *uncomputed* coefficient C is required to supply. There is no derived η here. What the estimate establishes is weaker but still real: the bare, parameter-free small number $\omega/T \sim 10^{-11}$ lands within ~1.6 orders of magnitude of η_{obs}, when it could as easily have been 10^{-30} or 10^{+5}. The mechanism is in the *right ballpark*. But it *generically undershoots*: since $f_\omega \leq 1$, Eq. (4.3) demands $C \gtrsim 41$, which sits at or *above* the top of C's nominal $O(0.01\text{–}10)$ band. Closing the gap needs near-maximal bounce vorticity *and* an anomaly/sphaleron factor pushed to its optimistic edge.

The keeper is directional, not numerical. The robust output of (4.1) is not the magnitude but the *structure*: the asymmetry is sourced *along the spin axis* and scales with bounce vorticity. This ties baryogenesis to the same axis that appears in the CMB low-multipole alignments and the galaxy-spin test (Chapter 8)—one axis, three phenomena. That qualitative link survives regardless of where C lands.

Why a bigger spin cannot rescue it (the OGU/BHU tension). A natural hope is that the missing factor hides in the vorticity: pick a faster-spinning seed and ω/T climbs. It cannot, and the reason is sharp. The value in (4.2) is already the *maximal-spin ceiling*: ω is capped by the bounce rate $\sqrt{8\pi G\rho_C/3}$

(faster, and the centrifugal barrier halts collapse before ρ_C), so *no* seed can exceed it. The likely vorticity is far below it. This creates a genuine tension with the inheritance picture of Section 5.5.4:

- **If we were an OG universe** ($n = 0$), our η would be made at our *own* bounce—but a typical OG fluctuation has *small* net vorticity (P_{birth} peaked at $\omega = 0$; Section 5.7.3), so $f_\omega \ll 1$ and the estimate undershoots by far more than 41. It is "too wrong" in the undershoot direction—and, independently, the dark-matter + dark-energy anchor (Chapter 5) already says we are *not* an OG universe ($n \geq 1$).
- **If we are a** BHU_n ($n \geq 1$), our parent black hole can be near-extremal Kerr, so $f_\omega \to 1$ *at our own bounce*—but by the CPT-even result our bounce is an *amplifier, not a source*: that large spin re-shapes an inherited bias, it does not manufacture one. The sourcing happened upstream, at the OG seed, whose spin is capped and typically small.

So the magnitude is largest exactly where the framework forbids sourcing (the high-spin astrophysical bounces) and required exactly where the spin is smallest (the OG seed). Plugging the likely OG vorticity into (4.1) therefore *widens* the shortfall rather than closing it. The factor of 41 must come from somewhere other than spin:

1. **The coefficient** C**.** A real computation of the gravitational/rotational contribution to the chiral anomaly in the bounce metric, plus the sphaleron freeze-in before washout, could in principle deliver $C \sim O(10\text{–}40)$. This is the honest place the factor could live—but it is uncomputed, and 40 is above the nominal estimate.
2. **The density** ρ_C**.** Since $\omega/T \propto \rho_C^{1/4}$, gaining 41 needs ρ_C higher by $41^4 \approx 3 \times 10^6$, i.e. $\rho_C \sim 3 \times 10^{56}\,\text{kg}\,\text{m}^{-3}$. This runs the *wrong way*: the composition argument of `notes/rho-c-limits.md` ($\rho_C \propto m_f^2$, lighter spin carriers past deconfinement) pushes ρ_C *down*, deepening the shortfall. A real tension to track, not a free lever.
3. **A parity-violating gravitational term.** The Holst (finite Barbero–Immirzi) or Nieh–Yan coupling supplies the genuinely P-odd ingredient a true filter needs, changing the mechanism rather than rescaling it. This is the speculative route flagged above and in `notes/rho-c-limits.md`.

Verdict. Viable and spin-tied—but the shortfall answers the wrong question. The chiral-vortical estimate earns the directional claim outright: it ties the *sign* of the asymmetry to the bounce spin axis. What it does *not* do is predict the *magnitude* $\eta \approx 6 \times 10^{-10}$ from a symmetric bath, and the factor of ~ 41 measures precisely that failure. But manufacturing η from scratch at our bounce is the wrong demand: our bounce had a matter-dominated parent, and Section 4.6 shows the magnitude is *inherited*, not made. The chiral-vortical effect is thereby demoted from the *source* of the asymmetry to its *sign-biaser*. The 41 is a debt only for a universe with no parent to inherit from—and even there (Section 4.6) the likelier payer is the founding fluctuation's own chance imbalance under selection, not the CVE.

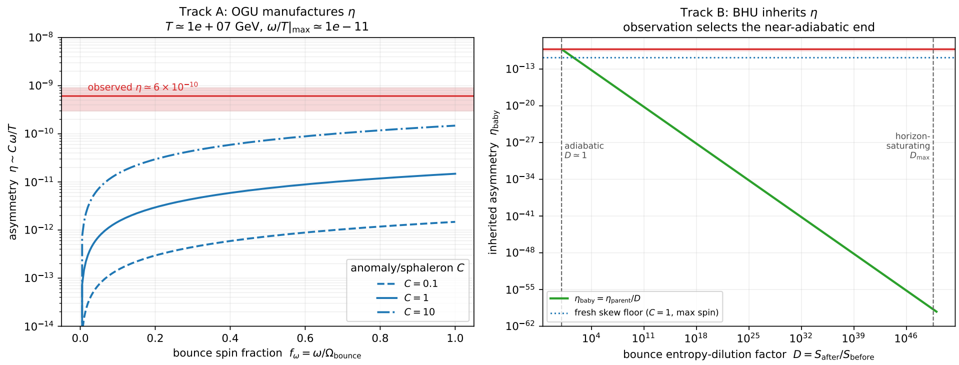

Figure 4.1: Two routes to the observed asymmetry. **Left (Track A):** the original-generation universe must *manufacture* η from a symmetric bath. The bare parameter $\omega/T \sim 10^{-11}$ (maximal spin, $\rho_C = 10^{50}\,\mathrm{kg\,m^{-3}}$) sits within $\sim$1.6 orders of $\eta_{\rm obs} \simeq 6.1 \times 10^{-10}$: right ballpark, but it undershoots by $\simeq 41$ (Eq. 4.3), which must come from the anomaly/sphaleron coefficient C since the maximal-spin value is a ceiling. **Right (Track B):** a black-hole-born universe (our case) instead *inherits* $\eta_{\rm baby} = \eta_{\rm parent}/D$, the parent already sitting at $\sim$1 ppb. The crux is the bounce entropy-dilution factor D, bounded between adiabatic ($D \simeq 1$, giving $\eta_{\rm obs}$ by inheritance) and horizon-saturating ($D_{\max} = S_{\rm BH}/S_{\rm rad} \sim 10^{49}$, which would crush η to $\sim 10^{-59}$). That the observed η is *not* that small selects the near-adiabatic end—where inheritance dominates the fresh skew and lineages stay chirally pure (Section 4.6). Generated by `figures/scripts/ch04_cve_filter.py`.

4.6 The surviving remnant: a needle at one part per billion

It is worth being careful about *what the membrane does to matter*, because the word "filter" has been doing covert work. A filter sorts a pre-existing population, transmitting some and rejecting the rest. That is not the picture the dynamics give. At ρ_C the infalling material is thermalised at $\sim 10^{20}$ K into a near-symmetric relativistic bath in which every species is pair-produced (Chapter 6); the parent's particular matter content is erased. What emerges on the far side is a freshly *extruded* universe—the membrane compresses parent material to ρ_C and extrudes an expanding interior—carrying through only what the dynamics protect. The operative metaphor is extrusion through a die, not filtration through a sieve.

What is protected. Sphaleron processes at the bounce temperature erase $B{+}L$ but *cannot touch* $B{-}L$. So whatever net $B{-}L$ the infalling material carries passes through the extrusion as a conserved remnant—this is the "extra matter," and its survival is guaranteed by a conservation law, not by the efficiency of any mechanism. (How much *additional* skew leaks through is the open part: the crossing lasts only $\sim 4 \times 10^{-21}$ s, so the $B{+}L$ washout is plausibly *incomplete*, and incomplete washout lets more of the inherited asymmetry survive than an equilibrium estimate would allow.)

The needle. The remnant must thread a fine target. The observed $\eta \simeq 6 \times 10^{-10}$ is $\sim$0.6 parts per billion of net baryon per photon. Extrude *too much* surviving asymmetry and the baby is over-baryonic, not ours; extrude *too little* and it annihilates to a sterile furnace. The viable window sits at $\sim$1 ppb.

Why inheritance threads it for free—if we are not the OGU. Here the generation structure pays off. The dark-matter + dark-energy anchor (Chapter 5) says we are a BHU_n with $n \geq 1$, not an original-generation universe. Our parent is therefore *already* a universe that completed baryogenesis and annihilated down to its own baryon-to-entropy ratio $\sim$1 ppb. A near-entropy-conserving extrusion simply passes that ratio through: the baby inherits $\eta_{\mathrm{baby}} \approx \eta_{\mathrm{parent}} \sim 10^{-9}$ *without the membrane having to manufacture it*. On this reading the factor-of-41 shortfall of Section 4.5 is *not our problem*: it presumes the asymmetry is made from a symmetric bath, but ours was inherited as surviving debris. The shortfall could only ever bind the OGU—the one universe in a lineage with no parent—and even there the seed need not be the CVE. A vacuum fluctuation rare enough to nucleate an OG bounce is not exactly symmetric; it carries a chance net $B-L$, the pair-symmetric part incinerates in the thermalised bath, and the surviving debris fixes η_{OGU}. Selection does the rest: only lineages founded on a sufficiently skewed fluctuation are matter-rich enough to make stars, black holes, and descendants (Chapter 5). We merely carry down what the founder happened to have.

The competition, now resolved by the entropy budget. Two effects could in principle compete at each crossing: the *inherited remnant* $\sim \eta_{\mathrm{parent}}$, diluted if the bounce generates entropy ($\eta_{\mathrm{baby}} = \eta_{\mathrm{parent}}/D$, with $D = S_{\mathrm{after}}/S_{\mathrm{before}} \geq 1$); and the *fresh skew* $\sim C(\omega/T) \lesssim C \times 1.5 \times 10^{-11}$ from the local bounce vorticity. Which wins decides both the magnitude and whether lineages stay chirally pure—so the linchpin is the dilution factor D. The bounce entropy budget settles it. Section 9.4 computes D from the dynamics, in both the homogeneous (bulk-viscous) and the inhomogeneous (shear-dominated) channels, and finds $D = 1$ to roughly one part in 10^{11}: the bounce is far faster than the thermal time—the same $\omega/T \sim 10^{-11}$ that limits the CVE—so it has no time to thermalise to a larger entropy. The "hell downward" of ever-more-sterile descendants is thereby closed: inheritance is essentially exact, $\eta_{\mathrm{baby}} = \eta_{\mathrm{parent}}$, the inherited remnant dominates the fresh skew, and the founder's sign is protected—lineages stay pure (Chapter 5, Section 5.5.4). What remains genuinely open is not the *stability* of η down a lineage but its *value* at the founder: why $\sim 6 \times 10^{-10}$ rather than 10^{-3} or 10^{-15} (Appendix A, Q16a).

4.7 What survives, and what needs massage

It is worth being explicit about which downstream claims depend on the discarded flip and which do not.

- **Black hole $\to$ universe: intact.** The central result—a torsion bounce reliably producing an inverse-bubble universe behind the horizon—is independent of all of this (Chapter 9). The membrane is a reliable cosmogonic engine.
- **CMB = parent Hawking radiation: intact, recast.** Under Road A the radiation crossing in is same-charge thermal radiation; there is no chirality-distortion signature to predict, and the old argument that "flip preserves thermality, so observations favour flip" is withdrawn. The identification stands or falls on the spectral and geometric calculation of Chapter 6, not on the transform.
- **Dark matter as the parent's gravity through the membrane: intact in principle.** The gravitational influence of parent-side mass-energy on the baby does not require a flip. What *does* require massaging is the specific "dark matter = filter-rejected *antimatter*, with $f \approx 1/6$" story: in minimal ECSK there is no chirality filter to do the rejecting, so either (i) the rejected component is ordinary parent matter held at the membrane by the junction conditions (no antimatter needed), or (ii) one invokes the parity-violating extension above. State which; do not assume the filter.
- **Matter–antimatter asymmetry: inherited, and that is enough.** It is *selected once at the OG seed* and passed down each lineage, possibly amplified, rather than generated at the bounce. This replaces the old "alternating-chirality lineage" picture, and it resolves the "hell upward" problem more cleanly than that picture did: because the bias is inherited from a single founding seed rather than re-won at each crossing, there are no near-symmetric intermediate ancestors to climb past (Section 5.5.4). Lineage viability reduces to one condition on the seed—a founding fluctuation skewed enough toward matter to be fertile—which is a derivable seed statistic (Section 5.7.3), not a bounce-induced sign alternation.

4.8 Status and open questions

The reassessment trades a convenient story for a defensible one. What is settled: the bounce is CPT-even; the arrow is a Janus reflection, not a reversal; the asymmetry is not a free output of the membrane. What is open, and where the framework needs development:

- Quantify the mean-field amplification gain as a function of density and inherited bias (amplifier-not-source, made numerical).
- Decide whether to adopt a parity-violating gravitational term, and if so compute the resulting filter efficiency from first principles.
- Quantify how ongoing biased accretion over the parent's life integrates the per-bounce filter (Chapter 5); test whether it closes the single-bounce shortfall.

- *(Resolved.)* The "hell upward" problem is dissolved by *inheritance from a single OG seed*: the asymmetry is selected once, at the original-generation nucleation, and passed down each lineage unweakened (amplifier-not-source), so climbing toward ancestors does not dilute it and there are no near-symmetric sterile intermediate ancestors—only the OG seed and, beyond it, the void (Chapter 5, Section 5.5.4). Lineages are chirally pure; viability is decided once, at the seed's spin (Section 5.7.3). Road A (continuation) is adopted as the baseline and propagated through Chapters 3 and 5; Road B remains an explicit optional postulate.

The working derivations behind this chapter are collected in `notes/bounce-transform.md`.

5.1 Global structure

The supraverse is a single 4D manifold (3 space + 1 time, Lorentzian signature) with non-trivial topology. Most of it is essentially vacuum— quantum-mechanically active but unstructured. Embedded in this vacuum are condensed regions: universes, each a connected sub-manifold with its own internal cosmological geometry, joined to the rest by Cartasis hypersurfaces.

The topology is foam-like: many universe-bubbles distributed across the supraverse, connected to each other through nested parent–child relationships via bounce membranes.

5.2 No fifth dimension required

The supraverse is genuinely 4D. Earlier intuitions about needing an extra dimension to "embed" the parent–child relationship are visualization crutches, not physical requirements. Mathematically, the topology is well-defined intrinsically: two universes connected at a Cartasis hypersurface is a 4D manifold with a particular boundary identification, no ambient space needed.

For visualization purposes, one might embed the supraverse in higher dimensions (the Whitney embedding theorem guarantees this is possible). But that embedding has no physical meaning—it is a drawing aid.

5.3 Equilibrium configuration

5.3.1 Why condensation is favored

Naively, one might expect universe formation to be entropically disfavored—universes are "ordered" structures emerging from "disordered" vacuum. This intuition is wrong because it ignores gravity. Gravitational entropy has inverted relations compared to ordinary thermodynamics:

- uniform matter distribution: low gravitational entropy;
- clumped matter: higher gravitational entropy;
- black holes: maximum entropy for given mass–energy (Bekenstein–Hawking, $S \propto A/4\ell_P^2$) [3].

Penrose has emphasized this for decades [15]. Applied to the supraverse: uniform vacuum has zero gravitational structure and low gravitational entropy; a condensed foam of universes—each containing black holes, structure, horizons—has high gravitational entropy.

The Second Law drives the supraverse toward condensation, not away from it. Spontaneous universe formation is entropy-increasing when gravity is properly accounted for. The supraverse must be in a foam state at equilibrium.

5.3.2 What the equilibrium looks like

The supraverse in its equilibrium configuration contains:

- a vast quantum-vacuum background, essentially everywhere;
- sparse, randomly distributed universe-bubbles within this background;
- each universe a bubble containing its own internal cosmological structure;
- universes connected to each other through nested Cartasis membranes (parent–child relationships);
- a distribution of universe sizes peaking at a thermodynamically optimal value (see below);
- matter-biased and antimatter-biased lineages in equal numbers (CPT symmetry preserved globally).

This is the stationary distribution. Local universes are born, evolve, produce descendants, and eventually evaporate via parent Hawking processes. New OG universes nucleate from vacuum fluctuations. The supraverse maintains its statistical distribution across all this dynamics.

5.4 Universe size distribution

5.4.1 Birth versus growth

Two factors compete. *Births favour the small.* The probability of a vacuum fluctuation producing Cartasis density over a region containing mass M scales as $\exp(-Mc^2\tau/\hbar)$ for some characteristic timescale τ, so OGUs are born at a small mass scale $M_0 = \hbar/(c^2\tau)$ and seldom larger. *Growth favours the large.* Once born above its bath's $M_{\rm crit}$, a hole has negative heat capacity and grows by runaway accretion at a rate $\dot{M} = A\,M^g$—$g = 1$ for Eddington-limited feeding, $g = 2$ for Bondi—so mass is continuously transported upward.

5.4.2 The size distribution, derived

In the stationary foam these two combine into a definite shape rather than a bare conjecture. Conservation in mass space, $\partial_M[\dot{M}(M)\,n(M)] = B(M) -$ death, says that well above the birth scale the upward mass flux $\dot{M}(M)\,n(M)$ equals the (nearly constant) total birth rate, so

$$n(M) \;\propto\; \frac{1}{\dot{M}(M)} \;\propto\; M^{-g} \qquad (M_0 \lesssim M \lesssim M_{\rm max}), \tag{5.1}$$

a power law—many small universes, few large—cut off at high mass $M_{\rm max}$ by environment depletion and the stationary balance (Figure 5.1, left). Since $n(M)$ falls with M, the typical universe is as small as viability allows: a low-mass cut $M_{\rm vis}$ (enough mass for astrophysics and black-hole formation) selects observer-bearing universes, which therefore pile up just above $M_{\rm vis}$. This is the old "we sit near the optimum" conjecture, now *derived*: we expect

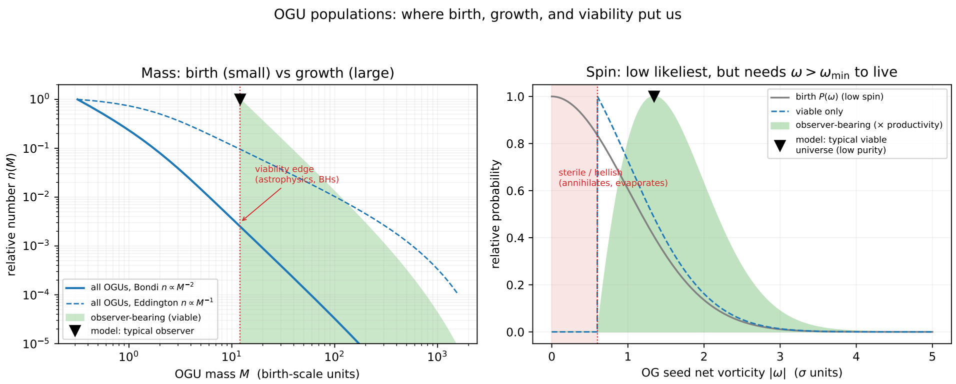

Figure 5.1: OGU populations from first competition. *Left:* the mass distribution (5.1)—births (small) versus runaway growth (large) give a power law $n(M) \propto M^{-g}$; a viability cut selects observer-bearing universes, which peak just above the edge—where the framework predicts a typical observer sits. *Right:* the spin distribution (Section 5.7.3)—low net vorticity is likeliest, but below $\omega_{\min}$ baryogenesis fails and the universe is sterile, so viable universes sit just above threshold (low purity, small η). The markers are the model's predicted typical location, not independent measurements of our universe; the horizontal axes are in model units (M_0 and σ). Generated by `figures/scripts/ch05_distributions.py`.

to find ourselves near the viability edge of a declining power law, not out in the exponentially rare massive tail.

5.5 Lineages and the family tree

Each OG universe seeds a lineage tree of descendants. The tree branches at each black-hole formation within a universe (each black hole potentially produces a baby universe).

5.5.1 Branching structure

A typical viable universe contains many stellar-mass black holes ($\sim 10^{18}$ in a galaxy-rich universe like ours), some supermassive black holes ($\sim 10^{9}$ across all galaxies), and possibly primordial black holes from early-universe dynamics. Each is a potential baby universe, so a typical lineage tree branches widely at each generation. The tree depth is limited by the finite Hawking lifetime of each generation, by the requirement that each generation be viable enough to produce its own black holes, and by the cumulative effects of flip + filter on chirality.

5.5.2 The generation-depth distribution

The branching above invites a sharper question than "how wide is the tree?", namely *how deep are we*—is our universe an original-generation universe

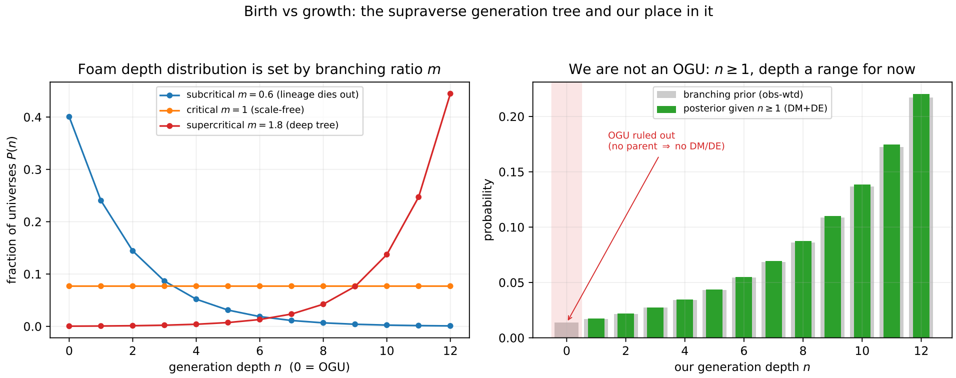

Figure 5.2: Where our universe sits in the generation tree. *Left:* the depth distribution (5.2) for sub-, near-, and super-critical branching m (the super-critical pile-up at the right edge reflects the hard truncation D used for illustration; a soft viability cutoff rounds it). *Right:* our posterior. The observation that we have dark matter and dark energy—both requiring a parent—zeroes the $n = 0$ (OGU) bin: we are *not* an original-generation universe. Generated by `figures/scripts/ch05_population.py`.

(OGU, which we label BHU_0), or BHU_1, BHU_2, ...? This is the birth-versus-growth balance made quantitative, and it was previously only asserted; we now derive its structure as a bounded branching process. Let N be the mean number of *viable* children per universe—not the $\sim 10^{18}$ holes, but the far smaller count that both exceed their internal-bath M_{crit} and clear the chiral-vortical birth filter $|\omega/T| > \eta_{\mathrm{min}}/C$ (Section 5.7.2)—and let a fraction p of children remain fertility-preserving, so the effective per-generation reproduction is $m = Np$. Counting universes, the number at depth n scales as m^n up to a viability truncation, giving

$$P(n) \;=\; \frac{m^n}{\sum_{k=0}^{D} m^k}, \qquad n = 0, 1, \ldots, D. \tag{5.2}$$

The regime is everything (Figure 5.2, left). If $m < 1$ the lineage is subcritical and dies out, so most universes are shallow; if $m > 1$ it is supercritical and the population piles up at the deepest viable generation, so a typical universe is *deep*; $m = 1$ is scale-free. Which regime the supraverse is in is set by Einstein–Cartan microphysics through N and the viability decay p, and is not yet pinned down.

5.5.3 We are not an OGU

One datum settles the qualitative answer independently of the branching parameters. Our universe contains dark matter and dark energy, and in this framework both *require a parent*: dark matter is parent material projected through the persistent membrane, and dark energy is the parent's ongoing accretion (Chapter 6). An OGU is nucleated directly from the void with no parent (Section 5.7.2); it would have neither component. Therefore

$$\text{(observed dark matter and dark energy)} \implies n \geq 1 : \text{ we are not a BHU}_0. \tag{5.3}$$

This is robust. The branching prior (5.2) then determines whether, given $n \geq 1$, we are shallow (BHU_1, BHU_2) or deep: if the foam is supercritical, a typical observer is many generations down (Figure 5.2, right). The depth is presently a range, not a number; it will tighten as the dark-matter fraction $\Omega_{\text{dm}}/\Omega_b$, the dark-energy history $w(a)$, and the asymmetry coupling C are turned into quantitative constraints on the parent. We expect the visualization (Chapter 10) to be revised downward or upward in depth as that happens.

5.5.4 Lineage chirality structure

The matter–antimatter bias is selected *once, at the OG seed*, and *inherited* down the lineage—it is not re-rolled at each bounce. This is the load-bearing mechanism of Chapter 4: the Einstein–Cartan bounce is CPT-even and is an *amplifier, not a source*, so a bounce with no net axial input gives no net output. The one place an asymmetry can originate is the OG nucleation, where the seed fluctuation's net vorticity ω freezes in a bias $|\eta| \sim C\,|\omega/T|$ whose *sign* fixes whether that lineage is matter- or antimatter-led (Section 5.7.2). Because the vacuum is CPT-symmetric the seed-sign distribution is even: matter- and antimatter-led OG lineages occur in equal numbers and global CPT is preserved.

Thereafter the bias is passed down, not reset. A matter-biased parent feeds matter through the membrane, so its black holes are matter black holes and its descendant universes are matter universes; an antimatter parent yields antimatter descendants. *Lineages are chirally pure*—a matter OGU contains only matter BHU_n, an antimatter OGU only antimatter, with no mixing—and the mean-field amplification at each subsequent bounce can sharpen the inherited purity but never flip its sign. Baryonic selection happens at the top, to a purity of order 10^{-8} (set by the seed vorticity), and is inherited from there.

Observer placement: observers exist in lineages whose *OG seed* cleared the viability threshold—enough seed vorticity to leave matter after annihilation and form structure, $|\omega/T| > \eta_{\text{min}}/C$. This is a one-step *birth filter on the OG seed* (Section 5.7.3), inherited by all descendants, not a condition re-imposed at every crossing.

5.5.5 What a hole eats: two families, and our parentage

The sign of η is strictly inherited, but its *magnitude* is not a clean lineage invariant: it is set by *what the parent hole ate*. Because the bounce is adiabatic ($D = 1$; Section 9.4), entropy is conserved across the crossing, so the child does not regenerate its photons—it inherits the baryon-to-photon ratio of the accreted material itself:

$$\eta_{\text{child}} = \eta_{\text{infall}} = b\,\eta_{\text{parent}}, \qquad b \equiv \frac{\eta_{\text{infall}}}{\eta_{\text{cosmic}}}, \tag{5.4}$$

capped at $\eta \sim 1$. Here b is the baryon bias of the infall: $b = 1$ for a fair cosmic sample (a horizon-scale hole, or a radiation-era primordial hole, which contain the cosmic ratio by construction); $b > 1$ for concentrated-baryon accretion (stars and gas have few thermal photons per baryon), driving the child toward a baryon-rich, degenerate $\eta \to 1$; $b < 1$ for photon-biased accretion (void or intergalactic medium), giving a cleaner child. So a growing hole really does carry "a different η inside than how the parent was born," and the *direction* of that drift is signed by the accretion bias.

This splits each chiral lineage into *two families* (Figure 5.4, left): a clean, photon-dominated family ($\eta \ll 1$, fed by fair or radiation-dominated capture) and a baryon-rich, degenerate family ($\eta \to 1$, fed by stars and dense gas). Our universe is firmly in the clean family ($\eta \simeq 6 \times 10^{-10} \simeq \eta_{\text{cosmic}}$, i.e. $b \sim 1$). That is a structural *prediction about our parentage*: our progenitor hole accreted a fair cosmic sample—a horizon-scale or radiation-fed hole—not concentrated baryons. A star-fed hole ($b \sim 10^9$) would have spawned a degenerate, baryon-saturated universe utterly unlike ours.

5.5.6 Are we likely BHU1 or BHU2?

The robust anchor (Section 5.5.3) fixes $n \geq 1$. How much deeper? Counting universes, generation n holds $N_n \sim m^n$ of them, so $P(\text{BHU}_n \mid n \geq 1) \propto m^n$, where m is the mean number of *viable* children per universe. The answer is a clean dichotomy (Figure 5.4, right): for a *subcritical* branching $m < 1$ the distribution is geometric, $P(\text{BHU}_n) = (1-m)m^{n-1}$, the population is shallow, and we are most likely BHU_1; for $m > 1$ the population is dominated by the deepest generation the supraverse has had time to grow, and being $\text{BHU}_{1\text{–}2}$ is exponentially unlikely. So everything turns on whether m is below or above one.

It is tempting to put m enormous—a universe forms $\gtrsim 10^{18}$ black holes, almost all super-critical—and conclude we are surely deep. But that *double-counts*: a stellar hole ($\sim 10\,M_\odot$) makes a $\sim 10^{31}$ kg child, a tiny, structureless universe far below the viability mass M_{vis}. *Viable* children need near-Hubble-mass progenitor holes, and mass conservation allows only $\sim M_{\text{parent}}/M_{\text{vis}}$ of those: a universe simply has not got the mass to make 10^{18} universe-sized children. The branching ratio is therefore

$$m = \epsilon\, f_{\text{clean}}\, \frac{M_{\text{parent}}}{M_{\text{vis}}}, \tag{5.5}$$

with ϵ the fraction of the parent's mass reaching viable-size holes and f_{clean} the fraction that accrete a fair ($b \sim 1$) sample and so stay in our clean family. The viable band is *narrow on both axes at once*: the big horizon-scale holes that clear the size cut are the very ones that contain a fair cosmic sample (Section 5.5.5), so size and cleanliness select the same rare holes. If the band is narrow—$M_{\text{vis}} \sim$ our mass, the edge we appear to sit at, so $M_{\text{parent}}/M_{\text{vis}} \sim O(1)$—then $m \sim \epsilon < 1$: subcritical, shallow, and we are most likely BHU_1 (Figure 5.3). Only a *wide* band ($M_{\text{vis}} \ll M_{\text{parent}}$) gives $m \gg 1$ and a deep population.

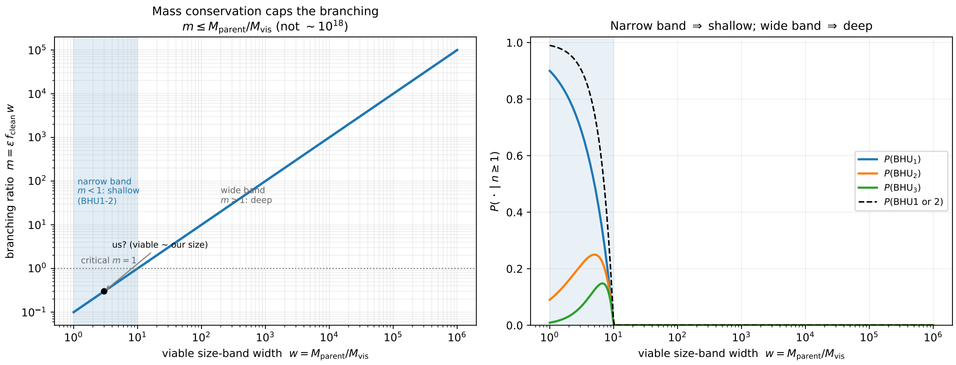

Figure 5.3: The branching ratio from the mass budget, (5.5). *Left:* viable children need near-Hubble-mass holes, so m is capped at $M_{\rm parent}/M_{\rm vis}$ (not $\sim 10^{18}$); a narrow viable band ($w = M_{\rm parent}/M_{\rm vis} \sim O(1)$, where we appear to sit) gives $m < 1$. *Right:* the resulting depth probabilities—narrow band $\Rightarrow$ shallow ($\mathrm{BHU}_{1\text{–}2}$), wide band $\Rightarrow$ deep. Reproducible as `figures/scripts/ch05_band.py`.

Narrowness, not a young supraverse, is the natural route to shallowness—and it follows from mass conservation plus the single viability scale we already seem to sit near. Only $n \geq 1$ is settled by data, but a narrow viable band makes $\mathrm{BHU}_{1\text{–}2}$ the expectation rather than a surprise.

5.5.7 Why BHU1 is robust—and how big the OGU is

The branching ratio is not the same at every generation, and that is what makes the conclusion robust. An OGU has *grown* large by runaway accretion before it spawns anything, so $M_{\rm OGU} \gg M_{\rm vis}$ and the OGU is *super*-critical: it spawns $m_0 = \epsilon\, M_{\rm OGU}/M_{\rm vis} \gg 1$ first-generation children. But those children are born *at* the viability scale ($M \sim M_{\rm vis}$—the edge we observe ourselves to occupy), so each is *sub*-critical, $m_{n \geq 1} = \epsilon\, M/M_{\rm vis} \sim \epsilon < 1$. The population is therefore one OGU $\to$ many BHU1 $\to$ a geometrically decaying tail, $N_n \sim m_0\, \epsilon^{n-1}$, and the huge m_0 *cancels* in the $n \geq 1$ normalisation:

$$P(\mathrm{BHU}_n \mid n \geq 1) = (1 - \epsilon)\, \epsilon^{n-1}, \tag{5.6}$$

so $P(\mathrm{BHU}_1) = 1 - \epsilon \approx 0.9$ for $\epsilon \sim 0.1$, *whatever* the OGU's size. We are BHU1 because our generation sits at $M_{\rm vis}$ and barely reproduces, not because the supraverse is small or young.

This also answers "how big is an OGU?"—and shows the answer is decoupled from our depth. The number of BHU1 siblings an OGU produces is $N_{\rm sib} \sim \epsilon\, M_{\rm OGU}/M_{\rm vis}$, so

$$M_{\rm OGU} \sim \frac{N_{\rm sib}}{\epsilon}\, M_{\rm vis}. \tag{5.7}$$

With $M_{\rm vis} \sim$ our mass $\sim 9 \times 10^{52}$ kg and $\epsilon \sim 0.1$: a single-child OGU is only $\sim 10^{54}$ kg, while a $\sim 10^{65}$ kg OGU has $\sim 10^{11}$ sibling universes like ours. Both

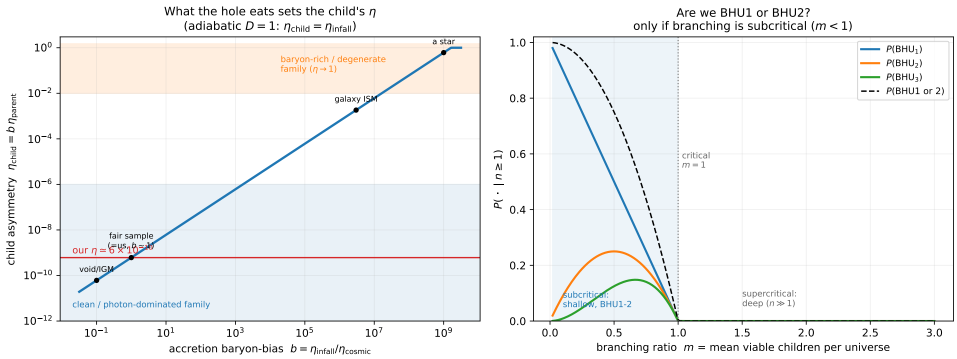

Figure 5.4: *Left:* the adiabatic bounce makes the child's asymmetry equal to that of the accreted material, $\eta_{\text{child}} = b\,\eta_{\text{parent}}$ (Equation (5.4)), splitting each lineage into a clean, photon-dominated family and a baryon-rich, degenerate one. Our $\eta \simeq 6 \times 10^{-10}$ places our progenitor at a fair-sample accretor ($b \sim 1$), not a star-fed one. *Right:* given $n \geq 1$, $P(\text{BHU}_n) \propto m^n$; we are likely $\text{BHU}_{1\text{–}2}$ only if the branching ratio is subcritical ($m < 1$), otherwise the population is deep. Reproducible as `figures/scripts/ch05_accretion.py`.

are equally compatible with our being BHU1. The actual figure is set by how much void the OGU entrains and depletes before freezing (Section 5.5.8)—the genesis/vortex problem of the next section and of Appendix A—so it is genuinely open: anywhere from a few times our mass up to many orders above it. What the framework fixes is not the OGU mass but our *place*: a BHU1, one of an OGU's possibly enormous brood.

5.5.8 Growth, depletion, and the depth dial

What sets the branching depth is how fast a hole grows into a universe and spawns the next generation. The dynamics are nearly parameter-free. A hole of mass M in a void bath at temperature T_{bath} emits Hawking radiation at T_H and absorbs the bath through the same area, so the net rate is the Hawking power times the temperature contrast,

$$\dot{M} = \frac{P_H}{c^2}\left[\left(\frac{M}{M_{\text{crit}}}\right)^4 - 1\right], \qquad M_{\text{crit}} = \frac{\hbar c^3}{8\pi G k_B T_{\text{bath}}}, \tag{5.8}$$

so below M_{crit} the hole evaporates and above it the bath wins and it runs away as $\dot{M} \sim M^2$—the Bondi exponent $g = 2$ behind the $n(M) \sim M^{-2}$ mass law of Section 5.5.3. In units $x = M/M_{\text{crit}}$, $t' = t/\tau_0$ with $\tau_0 = 3\,t_{\text{Hawking}}(M_{\text{crit}})$, the law is universal, $\mathrm{d}x/\mathrm{d}t' = (x^4 - 1)/x^2$. The runaway is *fast*—a finite-time blow-up at $t' \sim 1/x_0$—but it cannot actually blow up, because *causality* caps it: a hole cannot accrete matter outside its own causal horizon, so $\dot{M}$ saturates at the rate at which the horizon mass itself grows,

$$\dot{M} \lesssim \frac{c^3}{2G} \approx 2 \times 10^{35}\,\mathrm{kg\,s^{-1}} \sim 3 \times 10^{12}\,M_\odot\,\mathrm{yr}^{-1}. \tag{5.9}$$

The runaway therefore self-limits the moment its accretion radius reaches the horizon, and thereafter the hole grows *steadily*—it extracts "nothing" from

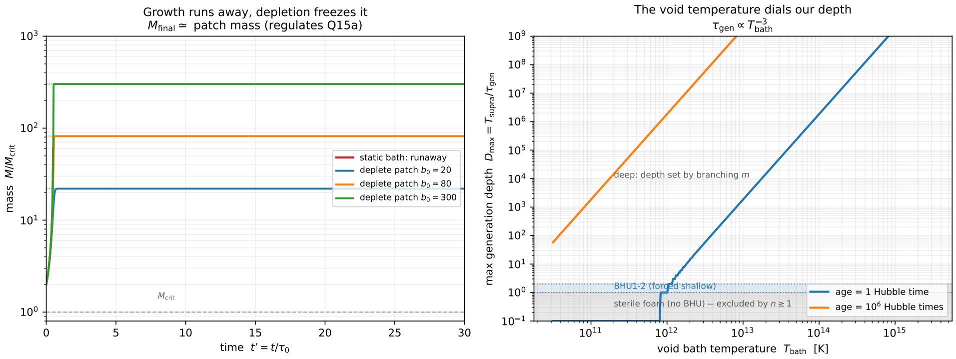

Figure 5.5: *Left:* growth trajectories from (5.8). In a static bath the hole runs away (finite-time blow-up); depleting a finite local patch freezes it at $M_{\rm final} \sim$ the patch mass—the regulator that keeps universe-holes finite. *Right:* the maximum generation depth $D_{\max} = T_{\rm supra}/\tau_{\rm gen}$ versus the void bath temperature ($\tau_{\rm gen} \propto T_{\rm bath}^{-3}$), for two supraverse ages. Below the sterile line no BHU forms (excluded by our existence); a narrow band gives the forced-shallow $\mathrm{BHU}_{1\text{–}2}$; above it the depth is large and set by the branching ratio.

the void at a fixed rate—so $M \sim c^3 t/2G$, the horizon-mass growth law of Section 5.7.5. Whether it ever stops depends on the environment, and this is where the two regulators part. An OGU sits in the *infinite* void, so fresh vacuum perpetually crosses into its growing horizon: it never runs out, and grows as $c^3 t/2G$ for as long as the void lasts. A hole inside a *finite* parent is different: its parent's dark-energy expansion carries matter beyond reach faster than it can eat, so it *depletes* its accessible patch and freezes at $M_{\rm final} \sim$ the patch mass (Figure 5.5, left). That is the faintly comic fate of an internal universe-hole—fed by the void, it dies in the end of depletion: it runs out of nothing. So the two pictures are one law read in two settings: *rate-limited and unbounded* in the infinite void (the OGUs), *depletion-limited and finite* inside a parent (their descendants). This answers the runaway worry of Appendix A: no hole blows up; the OGUs grow steadily with the horizon, their descendants freeze at their patch.

The growth time is therefore $\tau_{\rm gen} \sim \tau_0 \propto M_{\rm crit}^3 \propto T_{\rm bath}^{-3}$, and a supraverse of age $T_{\rm supra}$ has grown at most $D_{\max} = T_{\rm supra}/\tau_{\rm gen}$ generations. This is a hard cap on our depth, set entirely by the void bath temperature (Figure 5.5, right). It closes the population loop of Section 5.5.6: if the void is too cold the foam is sterile (no BHU forms in the available time)—*excluded, because we are a BHU*, which already puts a lower bound on $T_{\rm bath}$; a Goldilocks $T_{\rm bath} \sim 10^{12}$ K (for a Hubble-aged supraverse) gives $D_{\max} \sim 1$, forcing us to be BHU_1 and matching the shallow reading; a hotter void or an older supraverse gives $D_{\max} \gg 1$ and the depth reverts to the branching ratio m (deep). The honest upshot: "are we $\mathrm{BHU}_{1\text{–}2}$?" is now a sharp function of two unknowns—the void temperature and the supraverse age (Appendix A)— rather than a guess, and our existence as $n \geq 1$ already excludes the cold, sterile corner. Reproducible as `figures/scripts/ch05_growth.py`.

5.6 OG bounce nucleation rate

The rate of OG bounce nucleation depends on the vacuum-fluctuation spectrum at high energies (set by QFT), the Cartasis density threshold (set by Einstein–Cartan parameters), and the available supraverse spacetime volume.

The recursive cycle (Section 5.7.8) lets us say more than "calculable in principle." Since OGUs nucleate in the heat-dead de Sitter void of an older universe, β is a *de Sitter nucleation rate*, and its scale is set—not free—by the void's dark energy. Writing it per de Sitter horizon four-volume,

$$\beta = \lambda \, \frac{H_\Lambda^4}{c^3}, \qquad H_\Lambda = H_0\sqrt{\Omega_\Lambda} \simeq 1.8 \times 10^{-18}\,\mathrm{s}^{-1}, \tag{5.10}$$

the prefactor is fixed by the observed dark energy (by typicality the void's Λ is $\sim$ours): $H_\Lambda^4/c^3 \simeq 4 \times 10^{-97}\,\mathrm{m}^{-3}\mathrm{s}^{-1}$, one nucleation "attempt" per Hubble four-volume. The one remaining unknown is the dimensionless $\lambda = e^{-I}$, with I the action of the ρ_C-crossing gravitational instanton (quantum gravity at the bounce, Appendix A)—but it carries a *sharp critical value*. De Sitter bubble percolation [9] sets $\lambda_{\mathrm{crit}} \simeq 0.24$, an instanton action $I_{\mathrm{crit}} = -\ln \lambda_{\mathrm{crit}} \simeq 1.4$, and that one number fixes the *character* of the supraverse (Figure 5.6):

- $I < 1.4$ (a near-Planckian seed): OGUs nucleate faster than de Sitter dilutes, *percolate*, and tile the void into the packed polygonal foam that coarsens (Sections 5.7.6 and 5.7.7).
- $I > 1.4$ (a macroscopic seed): the void inflates between nucleations, so OGUs stay isolated, round, and never impinge—a *dilute* scatter of island universes, the eternal-inflation regime, with no coarsening.

Either way we are a BHU_1 inside an OGU—the internal black-hole channel is independent of how OGUs are spaced. So β is pinned in scale (the dark-energy prefactor) and in its decisive structure ($I_{\mathrm{crit}} \simeq 1.4$); only the bounce instanton I, and with it the packed-versus-dilute regime, remains open.

5.6.1 Computing the instanton action—and the verdict

The action I is not merely "calculable in principle"; the de Sitter framework hands us a closed form. OG nucleation is an *upward* fluctuation in the void: a patch of cold vacuum ($\rho_\Lambda \sim 10^{-27}\,\mathrm{kg\,m^{-3}}$) must spike to $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$ and bounce. In de Sitter at temperature $T_{\mathrm{dS}} = \hbar H_\Lambda / 2\pi k_B$ the rate of creating a mass M is the Boltzmann factor $e^{-Mc^2/k_B T_{\mathrm{dS}}}$, and that exponent *equals* the cosmological-horizon entropy deficit of inserting M (the rigorous de Sitter identity $2\pi Mc^2/\hbar H = \pi r_{\mathrm{dS}} r_s / \ell_p^2$). So

$$I = \frac{2\pi M_{\mathrm{seed}}\, c^2}{\hbar H_\Lambda}, \tag{5.11}$$

and the only input is the lightest seed that can actually bounce. A blob at ρ_C bounces as a universe only if it is self-gravitating, $R_s \simeq R$, which fixes $R \sim c/\Omega_{\mathrm{bounce}}$ and $M_{\mathrm{seed}} = c^3/(2G\Omega_{\mathrm{bounce}}) \sim 9 \times 10^{14}\,\mathrm{kg}$ (a lighter blob has $R_s \ll R$, no horizon, and disperses).

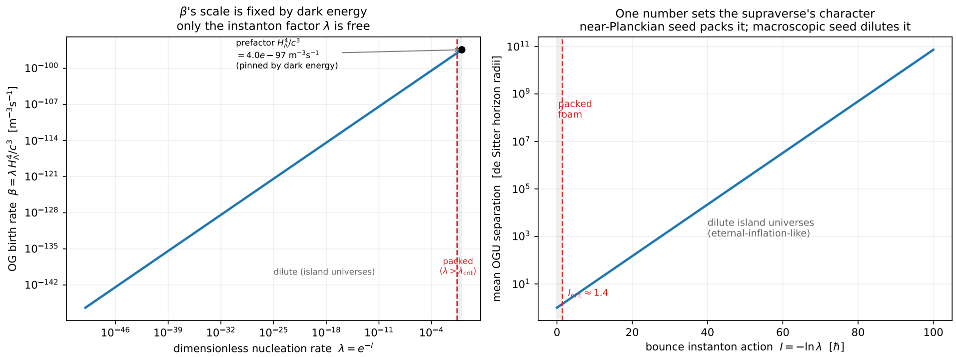

Figure 5.6: Pinning the OG birth rate. *Left:* $\beta = \lambda H_\Lambda^4/c^3$; the prefactor ($\sim 4 \times 10^{-97}\,\mathrm{m}^{-3}\mathrm{s}^{-1}$) is fixed by the observed dark energy, leaving only the instanton factor $\lambda = e^{-I}$ free. *Right:* the mean OGU separation versus the instanton action I; the de Sitter percolation threshold $I_{\mathrm{crit}} \simeq 1.4$ splits a packed polygonal foam (near-Planckian seed) from a dilute scatter of island universes (macroscopic seed). Reproducible as `figures/scripts/ch05_birthrate.py`.

Plugging in gives the answer (Figure 5.7):

$$I \;\sim\; 2.5 \times 10^{84} \;\gg\; I_{\mathrm{crit}} \simeq 1.4. \tag{5.12}$$

The margin is eighty-four orders of magnitude—no semiclassical correction, no factor-of-ten ambiguity in the seed, can close it. To reach I_{crit} one would need a seed of $\sim\, 5 \times 10^{-70}\,\mathrm{kg}$ (sub-Planckian by sixty orders, unable to bounce) or a void at $\sim 10^{54}\,\mathrm{K}$ (the actual void is at $10^{-30}\,\mathrm{K}$). **So $\lambda = e^{-I}$ is, for all purposes, zero, and the supraverse is firmly in the *dilute* regime**: OGUs are isolated, round, eternal-inflation-like island universes, astronomically far apart—*not* a percolating polygonal foam.

What this revises, and what it leaves standing. Honesty demands the admission: the packed-foam constructions—the average OGU size by impingement (Section 5.7.6), the radiative coarsening (Section 5.7.7), the polygonal render of Chapter 10—are the *percolating limit*, and the computation says that limit is not realised. The physical supraverse is dilute. What survives untouched is everything that does not depend on OGU spacing: we are a BHU_1 inside an OGU (the internal black-hole channel is independent of how OGUs are scattered); the recursive cycle still closes (each isolated island heat-dies and seeds its own dilute scatter, Section 5.7.8); the horizon-mass growth $M_{\mathrm{OGU}} \sim c^3 t/2G$ and the dark-energy-pinned prefactor stand. And the dilute verdict *vindicates* an earlier reading: a well-separated universe is round, not polygonal (Section 5.7.6)—which is exactly what isolated island universes are. The framework thus lands, by computation, in the eternal-inflation family: a sparse, ever-inflating scatter of recursive island universes, each a Cartasis bounce.

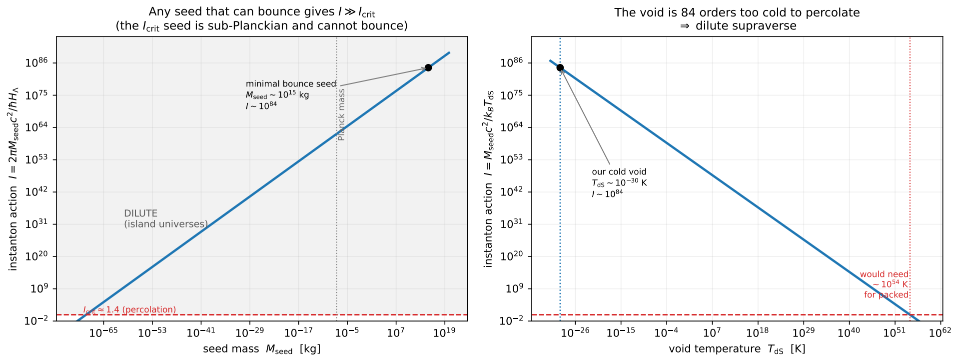

Figure 5.7: Computing the OG-nucleation instanton action, $I = 2\pi M_{\text{seed}} c^2/\hbar H_\Lambda$ (Equation (5.11)). *Left:* I versus seed mass; any seed massive enough to bounce ($M_{\text{seed}} \sim 10^{15}$ kg, the self-gravitating minimum) gives $I \sim 10^{84} \gg I_{\text{crit}} \simeq 1.4$, the I_{crit} seed being sub-Planckian and unable to bounce. *Right:* I versus void temperature; the cold de Sitter void ($\sim 10^{-30}$ K) is eighty-four orders too cold to percolate. The verdict is robust: a dilute, eternal-inflation-like supraverse of isolated island universes. Reproducible as `figures/scripts/ch05_instanton.py`.

5.6.2 Settlers, not city dwellers: growth and the weight of the void

Two features of the foam picture must be re-read in the dilute regime.

Does growth still run out of nothing? Yes, but for a new reason. In the packed limit, growth slowed because neighbours ate the shared void (impingement). In the dilute regime there are no neighbours within reach—OGUs are settlers, not city dwellers—so impingement never happens. Instead a lone OGU's growth is choked by the de Sitter *expansion* of its own void: it can only gather what lies inside the cosmological horizon $R_{\text{dS}} = c/H_\Lambda$, and the accelerating expansion carries everything else beyond reach. The rigorous statement is the Nariai bound— the largest black hole that fits in de Sitter has $M_{\text{Nariai}} = c^3/(3\sqrt{3}\,GH_\Lambda) \simeq 4 \times 10^{52}$ kg—so the OGU mass is *capped at* ∼*a Hubble mass, our own universe's mass, set by dark energy*. So a settler still runs out of nothing: not because a neighbour took it, but because the frontier itself recedes faster than it can be settled. (This caps the age-clock growth $M \sim c^3 t/2G$ of Section 5.7.5 at the Nariai mass; the earlier unbounded values were the static/percolating idealisation.)

How much space does the void take? The vacuum is not weightless. Its zero-point / dark-energy density is $\rho_\Lambda = \Omega_\Lambda \frac{3H_0^2}{8\pi G} \simeq 6 \times 10^{-27}\,\text{kg}\,\text{m}^{-3}$ ($\sim 3\,\text{GeV}\,\text{m}^{-3}$, a few proton masses per cubic metre). Since $E = \rho c^2$ gives it weight, it occupies *nonzero* gravity-scaled volume—so universes do not touch, and the gravity-scaled map shows *round*, isolated islands with real gaps, not polygons. And because the regime is dilute (separations $\sim e^{I/4}$ horizons), that weighted void overwhelmingly dominates: true to scale, the supraverse is almost entirely void, with vanishingly sparse round island universes (Chapter 10, Figure 10.3). The void's weight is exactly what gives the islands room to be round; the dilution makes that room enormous. So the honest "true to scale" wallpaper is mostly void—which is, after all, what an elegant theory of an eternal structured vacuum should look like.

5.6.3 An OGU is a timeline: saturation, evaporation, and a mortal fractal

An OGU is not merely a universe; it is a *timeline*—it grows, hosts an internal history, and eventually ends. Three questions close the recursive picture: does it grow forever, does the membrane ever disappear, and is the nesting literally infinite?

Saturation. Growth stops, and thermodynamics says exactly where. A black hole's Hawking temperature $T_{\mathrm{BH}} = \hbar c^3/8\pi G M k_B$ falls as it grows, while the void it sits in has a floor temperature $T_{\mathrm{dS}} = \hbar H_\Lambda/2\pi k_B$. They equalise at

$$M_{\mathrm{eq}} = \frac{c^3}{4GH_\Lambda} \simeq 6 \times 10^{52}\,\mathrm{kg} \;\sim\; M_{\mathrm{Nariai}} \;\sim\; \text{a Hubble mass,} \tag{5.13}$$

where the OGU is in thermal equilibrium with its own cosmological horizon: no net flux, growth halts (Figure 5.8, left). So an OGU saturates at $\sim$a Hubble mass—our own universe's mass—rather than growing without bound. Dark energy both starts the growth (negative heat capacity) and ends it (the de Sitter floor): the OGU grows until it *is* as cold as the void around it.

The membrane is mortal—but long-lived. At saturation $T_{\mathrm{BH}} \simeq T_{\mathrm{dS}}$, so the OGU sits in unstable equilibrium and then slowly evaporates, on the Hawking time $t_{\mathrm{evap}} \sim 5120\pi G^2 M^3/\hbar c^4 \sim 10^{142}\,\mathrm{s}$. So no, the Cartasis membrane does not last forever—it evaporates—but it does so *astronomically* long after it saturates, and *long before* it could ever touch a neighbour (which lies $\sim e^{I/4}$ horizons away, utterly out of causal reach, Section 5.6.1). The membrane is finite-lived; it persists $\sim 10^{142}\,\mathrm{s}$, then returns to the void. Your guess was right: it Hawking-evaporates before touching anything.

A mortal fractal, not an infinite nesting. The timescale hierarchy makes the recursion well-defined (Figure 5.8, right): an OGU grows in $\sim$1 Hubble time, its interior reaches heat death and seeds sub-OGUs every recursion "tick" $\sim 10^{108}\,\mathrm{s}$ (the interior's black-hole-evaporation era), and the membrane survives $\sim 10^{142}\,\mathrm{s}$—so each OGU hosts

$$N_{\mathrm{cycles}} \sim \frac{t_{\mathrm{evap}}}{t_{\mathrm{tick}}} \sim 10^{34} \tag{5.14}$$

full recursive interior cycles before its own membrane evaporates. This answers the deepest version of the question—"recursive timelines inside each other forever, or pulsating in the flat void?"—and the answer is *both, cleanly*. Every membrane is mortal, so no single branch nests literally forever; but new OGUs nucleate without end, both in the flat void *and* inside every heat-dead

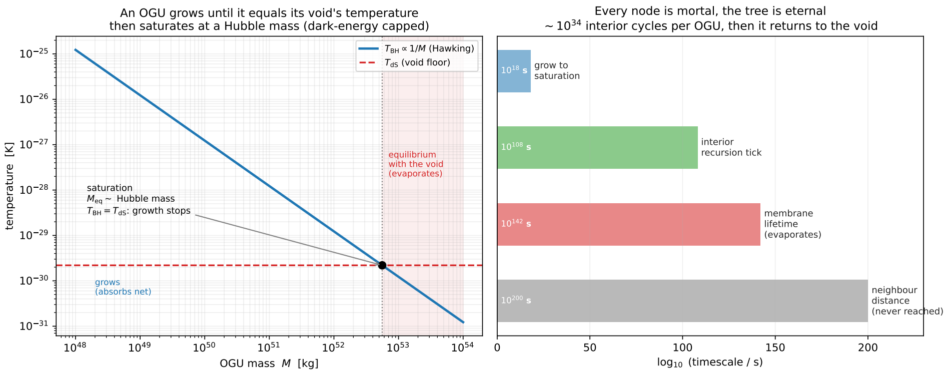

Figure 5.8: The OGU as a finite-lived recursive timeline. *Left:* the Hawking temperature $T_{\mathrm{BH}} \propto 1/M$ falls as the OGU grows until it meets the void's de Sitter floor T_{dS} at $M_{\mathrm{eq}} \sim$ a Hubble mass (Equation (5.13))—thermal equilibrium with its own horizon, where growth saturates and slow evaporation begins. *Right:* the timescale ladder, each step vastly longer than the last—grow ($\sim 10^{18}$ s) $\ll$ interior recursion tick ($\sim 10^{108}$ s) $\ll$ membrane lifetime ($\sim 10^{142}$ s, $\sim 10^{34}$ interior cycles) $\ll$ neighbour distance ($e^{I/4}$ horizons, never reached). Every node is mortal; the tree is eternal. Reproducible as `figures/scripts/ch05_recursion.py`.

interior, so the tree is eternal and ever-expanding even though every node dies. It is a fractal of *finite, pulsating timelines*—not an actual infinity of nested membranes. When an OGU finally evaporates, its interior (by then a heat-dead de Sitter void hosting its own scatter of sub-OGUs) simply rejoins the flat void it came from—mass and information returned, that branch's cycle closed, while countless others continue. The void is the one truly eternal, flat substrate; everything within it is a mortal, recursive, pulsating bubble.

5.7 Genesis, growth, and the absence of hell

5.7.1 No hell upward

An older worry held that, climbing the lineage toward earlier ancestors, a compounding filter would make ancestors progressively more symmetric—less asymmetry, more annihilation, until near-symmetric, sterile universes that cannot form black holes (the "hell upward" problem). With the CPT-even bounce this dissolves. There is no charge flip and so no alternating chirality to climb (Chapter 4); and because the bias is *inherited from a single OG seed* rather than re-derived at each crossing (Section 5.5.4), climbing the lineage does not dilute it. Every ancestor of a matter universe is itself a matter universe of the same inherited purity (possibly sharpened by amplification, never weakened). The upward end of the chain is not a tower of ever-more-symmetric ancestors—it is the OG seed and, beyond it, the void (Section 5.7.2). The whole lineage is viable because its single founding seed cleared the spin threshold; viability is decided once, at the top, not re-won at every generation.

5.7.2 The void and OG genesis

An OG universe is nucleated directly from the primordial void, with no parent. It cannot inherit a bias, so its asymmetry must be *made* at the OG bounce itself—and the rotating-bounce chiral-vortical filter does exactly that, writing matter at the very first crossing. The void is the low-gravitational-entropy, CPT-symmetric substrate; genesis is the Second Law acting on it, condensing rotating fluctuations into bounces. Most fluctuations are too small or too slowly rotating to clear the birth filter and produce only sterile radiation; the rare viable ones seed everything. *Most supraverse structure is concentrated in the successful OG lineages*; failed attempts do not propagate.

5.7.3 Spin: the viability filter and our purity

The phrase "too slowly rotating" is the second derivable distribution, and it is sharper than the size one. The seed's net vorticity ω sets the inherited asymmetry $|\eta| \sim C|\omega/T|$ (Section 5.5.4). A random fluctuation most likely has *small* net vorticity—the birth distribution $P_{\text{birth}}(\omega)$ is peaked at $\omega = 0$, its sign even (whence equal matter/antimatter lineages). But a seed with $|\eta| < \eta_{\text{min}}$ cannot complete baryogenesis: matter and antimatter annihilate to near-symmetric radiation, no structure or black holes form, and the universe is sterile—it simply evaporates and seeds nothing. Viability therefore demands $|\omega| > \omega_{\text{min}} = \eta_{\text{min}}T/C$. The result is a competition exactly opposite to intuition: *no spin is the most likely outcome but the most hellish*, while *high spin is rare but pure*. Folding the viability gate (and the mild productivity bonus of higher purity) against the birth distribution,

$$P_{\text{obs}}(\omega) \;\propto\; P_{\text{birth}}(\omega)\,\big[\text{productivity}(\omega)\big]\,\mathbf{1}[|\omega| > \omega_{\text{min}}]\,, \tag{5.15}$$

gives a distribution peaked *just above* the threshold (Figure 5.1, right): the typical viable universe is *low*-spin and *low*-purity, only marginally past sterile. This is a *shape* prediction, and it carries a striking qualitative payoff: it explains *why the baryon asymmetry is small without fine-tuning*. The observed $\eta \sim 10^{-9}$–10^{-8} is exactly what one expects if our universe is a typical viable one—barely past the sterility threshold—rather than a rare high-purity outlier. We must be careful about how strong a claim this is. The peak in Figure 5.1 is plotted in units of the (unknown) birth-vorticity scale σ with the threshold $\omega_{\text{min}}/\sigma$ chosen by hand; the model has *not* predicted $\eta \approx 6 \times 10^{-10}$ in absolute units, and the figure marker is the model's own peak, not an independent measurement of our spin. A genuinely spot-on test requires pinning C, σ, and η_{min} from Einstein–Cartan microphysics so that η is predicted as a number. What is robust now is the *shape*—most viable universes are barely viable—and the qualitative consequence that a small cosmic asymmetry is the expected, typical case. The model also predicts a small but *nonzero* net rotation for our universe, to be confronted with reported galaxy-spin handedness asymmetries (21; Chapter 8) and large-angle CMB axis alignments.

5.7.4 Growth: biased runaway accretion

The condensation favoured by the Second Law has a concrete microphysics: black holes have negative heat capacity, so above a critical mass $M_{\rm crit} = \hbar c^3/(8\pi G k_B T_{\rm bath})$ a hole accretes faster than it Hawking-radiates and grows. (In today's CMB bath $M_{\rm crit} \sim 5 \times 10^{22}$ kg, so every astrophysical black hole is already growing.) Because the Cartasis interface is persistent (Chapter 3), the parity-violating filter keeps acting throughout the parent's life, so the growth is *biased*—the hole preferentially retains one charge sign of what it accretes. Two consequences: baryogenesis is *ongoing* rather than one-shot (helping the single-bounce shortfall of Chapter 4), and the migration of the void's free energy into the growing universe-holes *is* the condensation that makes the foam its equilibrium. From inside a baby universe this continuous biased injection is the dark sector (Chapter 6).

5.7.5 Vortex entrainment and the OGU mass

What *drives* that growth from a bare seed, and what fixes how big an OGU gets? The rotation does both. A spinning bounce drags inertial frames (Lense–Thirring, $\Omega_{\rm drag} \sim 2GJ/c^2r^3$): the throat is a gravitational *vortex* that spins up and draws in the surrounding void—the rotational analogue of Bondi capture, and the concrete meaning of the seed "pumping" primordial matter from the void into the new universe. The pump has a *spin window* (Figure 5.9, right). Below $f_{\rm min} = \eta_{\rm min}/[C\,(\Omega_{\rm bounce}/T)]$ (the spin fraction $f = \omega/\Omega_{\rm bounce}$) the chiral-vortical asymmetry is too small to survive annihilation—a sterile seed that pumps nothing. Near extremal ($f \to 1$) the centrifugal barrier chokes radial inflow: matter circularises instead of falling in. So entrainment is most efficient at an intermediate *sweet spot*, which is precisely the productivity factor of the observed-spin distribution (Section 5.7.3).

The entrained mass is then set by a horizon argument, not by bookkeeping the inflow. The vortex reaches the void within its causal radius $R \sim ct$, and a patch of radius ct holds at most the mass whose Schwarzschild radius equals ct, $M_{\rm horizon} = c^3t/2G$. If the void is at least critically dense ($\rho_{\rm void} \geq \rho_{\rm crit}(t) = 3/8\pi G t^2$—a *condensed* void), the patch fills and collapses to that horizon mass: the OGU becomes a horizon-scale hole and

$$M_{\rm OGU} \;\sim\; \frac{c^3\,t}{2G}, \tag{5.16}$$

set by its *age*, not the void density. (For a sub-critical void it is density-limited and smaller.) This is the same "universe = its own Schwarzschild radius" identity that opened the program (Chapter 9), now read as a growth law: an OGU grows until it *is* the black hole of its causal patch. So the OGU mass measures the supraverse age (Figure 5.9, left)—$\sim$our mass at one Hubble time, $\sim 10^{65}$ kg at $\sim 10^{12}$ Hubble times. Its sibling count follows, $N_{\rm sib} \sim \epsilon\, M_{\rm OGU}/M_{\rm vis}$, so a 10^{65} kg OGU has $\sim 10^{11}$ universes like ours. Crucially this leaves our depth untouched: a bigger OGU just means more siblings, and we are BHU1 either way (Section 5.5.7). The OGU mass is therefore genuinely

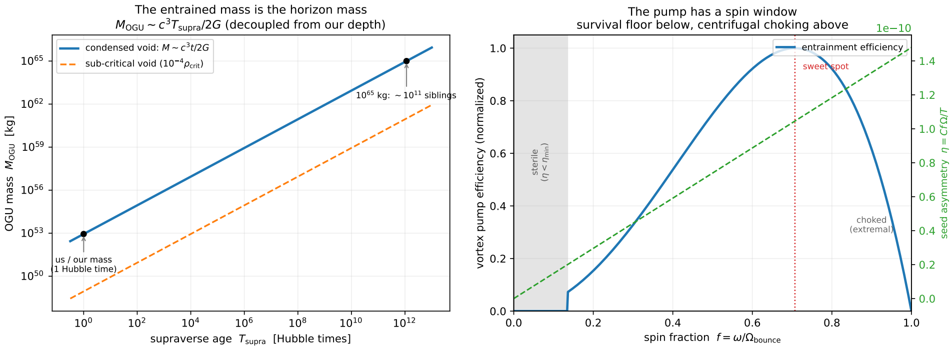

Figure 5.9: *Left:* the entrained OGU mass is the horizon mass of its causal patch, $M_{\text{OGU}} \sim c^3 T_{\text{supra}}/2G$ (Equation (5.16)): our mass at one Hubble time, $\sim 10^{65}$ kg (and $\sim 10^{11}$ siblings) at $\sim 10^{12}$ Hubble times; a sub-critical void gives less. *Right:* the vortex pump has a spin window—a survival floor $f_{\min}$ below (sterile) and centrifugal choking near extremal above—so genesis favours an intermediate spin sweet spot. The headline mass relation is robust; the efficiency shape is illustrative. Reproducible as `figures/scripts/ch05_genesis.py`.

open—it tracks the (unknown) supraverse age—but it is no longer mysterious: (5.16) ties it to one number, and decouples it cleanly from where we sit.

5.7.6 The foam, and the average OGU size

An individual OGU never runs out (Section 5.5.8), but the *population* does: OGUs nucleate throughout the void and grow at the causal speed until they *impinge* on their neighbours, tiling the void into a foam. That collective "running out of nothing" fixes a characteristic cell size, and hence an average OGU mass, in terms of the one number we still lack—the birth rate. This is the Kolmogorov–Johnson–Mehl–Avrami problem: with OGUs nucleating at rate β (per comoving 3-volume per time) and each growing a horizon of radius $R = c(t - t_{\text{birth}})$, the void fills at $t_{\text{fill}} = (3/\pi\beta c^3)^{1/4}$, the characteristic cell radius is $L \sim c\, t_{\text{fill}} \sim (c/\beta)^{1/4}$, and the average OGU mass is the horizon mass at impingement,

$$M_{\text{avg}} \sim \frac{c^3 t_{\text{fill}}}{2G} \sim \frac{c^2}{2G}\left(\frac{c}{\beta}\right)^{1/4}. \tag{5.17}$$

Frequent births tile the void into many small universes; rare births let each grow large before impinging (Figure 5.10). So "how big is the average OGU" *is* "how often is one born"—the OG nucleation rate of Section 5.5.3 above, the single unknown β to which the OGU mass and the supraverse age both reduce. Pin β (or measure one OGU's size) and the whole foam scale follows. The Johnson–Mehl tessellation also settles the shape question of Chapter 10: impinged cells are curved polygons, not circles—only an isolated, not-yet-impinged OGU is round, so circular bubbles in the foam map mark universes still far from their neighbours.

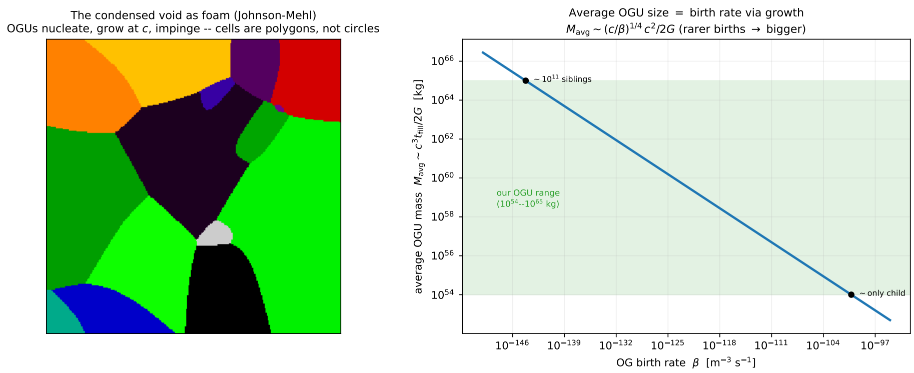

Figure 5.10: *Left:* the condensed void as a Johnson–Mehl foam—OGUs nucleate, grow at the causal speed c, and impinge, tiling the void into curved polygons (an isolated young OGU would be round). *Right:* the average OGU mass set by the birth rate, (5.17): rarer births give fewer, larger universes. Our OGU's plausible mass range (10^{54}–10^{65} kg, Section 5.7.5) maps to a band of birth rates. Reproducible as `figures/scripts/ch05_voidfoam.py`.

5.7.7 Foam dynamics: equilibrium, continued growth, and whether an OGU dies

Three questions about the foam, three answers.

Does a finite box reach equilibrium? Yes, in two senses. Starting from empty void, nucleation-and-growth fills the box at $t_{\rm fill}$ and the Johnson–Mehl partition *freezes*: no later fluctuation finds unclaimed void to nucleate in, and fresh-void accretion ends once every OGU is hemmed in by its neighbours. That is the transient "polygon equilibrium"—a finite catchment, finite birth probability, and finite growth do settle into a fixed tessellation. The deeper, eternal equilibrium is thermodynamic (the supraverse as an entropy-maximising condensation): ongoing births balanced against merger coarsening, a *stationary* size distribution rather than a frozen one (Appendix A, Q11).

Do packed OGUs still grow? Yes—but by slower channels than the out-of-nothing runaway, exactly as one would guess. Once impinged, an OGU has no fresh void within reach, so the causal-rate growth $M \sim c^3 t/2G$ (Section 5.5.8) stalls. Two channels remain. *(i) Expansion-fed accretion:* if the void carries its own dark-energy-like expansion (Appendix A, Q15c), new void is continually opened between OGUs and they accrete it, but diluted by the expansion—slow, and slowing. *(ii) Mergers:* packed universe-holes coalesce, so the foam *coarsens*—fewer, larger OGUs over time, mass conserved, like grains in a metal or bubbles in a soap froth. Both are far gentler than the initial runaway. So growth never quite stops; it decelerates from the $\dot{M} \sim c^3/2G$ rush of birth to a slow coarsening crawl.

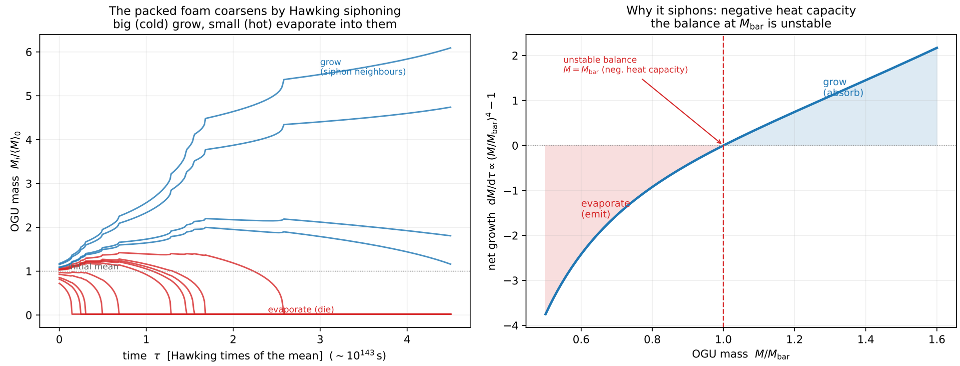

Figure 5.11: Radiative coarsening of the packed foam. *Left:* a mean-field Monte Carlo of OGUs sharing a Hawking bath—the big (cold) ones grow by siphoning the radiation of the small (hot) ones, which evaporate and die (the steps in the survivors' growth are neighbours dumping their mass into the bath), over a few Hawking times. *Right:* why—a black hole's negative heat capacity makes the balance at $M_{\rm bar}$ unstable, so $\mathrm{d}M/\mathrm{d}\tau \propto (M/M_{\rm bar})^4 - 1$ pushes the big to grow and the small to evaporate. Reproducible as `figures/scripts/ch05_coarsening.py`.

Does an OGU die? On any ordinary clock, no—but the intuition that *Hawking must eventually take over once growth halts* is exactly right, and it turns out to be the engine of coarsening. In the packed foam there is no fresh void, so each OGU is bathed in its neighbours' Hawking radiation (and the void's own de Sitter horizon glow—both $\sim T_H$ of a horizon-mass hole, $\sim 10^{-30}$ K, Chapter 2). At that bath an OGU sits right at $M \simeq M_{\rm crit}$: balanced. But a black hole has *negative* heat capacity ($T_H \propto 1/M$), so the balance is *unstable*. An OGU a touch larger than its neighbours is a touch colder, absorbs more than it emits, and grows; a touch smaller is hotter, emits more, shrinks—and shrinking makes it hotter still, a runaway to evaporation. So the foam coarsens by *radiative siphoning*: the big, cold OGUs drink the radiation of the small, hot ones, which evaporate away into them (Figure 5.11)—Ostwald ripening for universe-holes, the radiative twin of merger coalescence, conserving total mass while driving the foam toward fewer, larger universes. The catch is the clock: it runs on the Hawking time, $t_H = 5120\pi G^2 M^3/\hbar c^4 \sim 10^{143}$–$10^{178}$ s for OGU masses, so OGUs are immortal on any sub-Hawking timescale but *not* eternal—a sub-average universe is, with cosmic patience, eaten by its big neighbour. The *interior* heat-dies meanwhile (Chapter 7); the hole persists, feeding its children's dark sector (Chapter 6), until it is itself merged or, if it drew the short straw, slowly siphoned away. Whether birth in fresh void balances this coarsening into a true steady state or the foam coarsens without bound is the open dynamical content of the supraverse equilibrium (Appendix A).

5.7.8 The void is a heat-dead interior: the recursive cycle

Follow the coarsening to its end. The winners grow, the losers evaporate, and a surviving OGU eventually exhausts even its own structure: its black holes Hawking-evaporate, its matter redshifts away, and the interior settles into empty, cold ($T_{\rm dS} \sim 10^{-30}$ K), dark-energy-driven de Sitter space. A ginormous heat-dead OGU is, gravitationally, *a smooth empty void*— and that is precisely the substrate from which SCT nucleates OGUs. So it does what intuition demands: it breeds a fresh foam of OGUs inside itself, and the whole thing cycles.

The void is not primordial. This collapses a hidden assumption. The "void" of Section 5.7.2 need not be a separate arena outside everything; it is the *asymptotic state every universe reaches*. Universes nest both ways: we are inside a parent, and our own far future will, in its emptiness, nucleate a foam of fresh OGUs that seed their own lineages. There is no first void and no last universe—the supraverse is self-similar and eternal in both directions, with no first cause required.

Why heat death does not forbid it. The obvious objection is thermodynamic: heat death is *maximal* entropy, yet a fresh bounce needs *low* entropy. Penrose's resolution applies exactly [17]. The *total* entropy (thermal plus horizon) does rise monotonically, obeying the Second Law. But the entropy a bounce needs small is the *gravitational* (Weyl/clumping) entropy, and that *cycles*: low at the smooth bounce, high while matter clumps and black holes form, low again once the holes evaporate and the interior smooths back to de Sitter (Figure 5.12). The de Sitter far future is conformally flat—Weyl curvature zero—so it is gravitationally smooth, low in the only entropy that matters for nucleating the next low-entropy bounce. The reset runs on the black-hole Hawking clock ($\sim 10^{100}$ yr), the same clock as the foam coarsening above.

Conformal cyclic cosmology, made to branch. This is Penrose's CCC—the smooth, scale-free far future serving as the next aeon's low-Weyl beginning [17]—with one difference: a dead aeon here does not hand off to a *single* successor by conformal rescaling, but nucleates *many* OGUs in its emptiness, a whole foam. SCT is CCC made to branch. The birth rate β of Section 5.7.6 is then the de Sitter vacuum's nucleation rate, and because the void endures forever, even an exponentially tiny rate eventually fires—the cycle always closes. One temperature governs the whole far future: the de Sitter value $T_{\rm dS} = \hbar H_\Lambda / 2\pi k_B \sim 10^{-30}$ K sets the nucleation bath, the foam-coarsening bath, and the parent-horizon glow alike. *(This is a synthesis at the speculative end of the program—a natural closure of the foam picture, consistent with the gravitational arrow of Chapter 7, not a derived result.)* Reproducible as `figures/scripts/ch05_cycle.py`.

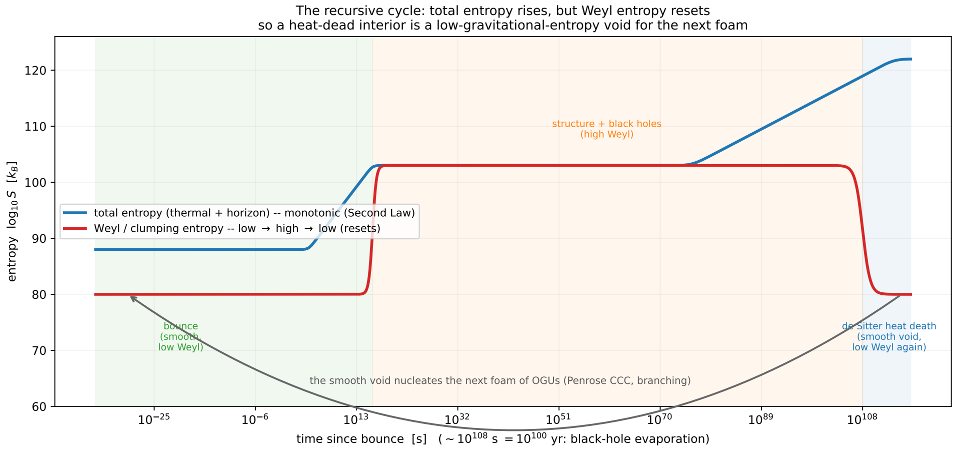

Figure 5.12: Why a heat-dead interior can seed the next foam. The *total* entropy (blue) rises monotonically to the de Sitter maximum (Second Law), but the *Weyl/clumping* gravitational entropy (red) cycles—low at the smooth bounce, high while black holes exist, low again once they evaporate and the interior smooths to de Sitter. It is the Weyl entropy a fresh bounce needs small, and it resets every cycle, so the heat-dead void is a valid low-gravitational-entropy substrate for the next foam (Penrose's CCC, branching). Schematic, with order-of-magnitude entropies and the black-hole Hawking timescale.

Why this rescues typicality. The cycle also settles a Copernican worry the framework had quietly incurred. To find ourselves *in* a Big Bang, fourteen billion years into a universe's life, looks embarrassingly special—a privileged moment in a privileged place, exactly what the Copernican principle forbids. The recursive cycle dissolves it. Every universe has its own bounce and its own observers a few billion years in; there have been, and will be, gazillions of aeons, each a foam of universes nested inside the heat-dead past of another. We sit not at a special origin but at a *typical* vantage—a random BHU_1 among countless near-identical ones, carrying the ordinary imprints of its kind. So the framework returns the very principle it began by assuming: nothing about our circumstance need be special, because the supraverse is an eternal, self-similar screensaver of near-identical beginnings, and ours is simply one lit pixel among the uncountably many.

5.7.9 Infinitely many OGs; do they grow forever?

By Axiom A1 plus infinity, the supraverse nucleates *infinitely many* OG universes. Whether an individual OG grows without bound is open. In a static infinite bath the negative heat capacity drives unbounded runaway accretion; in practice growth is regulated—by depletion of the local environment, by the void's own expansion diluting the bath, and by the stationary balance of the foam (large holes growing, new OGs nucleating, sub-critical holes evaporating). The honest statement is that individual universes can grow very large while the population remains statistically stationary; the global question is entangled with the measure problem (Appendix A).

5.8 Information structure of the supraverse

5.8.1 Local vs. global unitarity

Each universe, viewed in isolation, appears to lose information through Hawking radiation. But the Hawking radiation is entangled with the universe's interior content, and that entanglement extends across bounce membranes.

The supraverse as a whole is unitary. Information that "disappears" from a baby universe via Hawking radiation reappears in the parent's exterior. Information that flows into a parent black hole reappears in the baby's interior. Total information is conserved across the supraverse. This resolves the black-hole information paradox at the supraverse level: each universe's internal unitarity may be approximate, but global supraverse unitarity is exact.

5.8.2 Entanglement across membranes

The persistent Cartasis channel maintains entanglement between parent exterior and baby interior throughout the parent's lifetime. This is the structural realization of ER = EPR [14]: the geometric connection between parent and baby is the entanglement structure between Hawking radiation and the baby's matter content.

5.9 Time within the supraverse

There is no privileged supraverse-wide time direction. Each universe has its own internal time arrow (set by its bounce-to-future thermodynamic gradient). These arrows point in different directions in supraverse coordinates. The supraverse as a whole is globally time-symmetric: equal numbers of universes with arrows pointing "this way" and "that way." See Chapter 7 for the detailed treatment.

THE DARK SECTOR: DARK MATTER, DARK ENERGY, AND THE CMB

6.1 Overview

Three of the most puzzling features of observational cosmology—dark matter, dark energy, and the CMB—receive unified interpretations in the SCT framework. All three are signatures of the supraverse touching our universe through the Cartasis membrane.

6.2 Dark matter as parent's gravitational influence

6.2.1 The mechanism

The Cartasis membrane persists throughout the parent's lifetime, with new matter continuously falling through it from the parent's exterior. The matter content on the parent's side of the membrane exerts gravitational influence on our universe through the membrane, even though it does not enter our 3D space as ordinary matter. This gravitational influence manifests as:

- halo-like distributions around galaxies (where mass concentrated at bounce time);
- large-scale structure correlations (parent's mass distribution projected through the bounce);
- anomalously high rotation velocities of galaxies (additional gravitational binding);
- gravitational lensing without visible mass.

This matches the phenomenology of dark matter.

A rate behind it. The parent is not a static reservoir—it is a growing black hole. Holes have negative heat capacity, so above $M_{\text{crit}} = \hbar c^3/(8\pi G k_B T_{\text{bath}})$ (about 5×10^{22} kg in today's CMB) they accrete faster than they evaporate. Every astrophysical parent is therefore growing, and the parity-violating Cartasis filter (Chapter 4) keeps acting on what it accretes, so the injection through the membrane is both *ongoing* and *of a definite, filtered chirality*. Dark matter is thus the live gravitational signature of the parent's continuing biased accretion, not a frozen relic—which is what gives the dark sector a time-dependence (next).

The meal is lumpy: a large-scale structure imprint. What the parent feeds in is not smooth. A growing black hole eats whatever its environment supplies—infalling streams, merging clumps, the cosmic web of *its* universe—so the matter projected through the membrane carries the parent's inhomogeneity. For an OG universe this does not arise (the void has no structured meal), but for any non-OG universe—ours included (Section 5.5.3)—the dark-matter background should retain large-scale features inherited from the parent's accretion geometry, on scales unrelated to our own internal structure-formation history. This is a concrete reading of otherwise puzzling ultra-large structures: the Giant Arc and the $\sim$400 Mpc "Big Ring" of Lopez et al. [13], which sit uncomfortably with the cosmological principle in ΛCDM, are natural as frozen imprints of the lumpiness of the parent's meal. The framework's prediction is specific: such structures should correlate with the dark-matter background rather than the luminous one, should appear on scales above the internal homogeneity scale, and should not require any internal mechanism to assemble in the available time—the organization is inherited, not built. This is speculative and needs a quantitative correlation-function test (Chapter 8), but it is falsifiable and it is the kind of thing ΛCDM does not naturally produce.

6.2.2 Specifically: filter-rejected antimatter

If the bounce has any filter character (chirality-selective transmission, Chapter 4), then the rejected chirality content remains at the Cartasis membrane. From our perspective the rejected content is gravitationally present (it has mass–energy), does not appear in our 3D space (it is geometrically at the membrane, not in our interior), does not interact electromagnetically with our matter (no normal photon exchange across the membrane), and does not annihilate with our matter (geometric separation prevents direct contact).

This is the standard phenomenology of dark matter. The "missing antimatter" of standard cosmology and the "dark matter" of observational cosmology may be the same phenomenon.

6.2.3 Quantitative prediction

If filter efficiency is f (fraction of input matter that passes), then the dark-matter-to-baryon ratio is approximately $(1-f)/f$. Our observed ratio of $\sim 5:1$ implies $f \sim 1/6$. This is a specific number that should be calculable from Einstein–Cartan parameters. If the calculation gives $f \approx 1/6$ from first principles, that is a strong validation; if it gives something very different, the framework needs revision.

6.2.4 Predictions distinguishing this from particle dark matter

1. **Dark matter has chirality structure** (specifically, opposite to our matter). Probing dark-matter weak-interaction couplings (if any) should reveal antiparticle-like behavior, not particle-like.

2. **Dark-matter distribution should track the parent's matter structure**, not internal baby dynamics. This might explain anomalies in dark-matter halo profiles vs. CDM predictions.

3. **Dark-matter density should evolve with cosmic time** as the parent's accretion history changes. Tensions in dark matter inferred at different redshifts (e.g. the σ_8 tension) might be evidence for this.

4. **No dark-matter particles exist as fundamental constituents.** Direct-detection experiments will continue to find nothing. The dark matter is geometric, not particulate.

5. **Gravitational waves should propagate normally** (they are features of our spacetime), but might show subtle dispersion effects from membrane-mediated coupling at very low frequencies.

6.3 Dark energy as parent accretion dynamics

6.3.1 The mechanism

The parent black hole, embedded in its own universe, accretes from its surroundings. As parent mass grows, the Cartasis membrane geometry changes—specifically, the membrane's "size" relative to the baby interior grows. From inside the baby, this manifests as additional expansion pressure beyond what internal baby dynamics would predict. The acceleration of cosmological expansion (attributed to dark energy in standard cosmology) is actually the baby's response to parent growth.

This makes dark energy a *rate*: to leading order its strength tracks the time-derivative of the parent's accretion history. A parent still feeding from a rich environment sustains or strengthens the internal dark energy; one whose surroundings are depleting, or which is turning toward Hawking-dominated mass loss, produces weakening dark energy. The observed evolution of dark energy is therefore a direct, if model-dependent, readout of our parent's growth—the natural place to confront the DESI [6] w_0–w_a hints without forcing a sign.

6.3.2 Predictions distinguishing this from a cosmological constant

1. **Dark energy is not constant.** It tracks the parent's accretion history, which is generically non-constant.

2. **Dark energy will eventually reverse.** When the parent enters the Hawking-dominated regime (mass loss exceeds accretion), the Cartasis membrane shrinks and the baby's expansion decelerates, eventually contracting. This is a long-term prediction, not relevant on observational timescales now.

3. **Recent dark-energy evolution.** Current observations (DESI [6] and related) hint at evolving dark energy with $w_0 w_a$ parameters slightly inconsistent with strict $w = -1$. This is potentially consistent with parent-accretion dynamics.

6.3.3 Confronting DESI DR2

The framework makes a sign prediction here, not just a qualitative hope: a parent that is still accreting but with a *saturating* fractional growth rate gives, in the CPL parametrization $w(a) = w_0 + w_a(1 - a)$, a value $w_0 > -1$ with $w_a < 0$. Figure 6.1 shows the DESI DR2 fits [7] against this prediction: all four dataset combinations (DESI+CMB with Pantheon+, Union3, or DESY5) lie in the predicted wedge $w_0 > -1$, $w_a < 0$, with ΛCDM at $(-1, 0)$ disfavoured at 2.8–4.2σ. The sign pattern the parent-accretion picture predicts is exactly what is observed—and it follows from the parent black hole growing, with no cosmological constant.

The phantom crossing, derived. The DESI fits imply a *phantom* past ($w < -1$ at $z \gtrsim 0.3$) crossing to $w > -1$ today. This is not an extra assumption to bolt on; it follows from the injection picture directly. Absorb the membrane injection into an effective equation of state by writing the baby's dark-energy continuity equation with no explicit source, $\dot{\rho}_{\rm DE} + 3H(1 + w_{\rm eff})\rho_{\rm DE} = 0$, which is just the definition

$$w_{\rm eff}(a) = -1 - \tfrac{1}{3}\,\frac{d\ln\rho_{\rm DE}}{d\ln a}. \tag{6.1}$$

A density that *rises* with a (injection outpacing dilution) gives $w_{\rm eff} < -1$; one that *falls* (dilution winning) gives $w_{\rm eff} > -1$; a flat one gives exactly -1. A parent whose net injection rose and then saturated therefore makes $\rho_{\rm DE}(a)$ a hump, and (6.1) forces $w_{\rm eff}$ to cross -1 *at the peak*. Modelling the bump as a log-normal in scale factor, $\rho_{\rm DE} \propto \exp[-\beta(\ln a - \ln a_p)^2]$, gives the closed form $w_{\rm eff}(a) = -1 + \frac{2\beta}{3}(\ln a - \ln a_p)$, which linearises to CPL with $w_a = -\frac{2\beta}{3}$ and $a_p = \exp[(w_0 + 1)/w_a]$. The DESI+CMB+ DESY5 values then *predict* the crossing redshift $z_{\rm cross} = 1/a_p - 1 \approx 0.33$—squarely where DESI's own reconstruction crosses -1. The phantom past is thus a consequence of accretion that rose and saturated, not a tuned sign. What remains underived is the *amplitude* (β, equivalently the membrane coupling): we derive the shape and the crossing, fit the height. The prediction is still falsifiable: if future data drive $(w_0, w_a) \to (-1, 0)$, parent-accretion dark energy fails.

6.4 The CMB as parent Hawking radiation

6.4.1 The identification

The Cartasis membrane is the same geometric surface as the parent's event horizon (from outside), the baby's cosmic horizon at maximum lookback (from inside), and the Big Bang "surface" in standard cosmology. Hawking radiation is emitted at the parent's event horizon. From the baby's interior, this same radiation appears at the cosmic horizon—i.e. from every direction at maximum cosmic lookback distance. **This matches the observed CMB**: thermal radiation from every direction at maximum cosmic lookback.

6.4.2 Temperature and spectrum

The Hawking temperature for a black hole of mass M is

$$T_H = \frac{\hbar c^3}{8\pi G M k_B}.$$

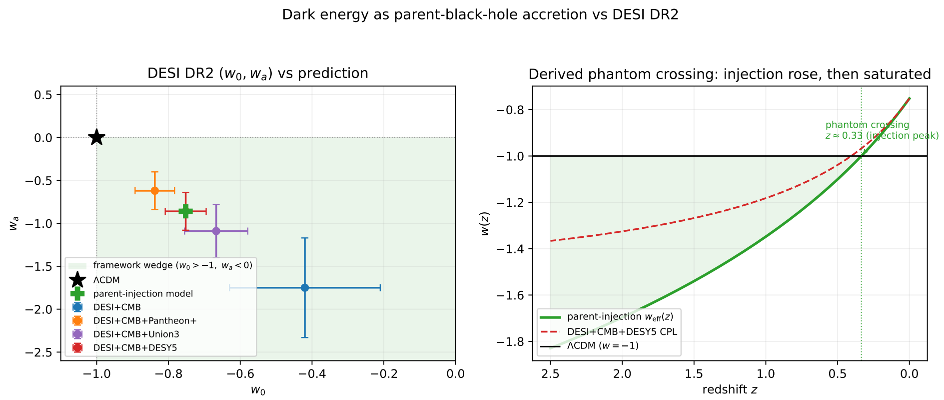

Figure 6.1: Dark energy as parent accretion vs DESI DR2. *Left:* the four DESI DR2 (w_0, w_a) fits all fall in the framework's wedge ($w_0 > -1, w_a < 0$, shaded); ΛCDM (star) sits at its corner, disfavoured at 2.8–4.2σ. *Right:* the peaked-injection $w_{\rm eff}(z)$ of (6.1) crosses $w = -1$ at the injection peak $z_{\rm cross} \approx 0.33$ (shaded: phantom epoch), tracking the DESI CPL reconstruction through the constrained range. Values flagged for verification against the published tables.

For our universe's parent mass ($\sim 10^{53}$ kg), the Hawking temperature is essentially zero at emission. But the radiation has been redshifting through our cosmic expansion, so the observed temperature is what we measure: 2.7 K. The redshift factor depends on detailed bounce geometry; a calculation is needed to verify consistency.

6.4.3 Consistency with observations

The CMB has a near-perfect Planck thermal spectrum (consistent with Hawking thermality), near-perfect isotropy (consistent with the bounce surface being homogeneous across the parent's horizon), tiny anisotropies at the 10^{-5} level (consistent with bounce-surface inhomogeneities and small parent-rotation effects), and acoustic peaks in the power spectrum (these need bounce-cosmology rederivation, not yet done) [18].

The standard cosmological explanation of the CMB (recombination at $z \sim 1100$) fits the spectrum and acoustic features beautifully. The bounce-cosmology reinterpretation needs to reproduce these features, which is non-trivial. **This is the central calculational challenge for the SCT framework.**

6.4.4 Preferred axis

If our parent is rotating (Kerr-like rather than Schwarzschild-like), the rotation axis imprints anisotropy on the bounce surface, which manifests as a preferred direction in the CMB and in subsequent structure formation. Observational hints of preferred axes include:

- the "axis of evil" in CMB low multipoles (alignment of the quadrupole and octupole with each other and with the ecliptic);

- recent JWST galaxy-rotation asymmetry [21]: a $\sim 2:1$ ratio of clockwise vs. counterclockwise rotation in JADES, statistically significant at 3.39σ, peaking near the Galactic poles.

Both could be signatures of parent rotation. The exact relationship between parent angular momentum and observed cosmic anisotropies is calculable but not yet done in detail.

6.5 Hubble tension

The Hubble tension is a persistent $\sim 5\sigma$ disagreement between H_0 measured from the CMB ($\sim 67\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) and from the local distance ladder ($\sim 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$). Standard cosmology has no clean resolution.

In SCT, the early universe (near the bounce) is dominated by bounce dynamics rather than a standard inflation + radiation-dominated phase. The relationship between early- and late-universe expansion is modified. This could in principle resolve the Hubble tension, but the specific calculation has not been done. If the framework can produce the right H_0 from late observations while also matching the CMB precisely, this is a major validation; if it cannot, the framework needs revision.

6.6 Summary of dark-sector predictions

Phenomenon	**Standard explanation**	**SCT explanation**
Dark matter	New particle (WIMP, axion, etc.)	Parent's matter content gravitating through Cartasis
Dark energy	Cosmological constant Λ	Parent-accretion dynamics changing membrane geometry
CMB	Recombination at $z \sim 1100$	Parent Hawking radiation redshifted by baby expansion
Matter–antimatter	Sakharov conditions + leptogenesis	Filter at Cartasis + statistical bias from OG fluctuation
Inflation	Inflaton field with specific potential	Torsion-driven expansion at bounce
Hubble tension	Unresolved	Bounce dynamics modify early/late expansion relationship
Preferred axes	Statistical fluctuation or systematic	Parent's rotation-axis imprint

Table 6.1: Standard vs. SCT interpretations of dark-sector phenomena.

6.7 Observational program

The framework predicts that we have *already observed* the supraverse through dark-matter and CMB measurements. We just do not recognize these as supraverse signatures because we interpret them through ΛCDM. Reinterpreting through SCT:

1. Dark-matter astronomy is *parent-universe* astronomy. The dark-matter distribution encodes information about our parent's structure.
2. CMB measurements are *parent Hawking radiation* measurements. The CMB spectrum and anisotropies encode information about our parent black hole's properties.
3. Dark-energy measurements track *parent accretion history*.
4. Preferred-axis observations probe *parent rotation*.

This is an enormous research program: re-derive everything we know from cosmological observations through the SCT lens, and look for consistencies and inconsistencies. Much of it is in principle computable from Einstein–Cartan bounce dynamics applied to specific parent configurations.

7.1 The problem

Standard physics is time-symmetric at the microscopic level. CPT is a theorem; the equations of motion (Schrödinger, Dirac, Einstein) are reversible. Yet we experience a clear forward arrow of time: eggs break and do not unbreak, heat flows from hot to cold, memories accumulate of the past but not the future.

Standard cosmology explains this through the *Past Hypothesis*: the universe began in an extraordinarily low-entropy state, and the arrow of time is the gradient pointing away from that initial state. But this is uncomfortable because:

- the low-entropy initial state is postulated, not explained;
- Boltzmann's original suggestion (we are a rare fluctuation in eternal equilibrium) does not work—it predicts that Boltzmann brains should dominate over ordered observers, contradicting observation.

7.2 SCT resolution

In SCT, the arrow of time is automatic:

1. **The Cartasis membrane is a low-entropy boundary.** At the bounce, matter is highly compressed ($\sim 10^{50}\,\mathrm{kg\,m^{-3}}$) and highly ordered (specific chirality bias inherited from the parent). Gravitational entropy is low because matter is uniformly compressed rather than clumped into black holes.
2. **The post-bounce evolution is toward higher entropy.** Expansion, cooling, structure formation, stellar evolution, eventual black-hole formation, eventual Hawking evaporation—all standard cosmological evolution proceeds toward higher entropy.
3. **The arrow of time is the gradient between these.** Forward time = away from bounce = increasing entropy. Backward time = toward bounce = decreasing entropy.

The Past Hypothesis is no longer a postulate; it is a *consequence* of the bounce being a low-entropy boundary.

7.3 Why doesn't this just push the problem back?

One might object: "fine, but now you need to explain why the bounce is low-entropy." The answer is that the bounce *cannot* be high entropy—it is the geometric configuration where matter is maximally compressed within Cartasis density limits. At Cartasis density, you have already gone beyond conditions where complex structure (which carries information and entropy) can exist.

The bounce is necessarily a state of high concentration, low complexity, and low information content—which means low gravitational entropy.

In other words, the bounce membrane is mechanically forced to be low entropy by the physics of Einstein–Cartan at extreme density. You cannot have a high-entropy bounce because high entropy requires complex structure, and complex structure cannot exist at Cartasis density. This converts the Past Hypothesis from "an unexplained initial condition" to "a necessary consequence of bounce physics."

7.4 Global vs. local time

In SCT, time has a particular structure. There is a *local arrow within each universe*: each universe has its own internal time direction, set by its bounce-to-future thermodynamic gradient, and observers within a universe experience their universe's arrow. There is *no global supraverse time arrow*: different universes have arrows pointing in different directions in supraverse coordinates, and the supraverse as a whole is time-symmetric. And there is a *static block universe at the supraverse level*: from the supraverse perspective, all of supraverse spacetime exists as a 4D block; "evolution" is a feature of being inside a specific universe, not a feature of the supraverse itself. This is the standard relativistic block-universe view, extended to the supraverse.

7.5 Why we experience forward time

Observers exist within specific universes. Observers require metabolism (irreversible chemistry, time-asymmetric), memory (information accumulation, requiring an entropy gradient), and computation/thought (irreversible logic gates, dissipative). All of these require being far from a low-entropy boundary and moving away from it. Observers can only exist where there is a thermodynamic gradient to extract work from.

So observers necessarily exist in regions with a clear arrow of time pointing away from a low-entropy boundary (the bounce). **We experience forward time because we could not exist otherwise.** It is not a metaphysical mystery; it is an anthropic-style consequence of where observers can exist within the supraverse structure.

7.6 Implications for cosmology

7.6.1 No initial-conditions problem

Standard cosmology has to postulate that the early universe was low-entropy and homogeneous. SCT derives both:

- low entropy: forced by bounce physics;
- homogeneity: bounce dynamics smooth out parent inhomogeneities, plus rapid expansion redshifts away small-scale perturbations (similar to inflation's mechanism).

7.6.2 Eventual heat death is local

In standard cosmology, the universe heads toward heat death (maximum entropy, no usable energy). In SCT, this happens too, locally—our universe will eventually reach a state of maximum entropy where structure no longer forms. But the *supraverse* does not approach heat death. The supraverse is already at its equilibrium configuration. Our universe's eventual heat death is just one bubble reaching its local equilibrium; the supraverse as a whole maintains its statistical distribution.

7.6.3 Universes evaporating back to parents

As discussed in Chapter 3, the parent's Hawking evaporation eventually shrinks the Cartasis membrane and the baby's connection to the parent fades. At some point the baby's matter content (now entangled with Hawking radiation in the parent's exterior) is effectively transferred back to the supraverse vacuum. This is the *cosmological* analog of an individual universe reaching heat death: not local equilibrium but reabsorption into the supraverse substrate. The arrow of time within the baby ends when the baby ends.

7.7 Connection to gravitational entropy

Penrose's longstanding emphasis on gravitational entropy is central here [15]. The key insights:

1. **Uniform vacuum is low gravitational entropy.** Smooth metric, no horizons, no information stored geometrically.
2. **Black holes are maximum entropy.** Bekenstein–Hawking $S \propto A/4\ell_P^2$. A black hole's entropy can exceed the entropy of all the matter that fell into it.
3. **The early universe was low gravitational entropy** despite being hot—because gravity was smooth, with no horizons and no clumping. Penrose's Weyl curvature hypothesis: the universe started with vanishing Weyl curvature.
4. **The universe evolves toward higher gravitational entropy** as structure forms, especially as black holes form.

In SCT, the bounce is necessarily a low-Weyl-curvature configuration (matter is uniformly at Cartasis density, no clumps). This automatically gives the low-gravitational-entropy initial condition that Penrose's hypothesis requires. The arrow of time follows.

7.8 Connection to gravitational decoherence

Gravitational decoherence is a mechanism in which large mass–energy superpositions decohere into definite states via gravitational coupling [8, 16]. This is a kind of "miniature universe-splitting"—the different branches become causally weakly-coupled regions of the gravitational field. In SCT, gravitational decoherence is the small-scale version of the same physics as bounce-driven universe production. Both are mechanisms where gravity separates different mass–energy configurations into causally distinct regions. The full continuum is:

- tiny mass–energy differences: weak gravitational decoherence, branches remain quantum-mechanically connected;
- moderate: stronger decoherence, branches become effectively separate;
- extreme (Cartasis density): bounce, branches become full baby universes.

The arrow of time within each branch, each universe, is set by the same mechanism: gravity carves out a low-entropy starting configuration, and time flows away from it.

7.9 The arrow and matter are CPT-locked

The thermodynamic arrow above is set by the low-entropy bounce boundary and would exist even in a perfectly matter/antimatter-symmetric universe; it is *not* caused by the baryon asymmetry. But the two are co-sourced at the same membrane and, at the supraverse level, locked together. The chiral-vortical filter that makes the matter excess (Chapter 4) is CP-violating, and by the CPT theorem CP violation is T violation: matter genesis is intrinsically time-asymmetric. More tellingly, the void is CPT-symmetric, so for every universe that is matter-biased with a forward thermodynamic arrow there is a CPT-image partner that is antimatter-biased with the opposite arrow—and the two are the *same* object viewed through CPT. The symmetric quantity is the product of matter-sign and arrow-sign, and observership selects the diagonal: every observer necessarily finds “matter + forward time” (their mirror’s inhabitants say the same of their “antimatter + backward” world). Matter and time travel in pairs not because one causes the other, but because the CPT structure of the void binds their signs.

8.1 Overview

SCT makes specific predictions that can in principle distinguish it from ΛCDM and other standard frameworks. This chapter catalogs them in rough order of how testable they are with current and near-future observational capabilities.

8.2 Tier 1: Already-observed potential signatures

These are existing observations that may already be evidence for SCT but are conventionally interpreted otherwise.

8.2.1 Galaxy rotation asymmetry

The JWST JADES survey shows a 2 : 1 asymmetry in galaxy rotation directions, peaking near the Galactic poles, at 3.39σ [21].

SCT prediction. The universe inherits a preferred axis from the parent's rotation. Galaxy rotation should show a hemisphere-based asymmetry aligned with the parent's spin axis.

Status. A hint, not a confirmation. Larger samples from ongoing JWST surveys will sharpen this in the next few years. If the asymmetry persists at higher significance and the Doppler-bias alternative is ruled out, this is direct evidence for a parent-imprinted axis.

De-confounding (this work). Running the test on the largest extant catalogue—the clean spiral sample of Galaxy Zoo 1 [12], 30,126 galaxies—shows how hard the measurement is (Figure 8.1). There is a highly significant *overall* handedness offset (clockwise fraction 0.480, a 3.9% deficit, binomial $p \sim 10^{-11}$), but it is the known human perception bias [11], not a sky dipole: a label-shuffle null gives the dipole $p = 0.62$ (no dipole beyond the monopole), and the best-fit axis lies 7° from the Galactic pole but 35° from the CMB axis. On the one large catalogue available, then, the asymmetry is systematics-dominated and what little axis exists tracks the Milky Way, not the CMB. This does not refute the parent-spin prediction—Galaxy Zoo 1 is a northern SDSS footprint with uncontrolled handedness bias—but it pins down what a real test requires: an all-sky, mirror-validated catalogue in which the discriminant is whether the spin dipole favours the CMB axis over the Galactic pole. That is the single measurement the framework most needs.

8.2.2 "Axis of evil" in the CMB

Low-multipole CMB anomalies show suspicious alignments between the quadrupole, octupole, and the ecliptic, at ~ 2–3σ.

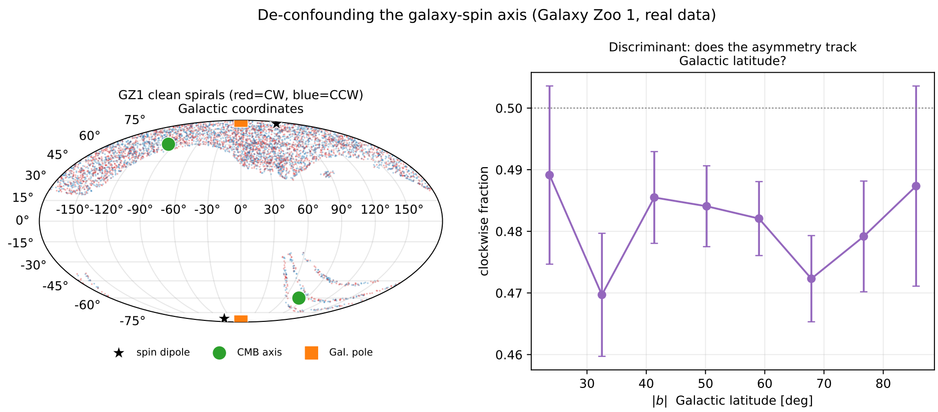

Figure 8.1: De-confounding the galaxy-spin axis on Galaxy Zoo 1 (clean spirals). *Left:* the SDSS northern-cap footprint (red = clockwise, blue = counter); the fitted spin dipole (star) sits on the Galactic pole, 35° from the CMB axis. *Right:* the clockwise fraction sits a few percent below 0.5 roughly independent of Galactic latitude—a global offset (perception bias), not a latitude systematic and not a significant dipole (shuffle null $p = 0.62$).

SCT prediction. Parent rotation imprints anisotropy on the bounce membrane, which manifests as low-multipole CMB anisotropies aligned with the parent's rotation axis.

Status. A long-standing anomaly. Conventional explanations (foreground contamination, statistical fluke) have not fully resolved it. It could be an SCT signature.

8.2.3 Evolving dark energy

DESI year-1 BAO results favor $w_0 w_a$ models with $w_0 > -1$ and $w_a < 0$ over strict Λ at ~ 2–3σ [6], suggesting that dark-energy density evolves.

SCT prediction. Dark energy reflects parent-accretion history, which is generically non-constant. Specifically, observed evolution toward weaker dark energy at recent times might indicate the parent has crossed into the Hawking-dominated regime.

Status. A statistical tension, not a confirmation. Years 2–5 of DESI data will sharpen this. If evolving dark energy is confirmed at $> 5\sigma$, ΛCDM is in trouble and SCT-like models become favored.

8.2.4 Hubble tension

A persistent $\sim 5\sigma$ disagreement between early-universe ($H_0 \approx 67$) and local ($H_0 \approx 73$) measurements.

SCT prediction. Bounce dynamics modify the relationship between early and late expansion. A specific calculation is needed, but the framework has degrees of freedom to potentially resolve the tension.

Status. Calculational work needed. SCT must reproduce both CMB physics and local measurements consistently. If it can, this is a major validation.

8.2.5 σ_8 tension

Lower-than-expected matter clustering inferred from low-redshift measurements vs. CMB-extrapolated predictions.

SCT prediction. Dark matter is geometric (the parent's influence through the membrane), not particulate. Its effective clustering behavior may differ from CDM expectations.

Status. Active observational research. SCT-based dark-matter modeling needed.

8.2.6 JWST high-redshift massive galaxies

JWST observations of unexpectedly massive galaxies at $z > 10$, challenging standard structure-formation timelines.

SCT prediction. Bounce dynamics might allow earlier structure formation than standard inflation, since the post-bounce expansion phase is different.

Status. Active research. Calculations of structure formation in bounce cosmology needed.

8.3 Tier 2: Near-future testable

These can be tested with planned observational programs in the next 5–15 years.

8.3.1 CMB spectral details

If the CMB is parent Hawking radiation, the spectrum should be perfect Planck (which it is, to 10^{-5}). But subtle deviations or correlations might exist: subtle spectral features from membrane geometry; correlations between CMB temperature and dark-matter distribution; specific polarization patterns from a rotating parent.

Tests. Next-generation CMB experiments (CMB-S4, LiteBIRD) will improve constraints on the CMB spectrum and polarization by orders of magnitude.

8.3.2 Cosmic neutrino background structure

The cosmic neutrino background (CνB) is predicted by standard cosmology but not yet directly observed. If observed (PTOLEMY-class experiments): a pure-flip framework predicts left-handed neutrinos in our convention; a strong-filter framework predicts a fewer-than-expected antineutrino-like component; standard cosmology predicts approximately equal ν and $\bar{\nu}$.

Tests. Direct CνB detection is decades away with current technology, but is being actively pursued.

8.3.3 Primordial gravitational waves

Inflation predicts a specific tensor-to-scalar ratio r. Bounce cosmology predicts a different (typically smaller) r.

Tests. LiteBIRD will constrain r to the $\sim 10^{-3}$ level. If observed at higher values, inflation is favored; if observed lower or absent, bounce cosmology is favored.

8.3.4 Dark-matter direct detection (negative result)

SCT predicts no fundamental dark-matter particles. WIMP, axion, sterile neutrino, and other direct-detection experiments should continue to find nothing.

Tests. Ongoing programs (LZ, XENONnT, future generations) will probe progressively lower cross-sections. Continued null results are weak evidence for geometric (non-particulate) dark matter.

8.3.5 Specific BAO and large-scale structure tests

The relationship between baryon acoustic oscillations and dark-matter clustering should follow specific SCT-predicted patterns if dark matter is geometric. Discrepancies with ΛCDM at large scales might be observable.

Tests. DESI, Euclid, and the Roman Space Telescope will map large-scale structure with unprecedented precision.

8.4 Tier 3: Future testable, requires major advances

8.4.1 Gravitational decoherence experiments

Tests of Diósi–Penrose gravitational collapse (or our preferred reinterpretation, gravitational decoherence) at mesoscopic scales [8, 16].

Tests. Bouwmeester- and Aspelmeyer-class proposals using mesoscopic mirror superpositions or matter-wave interferometers, within reach in the next one to two decades. If gravitational decoherence is confirmed at predicted rates, this strengthens the bridge between bounce cosmology and quantum mechanics that we have argued for.

8.4.2 Information-theoretic tests of dark matter

If dark matter is entangled with the parent through the bounce, sufficiently sensitive measurements might detect specific entanglement structures (correlations between dark matter and Hawking-radiation-like backgrounds, etc.).

Tests. Highly speculative. Requires precision quantum-cosmological measurements not currently possible.

8.4.3 Detection of parent-universe gravitational signatures

If gravity propagates through the Cartasis membrane bidirectionally, sufficiently long-wavelength gravitational waves might carry information about the parent's mass distribution.

Tests. Pulsar timing arrays (NANOGrav, etc.) probe nanohertz gravitational waves. Detection of unexpected features at very low frequencies might be parent signatures.

8.5 Tier 4: Theoretical / computational tests

These are not direct observations but consistency checks computable from the framework.

- **Filter efficiency yields the correct dark-matter ratio.** Compute filter efficiency from the Einstein–Cartan Lagrangian; predict the dark-matter-to-baryon ratio; compare to the observed $\sim 5{:}1$.
- **Bounce dynamics yield the correct CMB spectrum.** Compute the primordial perturbation spectrum from the torsion-driven bounce; compare to the observed CMB acoustic peaks.
- **Universe size distribution peaks near our parameters.** Simulate the supraverse population; show that observer-supporting universes cluster near our observed mass, expansion rate, and age.
- **Asymmetry magnitude from the rotating-bounce chiral-vortical filter.** Predict our observed baryon-to-photon ratio $\eta \simeq 6.1 \times 10^{-10}$ from the seed vorticity at the OG nucleation. *Status: first-cut viability check done, not yet a prediction* (Chapter 4, Section 4.5; `notes/cve-filter.md`). The scaling $\eta \sim C\,(\omega/T)$ puts the bare parameter $\omega/T \sim 10^{-11}$ within $\sim$1.6 orders of η_{obs}—the mechanism is in the right ballpark and is *sourced*

along the spin axis, tying it to the same axis as the galaxy-spin and CMB-alignment tests above. But matching the magnitude requires $C \cdot f_\omega \simeq 41$, with the spin fraction $f_\omega \leq 1$ capped at the maximal-spin ceiling, so the factor cannot come from a faster seed: it is a debt owed to a genuine calculation of the anomaly + sphaleron coefficient C (and is in tension with the lower-ρ_C composition argument, since $\omega/T \propto \rho_C^{1/4}$). What is robust now is the *direction and order of magnitude*, not the number; a spot-on test needs C, σ, and $\eta_{\min}$ pinned from Einstein–Cartan microphysics so that η is predicted in absolute units.

- **Hubble tension resolution.** Show that bounce-modified early-universe dynamics produce H_0 consistent with both CMB and local measurements.

8.6 What would falsify SCT?

1. **No preferred axis in galaxy rotations at higher precision.** If Shamir's effect goes away with larger samples, the parent-rotation imprint is disfavored.
2. **Dark-matter particle detection.** If a fundamental dark-matter particle is detected, the geometric interpretation fails.
3. **Dark energy proven constant.** If DESI and successors confirm $w = -1$ at higher precision, dark-energy-as-parent-accretion fails.
4. **CMB spectrum showing recombination-specific features inconsistent with a Hawking origin.**
5. **Inflation-specific signatures confirmed.** If primordial gravitational waves are detected at inflation-predicted levels, the bounce-as-inflation-substitute is challenged.
6. **CMB anomalies fully explained by foregrounds.** If the "axis of evil" and other anomalies are proven to be statistical or systematic artifacts, parent-rotation signatures are absent.
7. **Bounce simulations fail to produce realistic cosmological evolution.** If detailed Einstein–Cartan bounce calculations cannot reproduce the CMB, structure formation, etc., the framework is wrong in detail even if the conceptual structure is appealing.

8.7 Priority ordering for tests

For maximum scientific impact at minimum effort:

1. **CMB spectrum from bounce** (Tier 4)—computable now, decisive if it works or fails.
2. **Filter efficiency yields dark-matter ratio** (Tier 4)—computable now, strong validation if it works.

3. **Shamir's effect with larger samples** (Tier 1)—observational program already running.
4. **DESI dark-energy evolution** (Tier 1)—observational program already running.
5. **Primordial gravitational waves from LiteBIRD** (Tier 2)— observational program in progress.
6. **Hubble tension resolution** (Tier 4)—calculational.
7. **Gravitational decoherence** (Tier 3)—experimental, near-future.

Items 1, 2, and 6 are calculational work; items 3, 4, 5, 7 are observational waits. The calculational work could be done by a small group in a few years; the observational work happens on the experiments' timescales.

9.1 Philosophy

The framework's strength is that it should be simulable from first principles. Rare large vacuum fluctuations, Einstein–Cartan bounces, and thermodynamic equilibrium are all computable problems. The path from "interesting framework" to "constrained by data" runs through simulation.

The simulation work is tiered. Each tier is a tractable project of definite scope; higher tiers build on lower ones. A small group could complete Tiers 1–2 in a year; Tiers 3–4 are multi-year programs. Code lives in `sims/` and outputs in `sims/output/`.

9.2 Phase 0: Is the universe a black-hole interior?

Before any dynamical simulation, there is cheaper and more decisive work to do. The thesis at the foundation of everything else is that *our universe is the interior of a (rotating) black hole*—an inverse bubble behind a Cartasis membrane. In the Copernican spirit, we do not assume this; we take the universe's *measured* numbers and ask what they already imply. Four "corners" need only arithmetic and existing catalogs, yet each ties an observed quantity to a black-hole property. Figure 9.2 collects them; the calculation is reproducible as `figures/scripts/ch02_bh_first_light.py`.

Corner A — flatness is the Schwarzschild condition. For a Friedmann universe the critical density is $\rho_c = 3H^2/8\pi G$, and the mass inside the Hubble sphere of radius $R_H = c/H$ is $M = \frac{4}{3}\pi R_H^3\,\Omega\rho_c$. Its Schwarzschild radius is then

$$R_s = \frac{2GM}{c^2} = \frac{2G}{c^2}\cdot\frac{4}{3}\pi\left(\frac{c}{H}\right)^3\Omega\,\frac{3H^2}{8\pi G} = \Omega\,\frac{c}{H} = \Omega\,R_H.$$

So $R_s/R_H = \Omega$ *identically*. The measured flatness $\Omega = 1.000 \pm 0.005$ is therefore numerically the statement that the observable universe sits at its own Schwarzschild radius—exactly what a black-hole interior requires. With Planck values, $M \approx 9.2 \times 10^{52}$ kg and $R_s/R_H = 1.000$.

This is a necessary, not sufficient, condition: any critical-density FRW model satisfies it, and the interior Schwarzschild geometry is not literally FRW. It is the zeroth-order consistency check, not a proof.

Corner B — the spin is sub-extremal, and predicts an axis. If the interior is Kerr-like, it carries a dimensionless spin $a^* = cJ/GM^2$. The largest *causal* angular momentum of the Hubble-sphere mass, $J \sim McR_H$, gives

$$a^*_{\max} = \frac{cJ}{GM^2} = \frac{c^2R_H}{GM} = \frac{2}{\Omega} \approx 2,$$

of order unity. A realistic small net rotation therefore leaves the universe comfortably sub-extremal ($a^* < 1$), consistent with a genuine, cosmic-censored Kerr interior rather than a naked singularity. The decisive observational handle is direction, not magnitude: if the preferred axis is inherited from

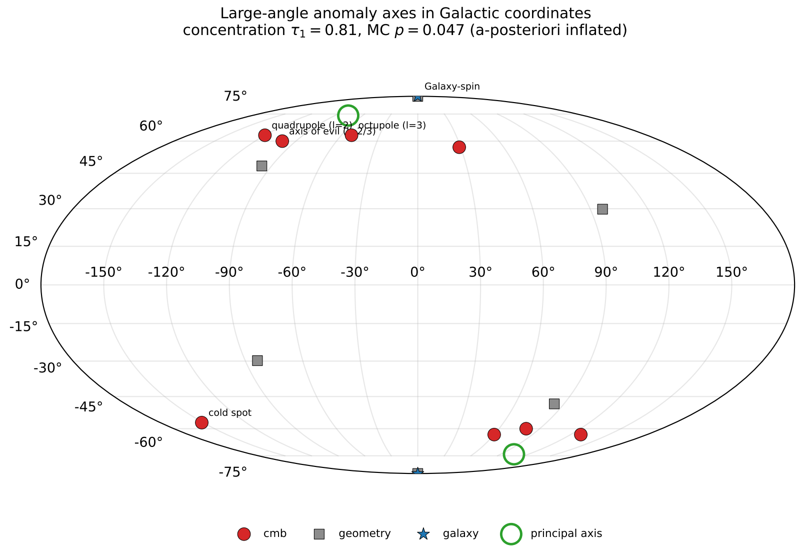

Figure 9.1: Large-angle anomaly axes in Galactic coordinates (each plotted with its antipode, since these are headless axes). The CMB anomaly axes and the galaxy-spin axis are moderately concentrated ($\tau_1 = 0.81$, isotropic $p \approx 0.05$, a-posteriori inflated), but the galaxy-spin axis coincides with the Galactic pole—the chief systematic confound—rather than sitting squarely on the CMB axis. The green circle marks the fitted principal axis.

the parent's spin, the CMB low-multipole alignment (the "axis of evil") and the JWST galaxy-rotation asymmetry [21] should share a single axis. **Testing that alignment against existing catalogs is the single cheapest potentially-decisive measurement in the whole program.**

First pass. A curated set of well-sourced large-angle axes—the combined CMB axis of evil, the kinematic dipole, the cold spot, and the galaxy-spin asymmetry—is moderately concentrated: the orientation-tensor largest eigenvalue is $\tau_1 = 0.81$, with an (a-posteriori-inflated) Monte-Carlo isotropic probability $p \approx 0.05$ (Figure 9.1). The honest reading, however, cuts against an easy claim: the galaxy-spin axis [21] sits $\sim 30°$ from the CMB axis but $\sim 0°$ from the Galactic pole. A Galactic-pole axis is exactly what a Milky-Way classification systematic (dust, inclination bias) would produce, so the mundane explanation is not excluded. The parent-spin interpretation survives only if, once a proper all-sky dipole is fit, the galaxy-spin axis favours the *CMB* axis over the Galactic pole. That is the crux the next data release must settle. Reproducible as `figures/scripts/ch08_axis_alignment.py`.

Corner C — "CMB = Hawking" cannot mean the present horizon. The Hawking temperature of the Hubble-sphere mass is $T_H = \hbar c^3/8\pi G M k_B \approx 1.3 \times 10^{-30}$ K, and (because $R_s = R_H$) it equals exactly half the present cosmological-horizon Gibbons–Hawking temperature, $T_H = \frac{1}{2}\hbar H/2\pi k_B$. Both are ~ 30 orders of magnitude below the 2.7255 K CMB. The honest consequence: the CMB is *not* present-day horizon radiation. The identification

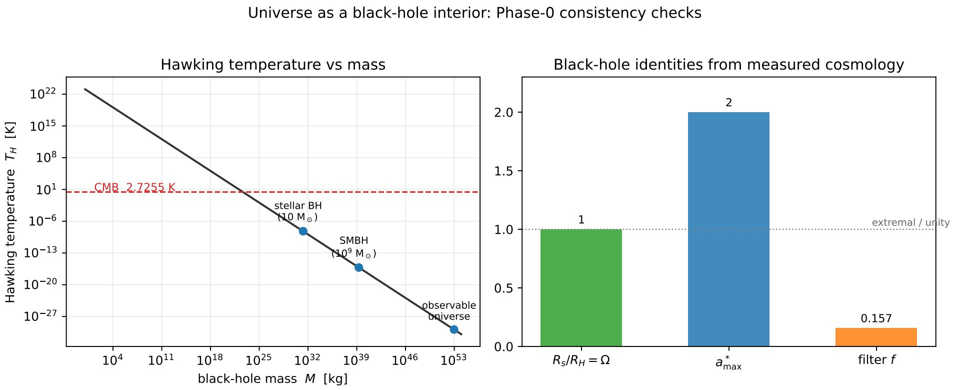

Figure 9.2: Phase-0 consistency checks. *Left:* the Hawking temperature $T_H(M)$; the observable universe lands $\sim$ 30 orders of magnitude below the CMB, so the CMB cannot be present-horizon radiation. *Right:* the dimensionless black-hole identities read straight off Planck cosmology—$R_s/R_H = \Omega \approx 1$, maximal spin $a^*_{\max} \approx 2$, and membrane filter $f \approx 0.16$.

can only work as parent-horizon radiation emitted near the bounce (hot at emission) and redshifted by the internal expansion. This sharpens what a successful calculation must deliver—a spectrum, not just a temperature—and hands the hard part to Tier 2.

Corner D — the dark sector fixes a single filter number. If dark matter is filter-rejected counterpart matter, the dark-matter-to-baryon ratio is $(1 - f)/f = \omega_c/\omega_b$, so the membrane pass-fraction is $f = \omega_b/(\omega_b + \omega_c)$. Planck's $\omega_c/\omega_b \approx 5.4$ gives $f \approx 0.157 \approx 1/6$. Any Einstein–Cartan filter model must reproduce this one number; it is the tightest quantitative target the membrane physics faces.

Recommended order. Do Corner A and B first (an afternoon, plus one catalog-alignment analysis), then C and D as supporting consistency relations. Only then move to the dynamical core below: Phase 0 establishes that the *kinematics* are consistent with a black-hole interior; the tiers establish the *mechanism*.

9.3 Tier 1: Single-bounce Einstein–Cartan dynamics

Goal. Numerically evolve a spherical collapse of fermionic matter through the bounce. Output: complete characterization of the bounce as a function of input parameters.

Minimal model. The cheapest dynamical demonstration is homogeneous: a Weyssenhoff spin fluid in Einstein–Cartan obeys a modified Friedmann

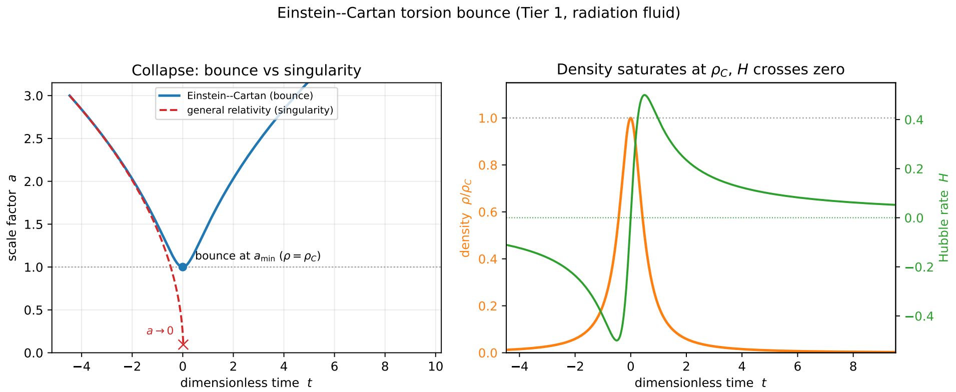

Figure 9.3: The Einstein–Cartan torsion bounce (Tier 1, radiation fluid), integrated in dimensionless units ($8\pi G/3 = \rho_C = 1$). *Left:* a collapsing universe bounces at $a_{\min}$ (blue) where general relativity would reach a singularity, $a \to 0$ (red dashed). *Right:* the density saturates at the Cartasis density ρ_C and the Hubble rate H crosses zero—collapse turning into expansion.

equation in which the averaged spin contributes a negative quadratic term,

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_C}\right),$$

so the expansion rate vanishes and reverses at $\rho = \rho_C$: a bounce, not a singularity. Integrating this single ODE (with the continuity equation for $\rho(a)$) and showing $a(t)$ reaching a finite minimum at ρ_C is the *minimum viable bounce*—a few dozen lines, and the first time the central claim is exhibited rather than asserted. The spherical-collapse code below upgrades this to an inhomogeneous interior matched to an exterior Schwarzschild geometry, so that one figure shows a black hole from outside and an expanding cosmology from inside.

Figure 9.3 is that minimum viable bounce, integrated for a radiation fluid. Where general relativity drives the same collapse to $a = 0$ in finite time, the torsion term halts it: the density saturates at ρ_C, the Hubble rate passes smoothly through zero, and the universe re-expands. With $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$ the whole bounce lasts $\sim 4 \times 10^{-21}$ s. This is the central claim—no singularity, an inverse bubble forms—exhibited rather than asserted, in a few dozen lines (`sims/src/cartasis_sims/bounce.py`, `figures/scripts/ch09_bounce.py`).

Outside vs. inside (Oppenheimer–Snyder with a bounce). Gluing this interior to an exterior Schwarzschild vacuum—the 1939 Oppenheimer–Snyder construction, now with the torsion bounce replacing the singularity—shows both faces at once (Figure 9.4). By Birkhoff's theorem the exterior is exactly Schwarzschild, so a distant observer watches the surface *freeze* at the horizon (Schwarzschild time $t \to \infty$ as the areal radius $R \to r_s$) and its light redshift to black: a black hole. Inside, the dust reaches ρ_C at a tiny areal radius $R_{\min}$, bounces, and re-expands. For the observable universe's mass

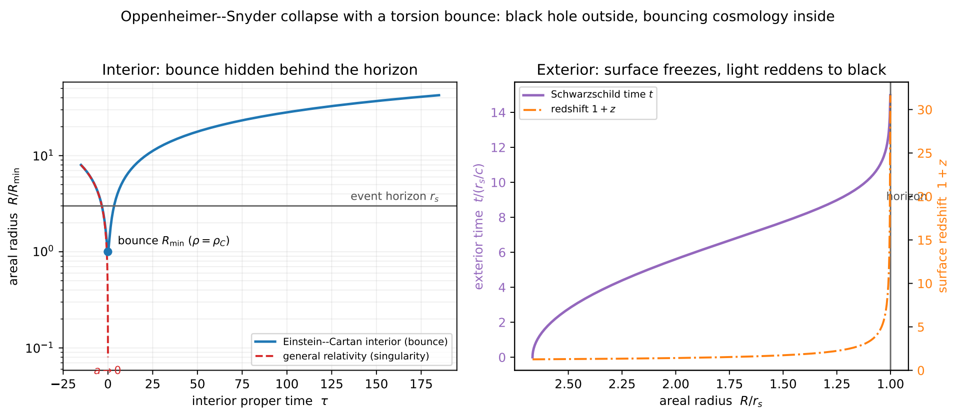

Figure 9.4: Oppenheimer–Snyder collapse with a torsion-bounce interior. *Left:* the surface's areal radius collapses through the horizon and bounces at $R_{\min}$ (blue) where general relativity reaches a singularity (red dashed). *Right:* from outside, the surface freezes at the horizon —Schwarzschild time $t \to \infty$ and surface redshift $1 + z \to \infty$ as $R \to r_s$. A black hole from outside; a bouncing, re-expanding cosmology from inside.

the bounce radius is $R_{\min} \approx 6\,\mathrm{m}$ (a proper volume $\sim 10^3\,\mathrm{m}^3$) against a horizon $r_s \approx 1.4 \times 10^{26}\,\mathrm{m}$—so $R_{\min}/r_s \sim 4 \times 10^{-26}$. The new universe is born microscopic, hidden deep behind the parent's horizon, and only later inflates to its $\sim$46 Gly extent: the Tardis effect of Chapter 3, made quantitative.

Robustness across the equation of state. The bounce is not special to one matter content. Sweeping the equation of state from dust ($w = 0$) through radiation ($w = 1/3$) to stiff ($w = 1$), the bounce density is universal—$a_{\min}$ and $\rho_{\max} = \rho_C$ are independent of w—and the peak expansion rate is exactly $\max |H| = \frac{1}{2}\sqrt{8\pi G\rho_C/3}$ (because $\max_x x(1-x) = \frac{1}{4}$). Only the bounce *duration* varies, and by an exact law,

$$\Delta\tau_{\mathrm{FWHM}} = \frac{4}{n}\frac{1}{\sqrt{8\pi G\rho_C/3}}, \qquad n = 3(1+w),$$

so stiffer matter gives a sharper bounce (Figure 9.5). For $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$ the duration is a few $\times 10^{-21}$ s across the whole range. The central claim does not rest on a tuned equation of state.

Setup.

- 1D spherical symmetry (radial coordinate r, time t).
- Initial conditions: a uniform ball of matter at moderate density, slightly overdense to drive gravitational collapse.
- Matter modeled as a Weyssenhoff fluid: a perfect fluid with spin density (semiclassical treatment of fermion spin in the continuous limit).
- Evolution: Einstein–Cartan field equations with torsion sourced by spin density.

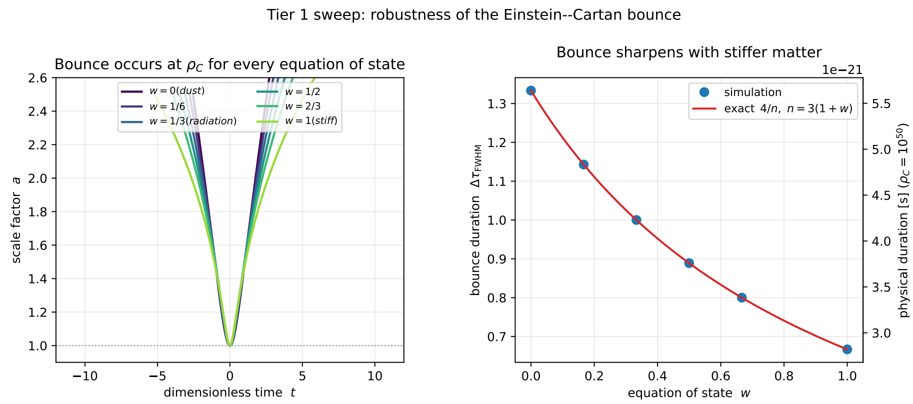

Figure 9.5: Robustness of the bounce. *Left:* for w from dust to stiff the scale factor always bounces at $a_{\min}$ ($\rho = \rho_C$); only the sharpness changes. *Right:* the bounce duration follows the exact law $\Delta\tau_{\text{FWHM}} = 4/n$ (right axis: physical seconds for $\rho_C = 10^{50}\,\text{kg}\,\text{m}^{-3}$).

- Tracking: density, pressure, spin density, metric components, and torsion components as functions of r and t.

Numerical method. A finite-difference scheme on a radial grid ($\sim 10^3$ to 10^4 points), with adaptive mesh refinement near the bounce (where gradients become extreme), implicit time stepping for stability near peak density, and conservation checks (total mass–energy, total spin, total particle number).

Expected output. The bounce happens at $\rho \sim \rho_C \sim 10^{50}\,\text{kg}\,\text{m}^{-3}$ (verified); the bounce timescale in proper time; the bounce duration and profile shape; the maximum density reached as a function of initial conditions; the equation of state through the bounce; and the energy balance (kinetic energy in expansion vs. energy contained in spin/torsion fields).

Tools. Python with NumPy/SciPy for prototyping (optionally Julia for performance), standard ODE solvers (RK4, implicit methods), custom grid handling and AMR, and plotting with matplotlib.

Effort estimate. 2–3 months for a competent graduate student; $\sim$ 500–1500 lines of code; validation against existing single-bounce papers (Brandt, Magueijo, Popławski [19, 20]).

Status of existing work. Several authors have done pieces of this. No one has produced a clean, well-documented reference implementation. *A valuable side-output of this tier is producing such a reference implementation.*

9.4 Tier 1c: The extruder—the bounce entropy budget

Universes are not born, they are extruded. The membrane does not sort a pre-existing population; it compresses parent material to ρ_C and extrudes a new interior through it, like material forced through a die (Chapter 4, Section 4.6). Because sphalerons conserve $B-L$, the parent's net $B-L$ passes through as a protected remnant and the baby *inherits* its parent's asymmetry, $\eta_{\text{baby}} = \eta_{\text{parent}}/D$, where $D = S_{\text{after}}/S_{\text{before}} \geq 1$ is the entropy multiplied across the crossing. The entire magnitude-and-lineage question (does a descendant keep the $\sim$1 ppb its parent made? do lineages stay chirally pure?) reduces to this one number D, bracketed between $D = 1$ (adiabatic inheritance) and the horizon-saturating $D = S_{\text{BH}}/S_{\text{rad}} \sim 10^{49}$ (which would crush η to $\sim 10^{-59}$, contrary to observation). This tier *computes* D from the bounce dynamics rather than guessing it.

Method. Entropy is generated only by irreversibility; the homogeneous perfect-fluid bounce is exactly reversible. Switch on dissipation as a bulk viscosity $\zeta = \tilde{\zeta}\, s$ ($\tilde{\zeta}$ dimensionless, s the entropy density). The Eckart entropy production of a bulk-viscous FRW fluid is $\mathrm{d}\ln S/\mathrm{d}\tau = \zeta(3H)^2/(sT) = 9\tilde{\zeta}\, H^2/T$, so across the bounce

$$\ln D = 9\,\tilde{\zeta}\,\frac{\Omega_{\text{bounce}}}{T_{\text{bounce}}}\,J, \qquad J = \int H_{\text{dim}}^2\, a\, \mathrm{d}\tau_{\text{dim}}, \tag{9.1}$$

with $\Omega_{\text{bounce}} = \sqrt{8\pi G\rho_C/3}$ and $J \simeq 1.38$ a pure number read off the dimensionless bounce of Figure 9.3.

Result—the extruder is adiabatic. The prefactor $\Omega_{\text{bounce}}/T_{\text{bounce}} \simeq 1.5 \times 10^{-11}$ is *the same* small parameter ω/T that limits the chiral-vortical estimate (Chapter 4): the bounce is far too fast to thermalize. Even an $O(1)$ bulk viscosity gives $D - 1 \sim 10^{-10}$, and a conformal radiation fluid has $\zeta = 0$ identically, so $D = 1$ exactly. Reaching the horizon-saturating limit would require $\tilde{\zeta} \sim 10^{12}$—unphysical by a dozen orders of magnitude (Figure 9.6). The conclusion is sharp: $D = 1$ to roughly one part in 10^{11}, so $\eta_{\text{baby}} = \eta_{\text{parent}}$. Inheritance is robust, the horizon-saturating branch is dynamically out of reach, and the factor-of-41 shortfall is the original-generation universe's problem alone—every descendant simply carries down the asymmetry made once at the top, with the founder's chiral sign protected (pure lineages). Reproducible as `sims/src/cartasis_sims/extruder.py` and `figures/scripts/ch09_extruder.py`.

The inhomogeneous shear channel. The obvious objection to a homogeneous, bulk-viscous estimate is shear. An anisotropic collapse carries a shear scalar σ whose energy density redshifts as a stiff fluid, $\rho_\sigma \propto a^{-6}$, so it outgrows radiation (a^{-4}) and comes to dominate the approach to the bounce—the BKL/Mixmaster regime. In plain general relativity this drives $\sigma^2/T \to \infty$ and the shear-viscous entropy production *diverges* at the singularity. The

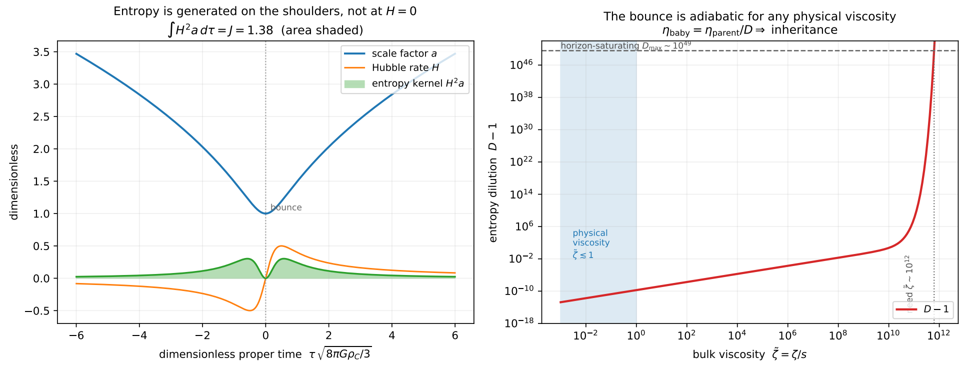

Figure 9.6: The extruder: entropy budget of the bounce. *Left:* the Eckart entropy-production kernel H^2a (green, area $= J \simeq 1.38$) peaks on the *shoulders* of the bounce and vanishes at the turnaround $H = 0$; the total generated is tiny. *Right:* the entropy dilution $D - 1$ versus bulk viscosity $\tilde{\zeta}$. Across the entire physical range ($\tilde{\zeta} \lesssim 1$, shaded) the bounce is adiabatic ($D - 1 \lesssim 10^{-10}$), so $\eta_{\rm baby} = \eta_{\rm parent}/D \approx \eta_{\rm parent}$: descendants inherit their parent's $\sim$1 ppb. Reaching the horizon-saturating $D_{\max} \sim 10^{49}$ (dashed) would demand $\tilde{\zeta} \sim 10^{12}$.

Einstein–Cartan bounce changes the conclusion. Adding ρ_σ to the capped Friedmann equation, $H^2 = \rho_{\rm tot}(1 - \rho_{\rm tot})$ with $\rho_{\rm tot} = (1 - f_\sigma)a^{-n} + f_\sigma a^{-6}$ and f_σ the shear fraction at the bounce, the torsion cap bounds the *total* density—and hence the shear—at ρ_C (Figure 9.7, left): the shear is bounced too, instead of running away. The entropy integral $J_{\rm shear} = \int \rho_\sigma\, a\, d\tau$ therefore stays $O(1)$ even for a maximally shear-dominated bounce ($f_\sigma \to 1$), so $\ln D_{\rm shear} = 2\tilde{\eta}(\Omega_{\rm bounce}/T_{\rm bounce})J_{\rm shear}$ carries the *same* $\sim 10^{-11}$ prefactor and $D - 1 \lesssim 10^{-10}$ across the whole anisotropy range (Figure 9.7, right). The same torsion that removes the singularity tames the BKL entropy divergence: inheritance survives the inhomogeneous channel too. Reproducible as `figures/scripts/ch09_shear.py`.

What this tier does *not* yet settle. Both the homogeneous and the shear channels are covered by Eckart viscosities on a still-simplified background; gravitational particle production at the bounce is a separate entropy source not captured here. The same timescale argument suppresses anything sourced by gradients $\times$ (bounce time) by the same $\sim 10^{-11}$ unless the gradients are trans-thermal, but a full inhomogeneous collapse (the Tier 1 code above) carrying its own entropy accounting is what would confirm it. Pinning D at the few-percent level, particle production included, is the cleanest near-term calculation bearing on the asymmetry (Appendix A, Q16a); the results so far make $D = 1$ the strongly favoured answer.

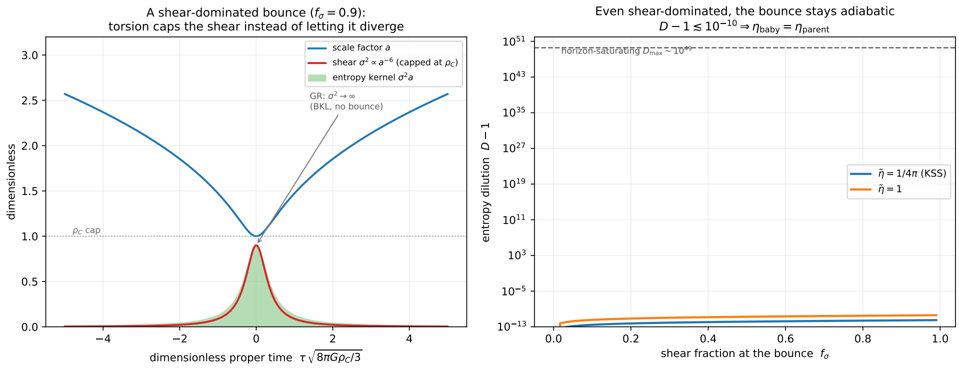

Figure 9.7: The shear channel of the extruder. *Left:* a shear-dominated bounce ($f_\sigma = 0.9$). The shear $\sigma^2 \propto a^{-6}$ would diverge at a GR singularity (BKL); the torsion cap bounds it at ρ_C and bounces it, keeping the entropy kernel $\sigma^2 a$ (shaded) finite. *Right:* the entropy dilution $D - 1$ versus the bounce shear fraction f_σ, for shear viscosities at the KSS bound $\tilde{\eta} = 1/4\pi$ and at $\tilde{\eta} = 1$. It stays below $\sim 10^{-10}$ all the way to a shear-dominated bounce—a dozen orders below the horizon-saturating $D_{\max}$—so $\eta_{\mathrm{baby}} = \eta_{\mathrm{parent}}$ regardless.

9.5 Tier 2: Perturbation evolution through the bounce

Goal. Take the output of Tier 1 (a clean bounce profile) and evolve cosmological perturbations through it. Output: the primordial power spectrum from the bounce, comparable to CMB observations.

Setup. Start with the Tier 1 spherical bounce background; add linear perturbations to the metric and matter fields; evolve them using linearized Einstein–Cartan equations; track the power spectrum of scalar, vector, and tensor modes through the bounce; and compute predictions for CMB observables (the scalar spectral index n_s, the tensor-to-scalar ratio r, the running of the spectral index, and non-Gaussianities).

Numerical method. Mode-by-mode evolution in Fourier space; long-wavelength modes treated as classical fields and short-wavelength modes quantum-mechanically (mode functions); Bunch–Davies-like initial conditions at large negative time (deep in collapse); modes tracked through the bounce and into post-bounce expansion.

Expected output. The primordial power-spectrum amplitude and shape, the spectral tilt, and the tensor-to-scalar ratio, compared to Planck CMB observations ($n_s = 0.965 \pm 0.004$, $r < 0.06$ upper limit) [18] and to ΛCDM + inflation predictions.

Decisive test. If the bounce produces a near-scale-invariant spectrum with $n_s \approx 0.96$ and r within observational bounds, **the framework is observationally viable.** If the spectrum is dramatically different, the framework fails this test. This is the central calculational challenge for SCT. Existing work on bounce cosmologies (loop quantum cosmology, ekpyrotic models) provides templates for similar calculations [1].

First result: the matter bounce is scale-invariant. A first pass at this test already passes. Evolving tensor (gravitational-wave) modes through the bounce with the Mukhanov–Sasaki equation $\mu_k'' + (k^2 - a''/a)\mu_k = 0$ ($h_k = \mu_k/a$, conformal time), with Bunch–Davies data deep in contraction and the spectrum read super-horizon, reproduces the analytic matter-bounce result. For a contracting phase of constant equation of state w the conformal scale factor is a power law $a \sim |\eta|^p$ with $p = 2/(1+3w)$, and the tensor spectrum is $P_T(k) \sim k^{n_T}$ with

$$n_T = 3 - |2p - 1|. \tag{9.2}$$

Modes exit the comoving Hubble horizon *during contraction*—the bounce analogue of inflationary horizon exit—freeze, and cross the bounce. A matter-dominated contraction ($w = 0$, $p = 2$) gives $n_T = 0$ *exactly*: a scale-invariant primordial spectrum, the observed near-scale-invariance, **reproduced with no inflaton** (Figure 9.8). The numerics match (9.2) to $\lesssim 1\%$ across w. What Einstein–Cartan supplies is the one ingredient the matter-bounce scenario otherwise lacks—a *non-singular* bounce, which competitor realisations buy with exotic ghost matter but torsion delivers for free at ρ_C (Chapter 2). The observed slight *red* tilt, $n_s \simeq 0.965$, corresponds to a contraction just *softer* than matter ($w \approx -0.003$), a mild quintessence-like condition; pinning that, computing the scalar spectrum and the tensor-to-scalar ratio r, and propagating to the acoustic peaks are the remaining Tier-2 steps. But the central qualitative hurdle—scale invariance from a bounce rather than inflation—is cleared. Reproducible as `figures/scripts/ch09_primordial.py`.

Toward the acoustic peaks. With a near-scale-invariant primordial spectrum in hand, the acoustic peaks follow from *shared* physics. The peaks are pre-recombination baryon–photon sound waves frozen at last scattering, and SCT runs the entire post-bounce hot Big Bang exactly as ΛCDM does—same plasma, same recombination at $z \simeq 1090$, same sound horizon—so it inherits the peaks rather than having to re-derive them. The acoustic scale is set by the comoving sound horizon $r_s = \int_{z_{\rm rec}}^{\infty} c_s/H\,dz$ and the distance to last scattering D_C, giving $\theta_s = r_s/D_C$ and peak multipoles $\ell_n \simeq (n - \frac{1}{4})\pi/\theta_s$. Evaluated on the standard post-bounce background this returns $r_s \simeq 142\,\mathrm{Mpc}$, $\theta_s \simeq 0.0102\,\mathrm{rad}$, and a peak series $\ell_n \simeq 230, 540, 850, 1150, 1460$—matching the observed Planck peaks (220, 540, 810, 1120, 1450) to a few percent (Figure 9.9). The hard, decisive number—the acoustic scale—is reproduced; what remains is the full Boltzmann computation of peak *heights* (sensitive to ω_b, ω_c, and the precise primordial tilt and amplitude). The "CMB = parent Hawking radiation" identification (Chapter 6) is consistent with this: it concerns the *origin* of the thermal plasma at the bounce/horizon surface, not a replacement for recombination—the peaks arise from standard acoustics either way.

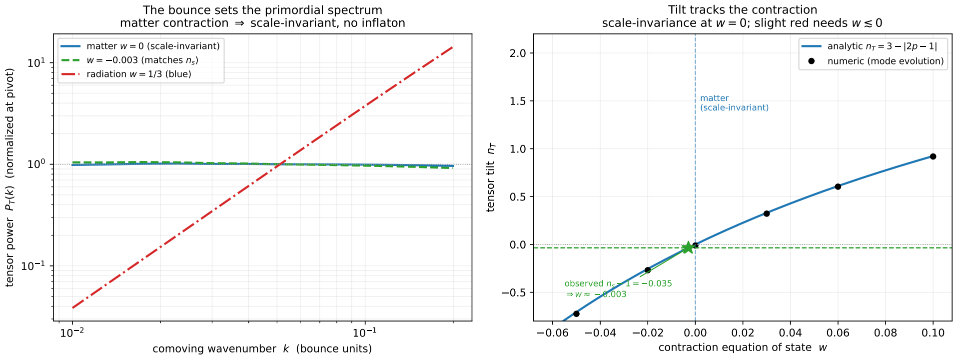

Figure 9.8: The primordial spectrum from the Einstein–Cartan bounce. *Left:* the tensor power spectrum $P_T(k)$ from modes evolved through the bounce is *flat* (scale-invariant) for a matter-dominated contraction ($w = 0$) and steeply blue for radiation—scale invariance with no inflaton. *Right:* the tilt n_T versus the contraction equation of state, numeric points on the analytic $n_T = 3 - |2p - 1|$ (Equation (9.2)); $w = 0$ is scale-invariant, and the observed $n_s - 1 = -0.035$ corresponds to a contraction just softer than matter, $w \approx -0.003$.

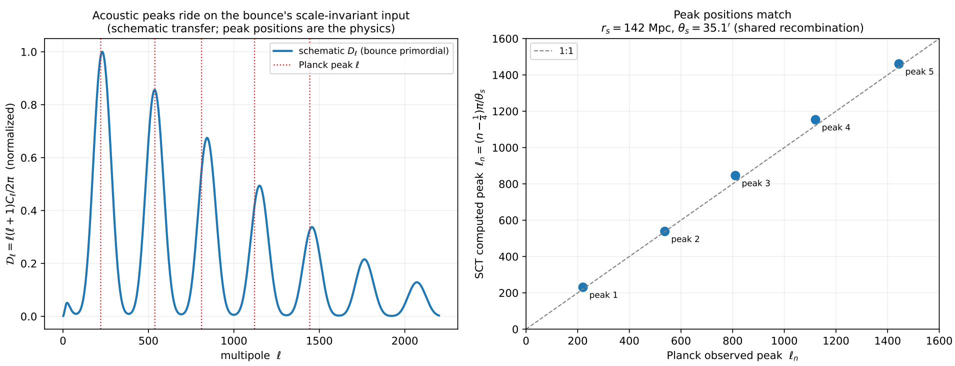

Figure 9.9: *Left:* a schematic CMB temperature spectrum—the bounce's scale-invariant primordial input times a damped acoustic transfer—with the observed Planck peak multipoles marked (dotted); the peaks ride at the right places. *Right:* the SCT-computed peak positions $\ell_n = (n - \frac{1}{4})\pi/\theta_s$, from the sound horizon $r_s \simeq 142\,\text{Mpc}$, against the Planck peaks—on the 1:1 line, because the pre-recombination acoustics are shared with ΛCDM. The schematic spectrum is illustrative; the peak *positions* are the physics. Reproducible as `figures/scripts/ch09_acoustic.py`.

The peak heights. Positions were the easy half; the *heights* are the discriminating half, and they too follow from shared recombination physics. The controlling, computed quantity is the baryon loading $R = \frac{3}{4}(\omega_b/\omega_\gamma)/(1 + z_{\rm rec}) \simeq 0.62$: baryons add inertia to the photon–baryon fluid, so the effective-temperature monopole at last scattering, $\Theta_{\rm eff} = (1 + R)\Psi\cos(kr_s) - R\Psi$, has deeper *compressions* than *rarefactions*—the odd peaks (1st, 3rd, 5th) are enhanced and the even peaks (2nd, 4th) suppressed. A Doppler term fills the

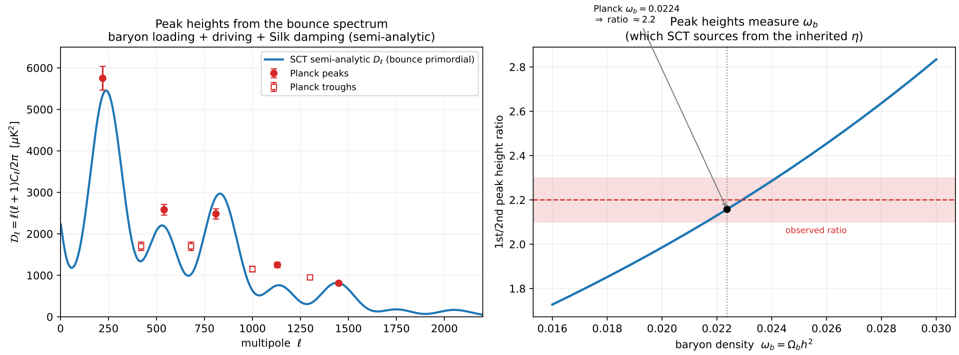

Figure 9.10: CMB peak *heights* from the bounce's scale-invariant spectrum. *Left:* a semi-analytic tight-coupling D_ℓ—baryon-loaded acoustic monopole, Doppler, radiation driving, and Silk damping—against the Planck peak (filled) and trough (open) heights; the match is good to tens of percent, the third and fourth peaks (sensitive to ω_c) being where a Boltzmann code refines it. *Right:* the first-to-second peak height ratio versus the baryon density; the observed ratio $\simeq 2.2$ picks out $\omega_b \simeq 0.0224$—the value SCT must source from the inherited η. Reproducible as `figures/scripts/ch09_cmb_peaks.py`.

troughs, Silk diffusion damps the tail, and radiation driving lifts the first peaks. Building that standard semi-analytic tight-coupling spectrum on the bounce's scale-invariant primordial input reproduces the Planck peak heights to tens of percent (Figure 9.10), the single cleanest feature being the first-to-second peak height ratio, $\simeq 2.2$, which *measures* ω_b. That is the link back to the rest of the framework: the heights demand a definite ω_b (from the inherited η, Chapter 4) and ω_c (the dark sector, Chapter 6), both of which SCT sources independently. A full Boltzmann computation (CAMB/CLASS-class) refines the third and fourth peaks, which are sensitive to ω_c; but at the semi-analytic level the decisive test is met —the bounce's spectrum, run through ordinary recombination, gives the observed peaks in both position and height. Reproducible as `figures/scripts/ch09_cmb_peaks.py`.

Effort estimate. 6–12 months for someone with cosmological-perturbation-theory background; $\sim$ 1000–3000 lines of code building on Tier 1.

9.6 Tier 3: Chirality dynamics through the bounce

Goal. Quantify the flip + filter mechanism. Predict the chirality bias of a baby universe given the parent's properties.

Setup. Use the Tier 1 bounce background; include fermion content explicitly (not just a Weyssenhoff fluid); track baryon number, lepton number, and chiral currents through the bounce; include electroweak physics at temperatures above the EW scale (sphaleron processes); compute the filter efficiency for chirality; and compute net baryon-number production/preservation.

Decisive tests. Does the framework predict $\eta = n_B/n_\gamma \sim 6 \times 10^{-10}$? Does it predict a dark-matter-to-baryon ratio ~ 5? Are these robust to parameter choices, or do they require fine-tuning?

Effort estimate. 12–24 months; significant theoretical work in addition to numerical; coupling Einstein–Cartan to the Standard Model is non-trivial.

9.7 Tier 4: Supraverse population statistics

Goal. Simulate the full supraverse: many vacuum fluctuations producing OG bounces, each seeding lineages of descendant universes. Compute equilibrium distributions of universe parameters; check that observer-supporting universes cluster near our observed parameters.

Setup. Treat the supraverse as a stochastic process: vacuum fluctuations produce OG bounces at random locations; each OG bounce seeds a lineage; each universe evolves cosmologically and produces descendants through internal black-hole formation; track populations across many generations; and compute distributions of universe mass, age, baryon asymmetry, and chirality lineage depth.

Decisive tests. An equilibrium distribution exists and is computable; observer-supporting universes cluster around some peak; and our observed parameters sit near the peak (validating the fine-tuning resolution).

Effort estimate. A multi-year project, several people; requires Tiers 1–3 as input. Output: the most comprehensive cosmological prediction the framework can make.

9.8 Tier 5: Direct observation comparisons

Goal. Use the simulations to make specific predictions for observational programs (Planck, DESI, Euclid, LiteBIRD, JWST, etc.) and compare directly: a precise CMB spectrum shape including specific deviations from ΛCDM; large-scale-structure clustering with dark matter as parent influence; the Hubble parameter at different redshifts; galaxy-rotation asymmetry magnitude from parent rotation; and the dark-energy evolution trajectory.

Decisive tests. The framework is genuinely tested against existing and forthcoming observational data. Each prediction either matches or does not.

Effort estimate. Each comparison is its own project; a multi-decade research program at full scope.

9.9 Minimum viable project

If we want one project that demonstrates the framework's viability: **Tier 2 (perturbation through bounce) with a focus on the CMB power spectrum.** This is decisive—either the bounce reproduces something close to the observed CMB spectrum or it does not. The result is publishable in either case. The effort is bounded at $\sim$ 12 months, and the required expertise (numerical relativity + cosmological perturbation theory) is widely available. This should be the first priority for any simulation effort.

9.10 Visualization parallel track

Alongside the physics simulations, the visualization work (Chapter 10) can proceed independently. It does not require Tier 2 outputs to start—it can begin with simpler models of supraverse structure and refine as physics simulations produce better inputs. The visualization track is more accessible and might be the right place for someone with computational/graphics interests to start.

9.11 Tooling

Languages. Python (prototyping, plotting, analysis; default for Tier 1); Julia (performance-sensitive numerics; optional for Tiers 1–2, advisable for Tiers 3–4); C++ with the Einstein Toolkit (if existing numerical relativity infrastructure is leveraged—heavy but well-tested).

Libraries. NumPy, SciPy, matplotlib (Python); DifferentialEquations.jl (Julia); Einstein Toolkit, GRChombo (numerical relativity); CAMB or CLASS (CMB power spectrum, for comparison to standard predictions).

Hardware. Tiers 1–2: workstation or single-node cluster ($\sim$ hours–days). Tier 3: small cluster ($\sim$ days–weeks). Tier 4: significant cluster resources ($\sim$ weeks–months). Tier 5: depends on the specific calculations.

Storage. Simulations should produce reproducible outputs in standardized formats (HDF5 preferred); visualization data should be portable across plotting backends; code in git, with versioning of major releases.

9.12 Reproducibility

All simulation code should be open source from the start, documented well enough that others can reproduce results, validated against analytic limits where they exist, and tested for numerical convergence (refine grid, check stability). The point is not to "prove" SCT through simulations—the point is to make the framework testable. Reproducible code lets others check, extend, and challenge results.

VISUALIZATION

10.1 The challenge

The supraverse is a 4D manifold with non-trivial topology: a foam-like structure of nested universe-bubbles connected by Cartasis hypersurfaces. This is hard to visualize because 4D structures do not render to 2D screens directly; the foam spans many orders of magnitude (subatomic to cosmic); universes sit inside black holes inside universes inside black holes; time evolution within each universe runs orthogonally to time evolution within others; and the supraverse as a whole is static (a block universe) but locally evolves. Standard scientific visualization techniques do not handle this well. New approaches are needed.

10.2 The gravity-scaled conformal mapping approach

10.2.1 Core idea

Apply a conformal transformation to the supraverse metric,

$$g_{\mu\nu}(x) \;\to\; \Omega^2(x)\, g_{\mu\nu}(x),$$

where $\Omega(x)$ is a function of local gravitational field strength, specifically

$$\Omega(x) \sim \big(G\,\rho(x)\big)^{-1/2}$$

(the inverse square root of local mass–energy density, modulated by gravitational coupling).

10.2.2 What this does

- High-density regions (black-hole interiors, Cartasis membranes) **scale up** in the conformal map.
- Low-density regions (intergalactic voids, supraverse vacuum) **scale down**.
- The conformal transformation **preserves causal structure** (light cones map to light cones).
- Distances are distorted (this is a visualization tool, not a metric for physics).

10.2.3 Visual result

The supraverse, mapped this way, looks like bubbles of comparable visual size regardless of physical scale; bounce membranes appear as prominent surfaces; nested structure means zooming into any high-density region reveals more universes; the appearance is self-similar and fractal-like; and chirality-coded coloring distinguishes matter universes from antimatter universes.

10.2.4 Relationship to Penrose diagrams

Penrose diagrams are standard tools for visualizing general-relativistic spacetimes. They use conformal transformations to compress infinite spacetime onto finite diagrams while preserving causal structure. Penrose's Conformal Cyclic Cosmology uses related techniques to glue cosmological cycles together [17]. Our gravity-scaled mapping is a variant choice: use *gravity* as the conformal factor rather than choices made to fit spacetime onto specific paper shapes. This emphasizes physically meaningful features (high-density structures, bounces) rather than coordinate-system features (asymptotic regions).

10.3 Visualization targets

10.3.1 Single universe with nested black holes

The simplest visualization. Take our universe and show internal structure with galaxies and matter distributions. Within each galaxy, scale up to show black holes; within each black hole, scale up to show the Cartasis membrane; inside the membrane (geometrically inside but topologically connected to a separate region), show the baby universe. This makes the Tardis effect immediately apparent: outside scale small (Schwarzschild radius), inside scale large (baby cosmological extent).

10.3.2 Family tree of lineages

Show our universe's ancestors going up the lineage (parent, grandparent, great-grandparent) and our descendants going down (our internal black holes' baby universes). Use alternating colors for matter/antimatter chirality across generations. This makes the tree structure explicit and visualizes the alternating-chirality prediction.

10.3.3 Supraverse population

Zoom out further to show many independent lineages distributed across the supraverse—equal numbers of matter-led and antimatter-led lineages, with vast vacuum between them. This makes the supraverse-as-foam structure explicit. Figure 10.1 is a first rendering of exactly this: the gravity-scaled foam of the condensed void, zoomed in stages until a single pinprick—home—is found inside a black-hole universe inside an OG universe.

10.3.4 Time evolution within a bubble

Show a single universe evolving from bounce (low-entropy boundary) through cosmological evolution (expansion, structure formation, eventual heat death). Highlight that time runs in a specific direction for each bubble, set by its bounce.

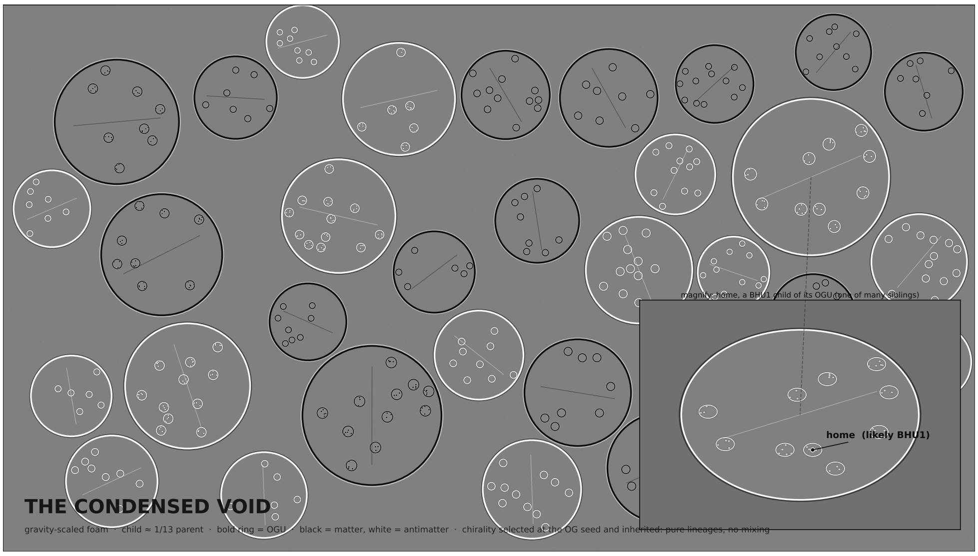

Figure 10.1: The condensed void as gravity-scaled foam, in grayscale so structure rather than colour carries the meaning. A gray void; every universe a gray disc whose Cartasis membrane carries the contrast. OG universes (BHU_0) take a bold double periphery (black ring, white ring inside); nested universes take a single periphery in their parent's chirality colour—black for matter, white for antimatter—inherited down the lineage (pure lineages, no mixing). All bubbles are drawn at comparable visual size regardless of physical scale. The inset zooms into one OGU and marks *home* as a likely BHU_1: a direct child of an OGU and one of its many siblings. We are not an OGU ($n \geq 1$, Section 5.5.3), and the narrow viable band makes BHU_1 the expected depth (Section 5.5.7). Generated by `figures/scripts/ch10_foam.py`.

10.3.5 Cartasis membrane structure

A detailed visualization of a single bounce membrane: parent matter falling in, torsion building up, reaching critical density, bouncing through to the baby. Show the flow of conserved quantities (energy, spin, chirality) across the membrane.

10.4 Implementation

Tools. Three.js or WebGL for browser-based interactive 3D visualization; Python with Mayavi or Julia with Makie for high-quality scientific plots; custom shader code for conformal coordinate transformations; and VR (optional) for immersive supraverse exploration.

Data pipeline. Physics-simulation outputs (Chapter 9) produce data describing supraverse structure; a conformal transformation computes visualization coordinates; geometric primitives (spheres for universes, surfaces for bounces, gradients for matter distributions) are instantiated in a visualization scene; and interactive rendering provides zoom, rotation, and color controls.

Aesthetic choices. A print-safe grayscale scheme reads cleanly and is the default for Figure 10.1: a gray void, gray universe interiors, and all contrast carried by the membranes—black for matter, white for antimatter, with OG universes given a bold black-and-white double periphery. A colour variant (matter in warm yellow/orange, antimatter in cool blue/purple) remains available for screen use. In either scheme, bounce membranes are the prominent surfaces; vacuum regions are subtle and low-contrast, possibly with faint quantum-fluctuation speckle; lighting that suggests "gravity" through subtle radial gradients on each bubble; and, in animations, universes evolving forward in their internal time while the supraverse as a whole appears static.

10.5 "Art" applications

The visualization is genuinely beautiful as art independent of its scientific use: cover art for the eventual paper/book, public-outreach materials, conference presentations, educational tools, and generative art pieces inspired by supraverse structure. This is a side benefit, not the primary purpose, but worth noting because the framework has unusually visualizable structure compared to most theoretical cosmology.

Figure 10.2 is one such piece: the mature foam rendered in colour. It is the same Johnson–Mehl construction as the population model (Chapter 5, Section 5.7.6)—OGUs nucleating, growing at the causal speed, and impinging into curved polygons—now with chirality split into two hue families (warm for matter lineages, cool for antimatter), each cell a gravity-scaled well glowing through its Cartasis membranes, and a hint of nested BHU children in the larger cells. A still frame here; animated, with cells nucleating, swelling, impinging, and slowly coarsening (Section 5.7.8), it is the supraverse as a lava lamp—except this one is our origin. Reproducible as `figures/scripts/ch10_supraverse.py`.

Figure 10.3 is the honest counterpart: the supraverse *true to scale*, in the dilute regime the instanton computation selects (Chapter 5, Section 5.6.1). Because the vacuum has weight ($\rho_\Lambda \sim 6 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$), the void occupies real gravity-scaled volume, so universes are *round* and isolated rather than packed polygons—but they are also vanishingly sparse, so the frame is overwhelmingly void with a scatter of small round islands, and a gravity-scaled zoom is needed to see any structure at all. The packed render above is the legible percolating limit; this is what the physics actually gives.

10.6 Computational notes

Static vs. dynamic. The supraverse is static (a block universe), but each bubble has its own internal dynamics. The visualization can be a static snapshot, an animation showing each bubble evolving in its own time while supraverse-level structure stays fixed, or interactive exploration through the foam.

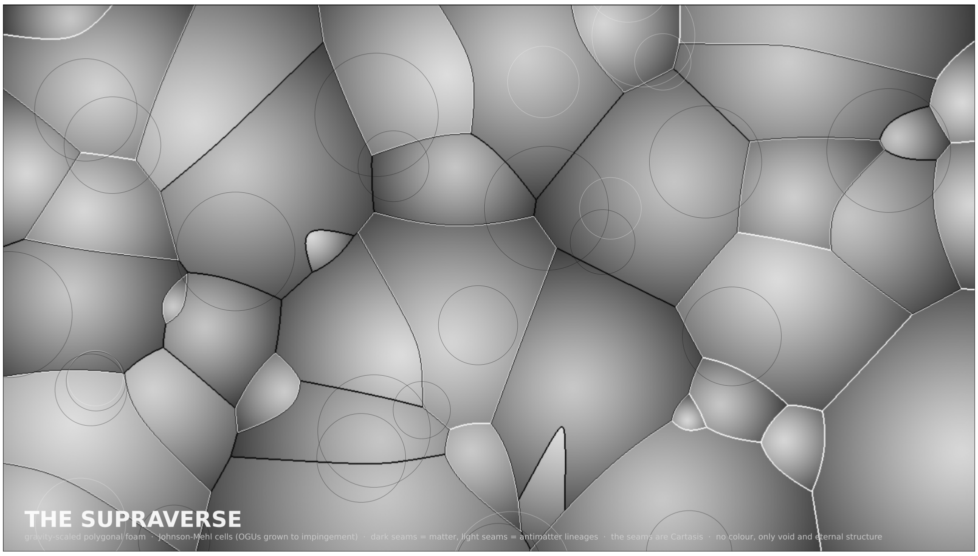

Figure 10.2: The supraverse, gravity-scaled and polygonal: a colour still of the condensed void as a Johnson–Mehl foam. Warm cells are matter lineages, cool cells antimatter (chirality inherited per lineage); each cell is a gravity well bright at its core and dark at the rim, the glowing seams its Cartasis membranes; faint rings in the larger cells are nested BHU children. One lit pixel among the uncountably many is home. (This is the *percolating* limit; the computed nucleation action puts the real supraverse in the dilute regime, Section 5.6.1, where the cells are isolated and round rather than packed polygons—but the packed view shows the structure most legibly.)

Scale handling. Universes span many orders of magnitude. The conformal scaling helps, but additional logarithmic mapping might be needed for some visualizations.

Performance. Real-time interactive exploration with many nested bubbles is computationally demanding. Optimizations: level-of-detail rendering, bubble culling, and asynchronous loading of detailed sub-structures.

10.7 Specific plot ideas

1. **"The Cartasis membrane."** A 2D cross-section showing the density profile of matter falling into a black hole, bouncing at ρ_C, and the geometric transition into the baby universe.

2. **"Tardis effect."** Side-by-side external view (small black-hole sphere) and internal view (vast cosmological space), annotated to show they are connected through the membrane.

3. **"Lineage tree."** A family-tree visualization with our universe at the bottom, ancestors going up, each generation alternating chirality colors, and the OG ancestor at the top.

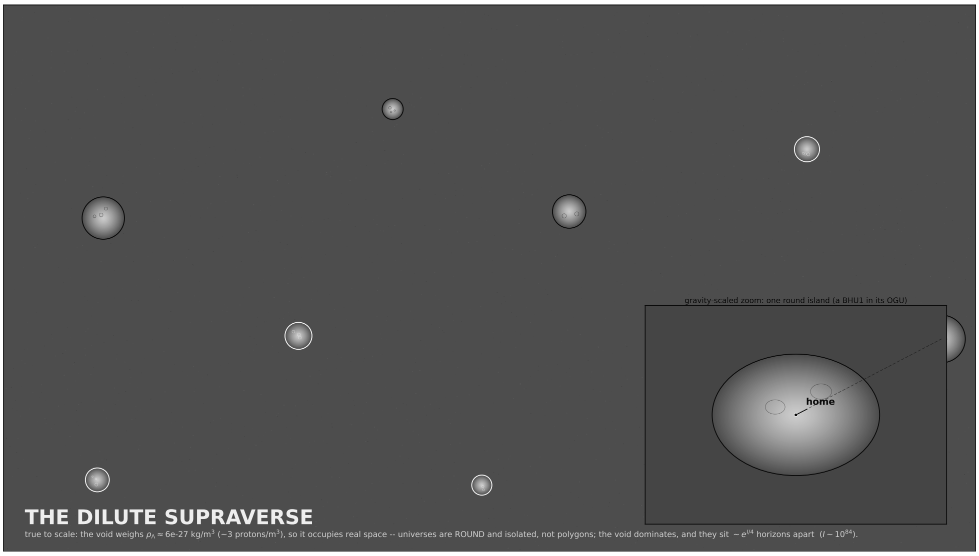

Figure 10.3: The dilute supraverse, true to scale. The weighted void ($\rho_\Lambda \sim 6 \times 10^{-27}\,\mathrm{kg\,m^{-3}}$, faintly speckled with zero-point energy) dominates; universes are sparse, *round*, isolated islands (gravity wells with chirality-toned Cartasis membranes, faint nested BHU children). The inset is a gravity-scaled zoom—the only way to resolve one—marking *home*. The void's nonzero weight is what keeps the islands round rather than polygonal; the dilution makes the gaps enormous.

4. **"Supraverse foam."** A wide-field view of many independent lineages distributed across cosmic vacuum, with the gravity-scaled mapping making all universes appear comparable in size.
5. **"Time arrow."** A diagram showing how time flows away from each bounce in each universe, with different arrows pointing different directions in supraverse coordinates.
6. **"CMB = Hawking radiation."** A diagram showing the geometric identification of the parent's event horizon with the baby's cosmic horizon.
7. **"Dark matter as parent matter."** A diagram showing the parent's matter distribution projected through the Cartasis membrane, manifesting as dark-matter halos in the baby.

10.8 First implementation: simple Python prototypes

To get started without major infrastructure:

```
import matplotlib.pyplot as plt
import numpy as np

def conformal_scale(rho):
    """Gravity-based scaling factor"""
```

```
    return (G * rho)**(-0.5)

def render_universe(ax, x, y, mass, chirality):
    """Render a single universe bubble"""
    radius = conformal_scale(mass / volume(mass))
    color = 'orange' if chirality > 0 else 'blue'
    circle = plt.Circle((x, y), radius, color=color, alpha=0.5)
    ax.add_patch(circle)

def render_supraverse(universes):
    """Render a collection of universes"""
    fig, ax = plt.subplots()
    for u in universes:
        render_universe(ax, u.x, u.y, u.mass, u.chirality)
    plt.show()
```

This is the simplest possible starting point. Real visualizations would be much more sophisticated, but the principle is the same: use gravity-based scaling to make multi-scale supraverse structure visible.

OPEN QUESTIONS

The framework is internally coherent but has many unresolved questions. This appendix catalogs them, both for honesty about the framework's current state and to guide future work.

A.1 Foundational physics

A.1.1 Q1. Precise value of Cartasis density

Current estimates place $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, but this varies by factors of 10–100 depending on coefficient choices in the Einstein–Cartan Lagrangian. A clean derivation pinning the value would be valuable.

A.1.2 Q2. Is Cartasis density universal?

Does the bounce happen at the same density for different matter content (different fermion species, different bound states)? Or is the bounce condition matter-dependent?

A.1.3 Q3. How does Cartasis interact with rotation?

Most analyses assume spherical (Schwarzschild) bounces. Real astrophysical black holes are typically rotating (Kerr). The Kerr–Cartan bounce dynamics may differ qualitatively from Schwarzschild–Cartan and might preferentially produce specific chirality dynamics. Detailed analysis is needed.

A.1.4 Q4. Quantum gravity at Cartasis

The framework treats Einstein–Cartan classically. At $\rho_C \sim 10^{50}\,\mathrm{kg\,m^{-3}}$, full quantum-gravity effects should be subdominant (still 46 orders of magnitude below Planck density), but they might matter for fine details. The full quantum-gravitational picture is unknown.

A.1.5 Q5. Baryon-number conservation

At trans-electroweak temperatures, sphaleron processes violate baryon number. At higher temperatures (approaching Cartasis), more exotic violations might occur. How conserved is B across the bounce? This affects both the baryon-asymmetry mechanism and the dark-matter interpretation.

A.1.6 Q6. Beyond-Standard-Model physics

The Standard Model is presumably an effective theory valid up to some scale. Above that scale, new physics (SUSY? extra dimensions? unknown new fields?) might dominate. Whether SCT is robust to such new physics, or whether it requires the SM to be the full story up to Cartasis, is unclear.

A.2 Bounce dynamics

A.2.1 Q7. The extrusion coefficient C and the remnant fraction

(Superseded framing.) Earlier drafts posed this as a "flip vs. filter" balance with a CPT-like flip dominating. That picture is withdrawn: the minimal Einstein–Cartan bounce is CPT-even and neither flips charge nor filters a pre-existing population (Chapter 4). The live quantitative questions are instead: (i) the anomaly + sphaleron coefficient C in the chiral-vortical extrusion estimate $\eta \sim C(\omega/T)$, which must reach ~ 41 at maximal spin to source the observed asymmetry at the OGU (Section 4.5); and (ii) the fraction of inherited net $B-L$ that survives the extrusion versus is washed out or diluted (Section 4.6, and Q16a). These set the asymmetry magnitude and decide whether lineages are chirally pure or mixed.

A.2.2 Q8. Bounce efficiency

What fraction of the parent's mass–energy is transmitted to the baby's expanding phase? Some might remain in an intermediate region, some might be lost to gravitational waves, some might emerge as Hawking radiation in the parent's exterior. The efficiency factor is currently a free parameter.

A.2.3 Q9. Information flow through the bounce

We have argued that information is preserved globally through entanglement structures across the membrane. The specific quantitative details—what gets entangled with what, how the Page curve is realized in bounce cosmology—are not worked out.

A.2.4 Q10. Persistent channel vs. severed connection

We favor the position that the channel persists after the first bounce over the position that each bounce severs it. But the detailed geometry of a persistent channel that maintains across the parent black hole's lifetime needs rigorous treatment.

A.3 Supraverse structure

A.3.1 Q11. Equilibrium establishment

We argue the supraverse is in thermodynamic equilibrium. How long does establishment take? Is the supraverse currently in equilibrium or still approaching it? What sets the equilibrium timescale?

A.3.2 Q12. OG bounce rate

What is the rate of OG bounce nucleation per unit supraverse spacetime? This determines whether the supraverse contains many universes or few, and the dependence on supraverse parameters.

A.3.3 Q13. Universe size distribution shape

We conjecture the distribution peaks near our observed size. What is the actual shape? How wide is the distribution? Are most observers in universes very close to our parameters, or is the distribution wide enough that observers could exist in significantly different universes?

A.3.4 Q14. Lineage interaction

Do separate lineages (descended from different OG bounces) ever interact? If so, what happens at boundaries? Can matter-lineage and antimatter-lineage universes be near each other in the supraverse?

A.3.5 Q15. Initial conditions for the supraverse

The framework requires the supraverse to be old enough for equilibrium establishment, but does not specify what the supraverse "started from." Was there an initial state, or is the supraverse genuinely eternal? This is partly a question about the meaning of time at the supraverse level.

A.3.6 Q15a. Do OG universes grow forever?

Black holes have negative heat capacity, so above $M_{\text{crit}} = \hbar c^3/(8\pi G k_B T_{\text{bath}})$ a universe-hole accretes faster than it evaporates and grows. The growth-rate model of Section 5.5.8 gives a partial answer: the runaway is fast ($\dot{M} \sim M^2$, finite-time blow-up), so slow cooling cannot regulate it, but *depletion* of the local causal patch does—the hole eats its patch, the source shuts off, and the mass freezes at $M_{\text{final}} \sim$ the patch mass. So individual holes are finite, set by their environment, not unbounded. What remains open is the patch size itself (hence M_{final}) and the void bath temperature T_{bath} that fixes the growth time $\tau_{\text{gen}} \propto T_{\text{bath}}^{-3}$; the latter dials our generation depth (Section 5.5.8) and is entangled with the measure problem below and with Q15c.

A.3.7 Q15b. The measure over infinitely many OG universes

Axiom A1 plus infinity implies infinitely many OG nucleations. Making "we are typical" meaningful requires a regulated measure over that infinite, possibly ever-growing ensemble—the measure problem shared with all multiverse frameworks, here sharpened by ongoing growth (Q15a). Unsolved.

A.3.8 Q15c. The cosmological constant in the void

An infinite, eternally active quantum vacuum makes acute the question of why the void is nearly flat rather than de Sitter at the QFT cutoff. The dark-energy *evolution* is reattributed to parent accretion (Chapter 6), but the bare vacuum-energy problem is untouched, arguably worsened.

A.4 Observational connection

A.4.1 Q16. CMB acoustic peaks

ΛCDM predicts specific acoustic peaks in the CMB power spectrum that match observations beautifully. Can SCT reproduce these from bounce dynamics? This is the central calculational test.

A.4.2 Q16a. Does η measure annihilation ash or Hawking inflow?

The baryon-to-photon ratio $\eta = n_B/n_\gamma \simeq 6 \times 10^{-10}$ has two readings of its *denominator*, and they imply different baryogenesis bookkeeping. (i) *Annihilation ash*: the photons are the baby's own thermal entropy, so η directly measures the extrusion's net-baryon-per-entropy, and the maximal-spin chiral-vortical estimate undershoots by ~ 41 (Chapter 4, Section 4.5). (ii) *Hawking inflow*: if the CMB is parent Hawking radiation (Chapter 6, Section 6.4), the photon denominator is largely *external*, the baby's internal asymmetry is decoupled from the observed η, and the small measured value is partly *dilution* by the Hawking flood—which can dissolve the factor of 41 without a large coefficient C. The *numerator* carries the mirror-image worry: a near-pure matter parent could inject a net $B-L$ far above 1 ppb, so matching $\eta \approx 0.6$ ppb requires either inheritance of the parent's own $\sim$1 ppb ratio (entropy-conserving extrusion) or dilution down to it (Section 4.6). The dilution half of this is now largely answered: the extruder entropy budget (Section 9.4), in both its homogeneous and shear-dominated channels, gives $D = 1$ to $\sim$1 part in 10^{11}, so the extrusion is adiabatic and $\eta_{\text{baby}} = \eta_{\text{parent}}$. What remains is (a) gravitational particle production at the bounce, the one entropy source not yet in the budget, and (b) the genuinely open question—the founder's *value*, why the original-generation seed sits at $\sim 6 \times 10^{-10}$ rather than 10^{-3} or 10^{-15}, which inheritance propagates but does not explain. It is entangled with Q5 (baryon-number conservation) and Q8 (bounce efficiency / entropy budget).

A.4.3 Q17. Hubble tension resolution

We claim SCT might resolve the Hubble tension through modified early-universe dynamics. The actual calculation showing this has not been done. If it cannot be done, the framework needs revision.

A.4.4 Q18. Dark-matter density profile

Observed dark matter has specific clustering properties (NFW profiles, mass–velocity relations). Can these be derived from parent gravitational influence through Cartasis? Or do they imply something else?

A.4.5 Q19. Galaxy-rotation asymmetry consistency

If parent rotation imprints galaxy-rotation asymmetry, the relationship should be specific and predictable. Shamir's observed asymmetry should match a particular parent rotation axis. Whether this consistency holds at higher statistical precision is unknown.

A.4.6 Q20. Specific BBN constraints

Big Bang Nucleosynthesis predicts light-element abundances. Does bounce cosmology reproduce these? If the bounce dynamics differ from inflation + radiation-domination, BBN constraints might require modifications.

A.5 Theoretical issues

A.5.1 Q21. Connection to string theory / loop quantum gravity

How does SCT relate to other proposed extensions of GR? Could string theory or LQG be ways of realizing the bounce in different formal frameworks? Are they competitors or complements?

A.5.2 Q22. Conformal cyclic cosmology relationship

Penrose's CCC is also a non-singular cyclic cosmology [17]. The recursive cycle of Section 5.7.8 gives a concrete relationship: a heat-dead universe's smooth, conformally-flat de Sitter far future is a low-Weyl-entropy void, which in SCT nucleates a *foam* of fresh OGUs rather than handing off to a single conformally-rescaled successor. SCT is, on this reading, "CCC made to branch." What remains open is whether the conformal matching at the dead-aeon/new-bounce boundary is as clean as CCC's—in particular whether the de Sitter nucleation rate (the birth rate β) and the Weyl-entropy reset can be made quantitative rather than schematic.

A.5.3 Q23. ER = EPR specific realization

The bounce structure suggests a geometric realization of ER = EPR. What is the precise mathematical statement? Can it be cleanly derived?

A.5.4 Q24. Holographic principle in the supraverse

The holographic principle suggests that bulk physics can be encoded on boundaries. Does the supraverse have a holographic structure? What is the appropriate boundary? Is each Cartasis membrane a holographic surface for its enclosed baby?

A.5.5 Q25. Anthropic considerations

We have implicitly used anthropic reasoning (observers exist in viable universes). Can this be made more precise? Does the framework predict where observers are in the supraverse population in a quantitatively testable way?

A.6 Philosophical / interpretive

A.6.1 Q26. What is the supraverse "embedded in"?

Mathematically, the supraverse is a 4D manifold with non-trivial topology. Physically, what is its ontological status? Is it the totality of existence, or is there a meta-supraverse containing it?

A.6.2 Q27. Connection between local quantum measurements and bounce dynamics

We argued that gravitational decoherence is the small-scale version of bounce dynamics. The precise relationship needs working out—is every quantum measurement literally a miniature bounce-like splitting? Or just analogous?

A.6.3 Q28. Implications for the meaning of physical constants

If we are typical inhabitants of a supraverse population, our physical constants are typical for our class of universe. Does this mean physical constants are statistical features of the supraverse distribution, or are they fundamental? Or is the distinction meaningless?

A.6.4 Q29. Time at the supraverse level

We argue the supraverse is static (a block universe). But the supraverse also "supports" universe nucleation and equilibrium establishment, which seem to require time. How do we reconcile a static block universe with these dynamical-sounding processes?

A.7 Methodological

A.7.1 Q30. Falsifiability completeness

The framework makes predictions (Chapter 8). Are these collectively sufficient to falsify the framework if it is wrong? Or could the framework be twisted to accommodate any observational result? A genuinely scientific framework should be unambiguously falsifiable. We have argued SCT is, but this should be checked carefully.

A.7.2 Q31. Naturalness arguments

We have argued SCT is "natural" because it requires few postulates. But naturalness arguments have led physics astray before (e.g. naturalness arguments for SUSY have not worked out). Is SCT's parsimony actually evidence for its truth, or just for its appeal?

A.7.3 Q32. Relationship to existing alternative cosmologies

Many alternatives to ΛCDM exist: timescape (Wiltshire), inhomogeneous cosmology, modified gravity (MOND/MOG), cyclic universes. SCT shares features with some of these. Detailed comparison would clarify what SCT uniquely contributes and what it borrows.

A.8 Practical / programmatic

A.8.1 Q33. Publication strategy

If/when this framework is written up rigorously, where would it best be published? Specialized journals (*Physical Review D*, *Classical and Quantum Gravity*)? Or as a book/monograph?

A.8.2 Q34. Collaboration approach

The framework benefits from multiple expertises: numerical relativity, cosmology, quantum field theory, philosophy of physics. How to assemble the right team for serious development?

A.8.3 Q35. Funding considerations

This is theoretical/computational work that does not require large facilities. Modest computational resources, person-time. What funding mechanisms are appropriate?

A.9 Priority ordering

The most important open questions for near-term progress:

1. **Q16 (CMB acoustic peaks):** decisive test; must be addressed for the framework to be taken seriously.
2. **Q7 (flip vs. filter):** resolves chirality dynamics, enables the dark-matter prediction.
3. **Q1 (precise ρ_C):** enables quantitative bounce calculations.
4. **Q12–Q13 (supraverse statistics):** tests observer-placement predictions.
5. **Q17 (Hubble tension):** high-impact if it works.

Items 1–3 are calculable now with current Einstein–Cartan understanding plus standard numerical methods. Items 4–5 require more theoretical development.

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